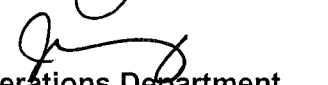


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Reactor Coolant System		
TASK:	Measure Leakage to the Containment Sump		
TASK NUMBER:	N0020080101		
JPM NUMBER:	15-01 NRC RO A1-1		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.25
APPLICABILITY:	IMPORTANCE FACTOR:		3.9
EO ☐	RO ☒	STA ☐	SRO ☐
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	S2.OP-AR.ZZ-0003, Overhead Annunciators Window C, Rev. 19 S2.OP-SO.RC-0004, Identifying and Measuring Leakage, Rev. 16 (Both Rev checked 8-9-16)		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	10 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-9-16
Validated By:	R Evans / N Mulford SME or Instructor	Date:	9-22-16 / 9-29-16
Approved By:	 Operations Training Manager	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

TASK NUMBER: N0020080101

SIMULATOR SETUP N/A

INITIAL CONDITIONS:

Unit 2 is operating at 100% power, steady state, BOL.

At 0015, OHA C-2 CNTMT SUMP PMP START unexpectedly annunciates, along with OHA A-41. The Aux typewriter shows 21 containment sump pump has started. 2 minutes later, OHA C-2 clears, OHA A-41 annunciates, and the Aux typewriter shows 21 containment sump pump has stopped. The plant remains stable, no indications of a RCS leak are apparent, and no operations have been performed which would cause containment sump level to rise.

INITIATING CUE:

You are the RO. Calculate the leak rate into the Containment Sump by performing Section 5.3 of S2.OP-SO.RC-0004, Identifying and Measuring Leakage.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Calculate leakage into Containment Sump of 0.85-1.0 gpm.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide blank copy of S2.OP-SO.RC-0004, and marked up copy of Attachment 1.			
*	SO.RC-4 5.3.1	<u>IF</u> either Containment Sump Pump AUTO starts, <u>THEN RECORD</u> the following on Attachment 1, Section 2.0: • Operating Containment Sump Pump Number • Time of pump start	Records <u>21 Cont Sump</u> Records time of pump start as <u>0015</u>		
*	SO.RC-4 5.3.2	CALCULATE elapsed time in minutes between start time and previous Containment Sump Pump stop time <u>AND RECORD</u> on Attachment 1, Section 2.0.	Calculates elapsed time in minutes between start time and previous Containment Sump Pump stop time and records on Attachment 1, Section 2.0 by using previous stop time of <u>1855 yesterday</u>, and with cue of <u>0015</u> today determines elapsed time is <u>320 minutes</u>.		
	SO.RC-4 5.3.3	ENSURE no draining, sampling, or liquid additions to Containment Sump have occurred during selected time frame.	Determines from initial conditions that no draining, sampling, or liquid additions to Containment Sump have occurred during selected time frame.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	SO.RC-4 5.3.4	CALCULATE Containment Sump Leak Rate using Attachment 3.	Calculates Containment Sump Leak Rate using Attachment 3 (page 1 of 2), and determines the 320 minute line crosses the 0.9 gpm leak rate line. Note: Due to no increment between 300 and 350 minutes, extrapolation of data may result in leak rate slightly above or below 0.9 gpm. However, the leakrate MUST be determined to be within the bounding numbers for actions required >0.85 gpm (to initiate Section 5.6) and >1.0 gpm (to initiate S2.OP-ST.RC-0008)		
	SO.RC-4 5.3.5	RECORD Containment Sump Pump stop time and date on Attachment 1, Section 2.0.	Records Containment Sump Pump stop time as 0017 and today's date on Attachment 1, Section 2.0.		
	SO.RC-4 5.3.6	<u>IF</u> leakage to Containment Sump exceeds 1.0 gpm <u>AND</u> Unit in Modes 1-4, <u>THEN</u> : A. INITIATE S2.OP-ST.RC-0008(Q), Reactor Coolant Water Inventory Balance. B. REFER to Technical Specification 3.4.7.2.	Determines leakage to Containment Sump does not exceed 1.0 gpm		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	SO.RC-4 5.3.7	IF leakage to Containment Sump exceeds 0.85 gpm, <u>THEN</u> INITIATE Section 5.6 of this procedure.	Determines leakage to Containment Sump does exceed 0.85 gpm, and initiates Section 5.6 of procedure.		
			Terminate JPM when procedure is returned to evaluator.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- W M ✓ 1. Task description and number, JPM description and number are identified.
- W M ✓ 2. Knowledge and Abilities (K/A) references are included.
- W M ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- W M ✓ 4. Initial setup conditions are identified.
- W M ✓ 5. Initiating and terminating Cues are properly identified.
- W M ✓ 6. Task standards identified and verified by SME review.
- W M ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W M ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 19 Date 9-22-16
16
- M ✓ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature]

SME/Instructor: [Signature] N. M. Ford

SME/Instructor: _____

Date: 9-22-16

Date: 9/29/16

Date: _____

INITIAL CONDITIONS:

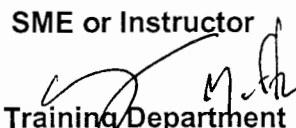
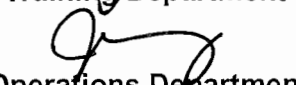
Unit 2 is operating at 100% power, steady state, BOL.

At 0015, OHA C-2 CNTMT SUMP PMP START unexpectedly annunciates, along with OHA A-41. The Aux typewriter shows 21 containment sump pump has started. 2 minutes later, OHA C-2 clears, OHA A-41 annunciates, and the Aux typewriter shows 21 containment sump pump has stopped. The plant remains stable, no indications of a RCS leak are apparent, and no operations have been performed which would cause containment sump level to rise.

INITIATING CUE:

You are the RO. Calculate the leak rate into the Containment Sump by performing Section 5.3 of S2.OP-SO.RC-0004, Identifying and Measuring Leakage.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Conduct of Operations		
TASK:	Calculate Shutdown Margin		
TASK NUMBER:	N1200030301		
JPM NUMBER:	15-01 NRC RO A1-2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.43
APPLICABILITY:	IMPORTANCE FACTOR: 4.1		
EO ☐	RO ☒	STA ☐	SRO ☐
EVALUATION SETTING/METHOD:		Classroom	
REFERENCES:	SC.RE-ST.ZZ-0002(Q), Rev. 23 SHUTDOWN MARGIN CALCULATION S2.RE-RA.ZZ-0016(Q), Rev. 8 CURVE BOOK (both checked 8-9-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	30 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	50 minutes		
Developed By:	G Gauding Instructor	Date:	8-9-16
Validated By:	R Evans / N Mulford SME or Instructor	Date:	9-22-16 / 9-29-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Conduct of Operations

TASK: Calculate Shutdown Margin

TASK NUMBER: N1200030301

INITIAL CONDITIONS:

- Unit 2 Reactor Power is 100%.
- All Control Rod Group Demand Counters are at 227 steps.
- Current boron concentration is 300 ppm.
- Current core burnup is 10,000 EFPH.
- Control Rod 1D5 was been declared INOPERABLE at 1000 today due to not moving out during a rod exercise test. You have been directed to perform a Shutdown Margin Calculation IAW SC.RE-ST.ZZ-0002 to satisfy the action requirement of TSAS 3.1.3.1 Action c.3.
- Control Rod 1D5 has been identified as trippable and is currently at 214 steps.

INITIATING CUE:

The current time is 1010.

This is a time critical JPM. Time will commence when you are provided with the applicable procedures.

When performing calculation, the CRS directs you to use only the tables as directed by SC.RE-ST.ZZ-0002, do not use Figures.

Successful Completion Criteria:

1. All critical steps completed
2. All sequential steps completed in order
3. All time-critical steps completed within allotted time
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Perform SDM calculation and determine SDM is -2441 (+/- 5 pcm) and SAT.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: Conduct of Operations

Task: Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Obtains copy of procedure SC.RE-ST.ZZ-0002(Q). Obtains copy of S2.RE-RA.ZZ-0016 Curve Book.	Provide Rev. 23 of surveillance. Provide Rev. 8 of Curve Book. Evaluator: MARK start time of JPM: _____		
	3.0	Reviews and signs off Precautions and Limitations Section 3.0	Reviews Precautions & Limitations and signs off Steps 3.1 – 3.7.		
	5.1	SELECT the applicable step below (e.g. 5.1.1.A):	Determines 5.1.1.C is the correct step for the current conditions, which is to complete Attachment 3.		
	5.1.1.C	<u>IF</u> SDM is to be verified (per T/S 4.1.1.1.1.a) with reactor critical (Mode 1 or 2) and one or more control rods are inoperable, <u>THEN COMPLETE</u> Attachment 3.	Refers to Attachment 3.		
	Att. 3 2.1	The reactor is in Mode 1 or Mode 2 with $k_{eff} \geq 1.0$	Determines Unit is in Mode 1.		
	Att. 3 3.0	Precautions and Limitations: Section 3.0	Reviews and signs steps 3.1-3.5.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

System: Conduct of Operations

Task: Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att. 3 Section 4.1	Section 4 contains the data points required and the calculations to determine shutdown margin.	Determines the following data and calculations: 4.1.1 100% 4.1.2 300 ppm 4.1.3 D @ 227 steps 4.1.4 10,000 EFPH 4.1.5 0 4.1.6 1		
*	Att. 3 Section 4.2		4.2.1 -3841 4.2.2 -3335 4.2.3 728 4.2.4 0 4.2.5 500 4.2.6 0 4.2.7 -5948		
*	Att. 3 Section 4.3		4.3.1 -5948 4.3.2 645 4.3.3 120 4.3.4 2742 4.3.5 -2441		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

System: Conduct of Operations

Task: Calculate Shutdown Margin

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	4.4	Att 4 Acceptance Criteria 4.4.1 IS THE SDM (Item 4.3.5) EQUAL TO OR MORE NEGATIVE THAN (-)1300 PCM?	Reviews required SDM in MODE 1 or 2 of -1300 pcm and initials 4.4.1. Determines calculated SDM is SAT and initials SAT.		
*			Evaluator: MARK stop time of JPM: _____ when procedures are returned to proctor. Total time from START to STOP is 50 minutes or less.		
			Terminating Cue: Once operator has returned procedures to proctor, terminate JPM.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- M ✓ 1. Task description and number, JPM description and number are identified.
- M ✓ 2. Knowledge and Abilities (K/A) references are included.
- M ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- M ✓ 4. Initial setup conditions are identified.
- M ✓ 5. Initiating and terminating Cues are properly identified.
- M ✓ 6. Task standards identified and verified by SME review.
- M ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- M ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 23 Date 8-22-16
8
- M ✓ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: 8-22-16

SME/Instructor: N. Mulford

Date: 9/29/16

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

- Unit 2 Reactor Power is 100%.
- All Control Rod Group Demand Counters are at 227 steps.
- Current boron concentration is 300 ppm.
- Current core burnup is 10,000 EFPH.
- Control Rod 1D5 was been declared INOPERABLE at 1000 today due to not moving out during a rod exercise test. You have been directed to perform a Shutdown Margin Calculation IAW SC.RE-ST.ZZ-0002 to satisfy the action requirement of TSAS 3.1.3.1 Action c.3.
- Control Rod 1D5 has been identified as **trippable** and is currently at 214 steps.

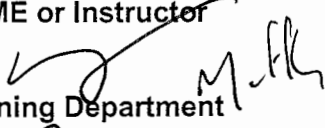
INITIATING CUE:

The current time is 1010.

This is a time critical JPM. Time will commence when you are provided with the applicable procedures.

When performing calculation, the CRS directs you to use only the Tables as directed by SC.RE-ST.ZZ-0002, do not use Figures.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM 1 & 2		
SYSTEM:	Chemical and Volume Control		
TASK:	Demonstrate the Operability of the RWST and/or the BASTs		
TASK NUMBER:	N0040420201		
JPM NUMBER:	15-01 NRC RO Admin A2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.2.37
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Classroom Simulator - Perform 9/11/16		
REFERENCES:	S2.OP-ST.CVC-0010, Rev. 10 (checked 9-28-16) Salem Tech Specs		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	11 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-9-16
Validated By:	S Harris / J Pierce SME or Instructor	Date:	9-28-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Demonstrate the Operability of the RWST and/or the BASTs

TASK NUMBER: N0040420201

SIMULATOR SETUP: N/A

INITIAL CONDITIONS:

- Salem Unit 2 is at 100% power.
- 21 BAT pump developed a leak that went unnoticed until the cross-connected BAST's reached their low level alarm setpoint.
- 21 BAT pump, and the leak, are isolated.
- 21 BAST level – 49.5%
- 22 BAST level – 49.5%
- 21 BAST temp – 98.2°F
- 22 BAST temp – 97.1°F
- RWST temperature – 70°F
- RWST levels are:
 - CH I – 41.4'
 - CH II - 41.4'
 - CH III – 41.4'
 - CH IV – 41.5'
- Chemistry reports current RWST and BAST boron concentrations are:
 - 21 BAST – 6610 ppm
 - 22 BAST – 6610 ppm
 - RWST – 2350 ppm
- Off Normal and Off-Normal Tagged list review has been performed SAT.

INITIATING CUE:

You are the Reactor Operator. Perform S2.OP-ST.CVC-0010, Borated Water Sources.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Complete Attachment 2 of S2.OP-ST.CVC-0010 per key and determine the Surveillance is SAT

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Demonstrate the Operability of the RWST and/or the BASTs

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide marked up copy of S2.OP-ST.CVC-0010, Borated Water Sources and Tech Spec 3.1.2.6 (2 pages)			
	3.0	Precautions and Limitations	Reads and initials steps.		
	5.1.1	COMPLETE Attachment 2 per the following: • RECORD Data. • RECORD Test Results by initialing SAT or UNSAT column using the stated Acceptance Criteria.	Completes Attachment 2 with the below data.		
*	Att. 2	RWST Level	Records RWST level with channel for which calibration data was obtained RWST levels are: CH I: 41.4 CH II: 41.4 CH III: 41.4 CH IV: 41.5 Initials SAT for RWST Level and inserts date.		
*	Att. 2	RWST Temp	Refers to initial conditions for RWST temperature 70°F . Initials SAT for RWST Temp and inserts date.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Demonstrate the Operability of the RWST and/or the BASTs

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att. 2	RWST Conc	Refers to initial conditions for RWST boron concentration and enters 2350 ppm. Initials SAT for RWST boron concentration and inserts date.		
*	Att. 2	21 BAST Vol <u>AND/OR</u> 22 BAST Vol	Refers to initial conditions for 21 and 22 BAST level and enters: 21 BAST vol – 49.5% 22 BAST vol - 49.5% Uses Tech Spec 3.1.2.6.a, 3.1.2.6.b, and Figure 3.1-2 and determines 21 BAST Vol <u>AND/OR</u> 22 BAST Vol is SAT . Initials SAT for BAST volume and inserts date.		
*	Att. 2	21 BAST Temp <u>AND/OR</u> 22 BAST Temp	Refers to initial conditions for 21 and 22 BAST temp and records: 21 BAST temp - 98.2°F 22 BAST temp - 97.1°F Initials SAT for 21 BAST Temp <u>AND/OR</u> 22 BAST Temp and inserts date.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Demonstrate the Operability of the RWST and/or the BASTs

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att. 2	21 BAST Conc <u>AND/OR</u> 22 BAST Conc	Refers to initial conditions for BAST boron concentrations and enters 6610 ppm for 21 BAST and 6610 ppm for 22 BAST. Uses Tech Spec 3.1.2.6.a, 3.1.2.6.b, and Figure 3.1-2 and determines 21 and 22 BAST boron concentration is SAT. Initials and inserts date.		
	5.3.1 5.3.2	This surveillance is considered satisfactory when Attachments 2 and 3 (as applicable) are complete with components listed meeting Acceptance Criteria stated in each Attachment. <u>OR</u> This surveillance is unsatisfactory. A. INITIATE Notification(s) to correct the unsatisfactory condition. B. RECORD Notification number(s) and reason for unsatisfactory completion on Attachment 4 in the Comments Section.	Initials step 5.3.1 N/A's 5.3.2.		
	5.4.1	COMPLETE Attachment 4, Sections 1.0 and 2.0, <u>AND FORWARD</u> this procedure to the SM/CRS for review.	Completes Attachment 4, Sections 1.0 and 2.0, and returns procedure to JPM Evaluator.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Chemical and Volume Control

TASK: Demonstrate the Operability of the RWST and/or the BASTs

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			JPM is complete when procedure is returned to evaluator.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- W ④ AS 1. Task description and number, JPM description and number are identified.
- W AS 2. Knowledge and Abilities (K/A) references are included.
- W AS 3. Performance location specified. (in-plant, control room, or simulator)
- W AS 4. Initial setup conditions are identified.
- W AS 5. Initiating and terminating Cues are properly identified.
- W AS 6. Task standards identified and verified by SME review.
- W AS 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W AS 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 10 Date 4/30/13
- W AS 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: ④ AS for J. Pierce

Date: 9/28/16

SME/Instructor: AS for S. Harris

Date: 9/28/16

SME/Instructor: _____

Date: _____

Original checklist inadvertently discarded. AS 10/11/16

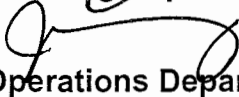
INITIAL CONDITIONS:

- Salem Unit 2 is at 100% power.
- 21 BAT pump developed a leak that went unnoticed until the cross-connected BAST's reached their low level alarm setpoint.
- 21 BAT pump, and the leak, are isolated.
- 21 BAST level – 49.5%
- 22 BAST level – 49.5%
- 21 BAST temp – 98.2°F
- 22 BAST temp – 97.1°F
- RWST temperature – 70°F
- RWST levels are:
 - CH I – 41.4'
 - CH II - 41.4'
 - CH III – 41.4'
 - CH IV – 41.5'
- Chemistry reports current RWST and BAST boron concentrations are:
 - 21 BAST – 6610 ppm
 - 22 BAST – 6610 ppm
 - RWST – 2350 ppm
- Off Normal and Off-Normal Tagged list review has been performed SAT.

INITIATING CUE:

You are the Reactor Operator. Perform S2.OP-ST.CVC-0010, Borated Water Sources Data.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ADMINISTRATIVE (ALARA)		
TASK:	Perform Stay Time Calculation for Emergency Condition.		
TASK NUMBER:	N1200100104		
JPM NUMBER:	15-01 NRC RO A3		
ALTERNATE PATH:	☐	K/A NUMBER:	2.3.4
		IMPORTANCE FACTOR:	3.2
APPLICABILITY:		RO	SRO
EO	☐	RO	☒
STA	☐	SRO	☐
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	Radiological Survey Maps, RP-AA-203 Rev. 6, NC.EP-EP.ZZ-0304 Rev. 16		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	10 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-9-16
Validated By:	R Evans / N Mulford SME or Instructor	Date:	9-22-16 / 9-29-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: ADMINISTRATIVE (ALARA)

TASK: Perform Stay Time Calculation for Emergency Condition.

TASK NUMBER: N1200100104

INITIAL CONDITIONS:

1. Unit 1 experienced Rx trip with a small RCS leak from 100% power.
2. The Rx failed to automatically trip on a valid trip demand, and was tripped successfully with the Rx trip handle.
3. Salem is currently in an Alert.
4. Your TEDE dose for the year is 1925 mrem.

INITIATING CUE:

You have been directed to perform a detailed inspection of 22 RHR pump room prior to starting the pump. You will **NOT** be going into the 22 RHR HX area. Your job is estimated to take 2 hours to complete.

Using the correct survey map, and conservatively using the **HIGHEST** dose rate anywhere in the pump room for your **ENTIRE** exposure, determine:

1. Can you complete the job without exceeding your **current** dose authorization
2. What would be your new TEDE dose for the year after performing the job (or as much as possible) **WITHOUT** exceeding your **current** dose authorization?

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:
Determines:

1. YES job can be performed
2. Calculates new TEDE dose of 2005 mrem.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: ADMINISTRATIVE (ALARA)

TASK: Determine Radiological Conditions For Personnel Exposure

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	1	Provide the attached survey maps of 21 and 22 RHR rooms, and S1 and S2 55 General area. Also have blank copies of RP-AA-203, Exposure Control and Authorization and NC.EP-EP.ZZ-0304, Operational Support Center (OSC) Radiation Protection Response if asked for.	Selects the Survey Map "S2 AUX 045' 22 RHR ROOMS" Map # 21045Z2		
		START TIME: _____			
		If asked, provide NC.EP-EP.ZZ-0304, OSC Radiation Protection Response.	Note: Contains guidance for automatic increase in allowed dose to 4500 mrem. (Section 5.0 Note.)		
			Determines dose limit is automatically raised to 4500 mrem upon the declaration of an ALERT or higher Emergency.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: ADMINISTRATIVE (ALARA)

TASK: Determine Radiological Conditions For Personnel Exposure

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	2	Determine highest dose rate in room.	Per the survey map, squares are dose rates in mrem/hr, and circles are smear locations. The highest square is located under the words "22RH37" on the map and indicates 40 mrem/hr. Note: The square with "250" inside it is in the RHR HX room, and is not in the area which the operator will go.		
*	3	Calculate TEDE dose for year	Subtracts year to date TEDE dose from Emergency Dose limit of 4,500 mrem, and gets 2575 mrem . Determines 2 hours of work will result in 80 mrem dose , and new TEDE dose is 2005 mrem , which does NOT exceed 4,500 mrem authorized in an Emergency.		
			Terminate JPM when candidate has returned paperwork.		
	4	STOP TIME: _____			

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- W M ✓ 1. Task description and number, JPM description and number are identified.
- W M ✓ 2. Knowledge and Abilities (K/A) references are included.
- W M ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- W M ✓ 4. Initial setup conditions are identified.
- W M ✓ 5. Initiating and terminating Cues are properly identified.
- W M ✓ 6. Task standards identified and verified by SME review.
- W M ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W M ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 6 Date 5-22-16
16
- M ✓ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: _____

Date: 9-22-16

SME/Instructor: M. J. Ford

Date: 9/29/16

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

1. Unit 1 experienced Rx trip with a small RCS leak from 100% power.
2. The Rx failed to automatically trip on a valid trip demand, and was tripped successfully with the Rx trip handle.
3. Salem is currently in an Alert.
4. Your TEDE dose for the year is 1925 mrem.

INITIATING CUE:

You have been directed to perform a detailed inspection of 22 RHR pump room prior to starting the pump. You will **NOT** be going into the 22 RHR HX area. Your job is estimated to take 2 hours to complete.

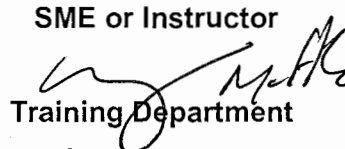
Using the correct survey map, and conservatively using the **HIGHEST** dose rate anywhere in the pump room for your **ENTIRE** exposure, determine:

1. Can you complete the job without exceeding your **current** dose authorization
2. What would be your new TEDE dose for the year after performing the job (or as much as possible) **WITHOUT** exceeding your **current** dose authorization?

1. _____

2. _____

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ADMINISTRATIVE		
TASK:	Review a completed surveillance to demonstrate Operability of 21 CFCU		
TASK NUMBER:	N1230300302		
JPM NUMBER:	15-01 NRC SRO Admin A1-1		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.25
APPLICABILITY:	IMPORTANCE FACTOR:		4.2
EO ☐	RO ☐	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	S2.OP-ST.CBV-0003, Rev. 19 (checked 8-10-16)		
TOOLS AND EQUIPMENT:			
VALIDATED JPM COMPLETION TIME:	12 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-10-16
Validated By:	R Boyer / P Williams SME or Instructor	Date:	9-28-16 / 10-4-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Administrative

TASK: Review a completed surveillance to demonstrate Operability of 21 CFCU

TASK NUMBER: N1230300302

INITIAL CONDITIONS: 21 CFCU was returned from corrective maintenance. S2.OP-ST.CBV-0003, Containment Systems - Containment Cooling was performed for the Operability Retest.

INITIATING CUE: You have been directed to review the surveillance for completeness and accuracy IAW Step 5.3.4.

If any discrepancies are identified, note them and any actions that you would take on Attachment 8, Section 1.0, Comments.

Successful Completion Criteria:

1. All critical steps completed
2. All sequential steps completed in order
3. All time-critical steps completed within allotted time
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Identify that the Minimum Flow Rate recorded on Attachment 1, Section 3.0, for 87 psid is incorrectly recorded as 1592. Identify the correct Minimum Flow Rate should be 1638.
2. Identify that the actual Cooling Water Flow Rate (1625) is below the Minimum Flow Rate required as shown on Attachment 1 and Exhibit 1, and the surveillance is UNSAT vs SAT.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Review a completed surveillance to demonstrate Operability of 21 CFCU

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Provide completed S2.OP-ST.CBV-0003 surveillance.		
			Reviews procedure.		
			Identifies that the Minimum Flow Rate recorded on Attachment 1, 21 CFCU Data Sheet, Section 3.0, for 87 psid is incorrectly recorded as 1592. (This was due to operator performing ST rounding down as opposed to rounding up IAW P&L 3.4.) Identifies correct value for Minimum Flow Required as 1638. Cue: IF candidate only determines flow is wrong, THEN ask candidate to determine correct flow. Identifies that the actual Cooling Water Flow Rate of 1625 gpm is below the Minimum Flow Rate of 1638 as shown on Exhibit 1 and the ST is UNSAT.		
			When candidate turns in JPM, ensure they have provided correct flow as per Cue above, then state JPM is complete.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 4

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ☒ 1. Task description and number, JPM description and number are identified.
- ☒ 2. Knowledge and Abilities (K/A) references are included.
- ☒ 3. Performance location specified. (in-plant, control room, or simulator)
- ☒ 4. Initial setup conditions are identified.
- ☒ 5. Initiating and terminating Cues are properly identified.
- ☒ 6. Task standards identified and verified by SME review.
- ☒ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ☒ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 19 Date 9/22/16
- ☒ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature] 130X101

SME/Instructor: [Signature] Williams

SME/Instructor: _____

Date: 9/28/16

Date: 10/4/16

Date: _____

INITIAL CONDITIONS:

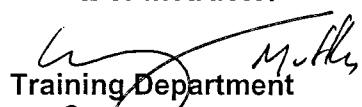
21 CFCU was returned from corrective maintenance. S2.OP-ST.CBV-0003, Containment Systems - Containment Cooling was performed for the Operability Retest.

INITIATING CUE:

You have been directed to review the surveillance for completeness and accuracy IAW Step 5.3.4.

If any discrepancies are identified, note them and any actions that you would take on Attachment 8, Section 1.0, Comments.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ADMINISTRATIVE		
TASK:	Direct Actions for Spent Fuel Movement (Determine equipment allowed to be removed from service while SFP manipulations are in progress)		
TASK NUMBER:	N1120760102		
JPM NUMBER:	15-01 NRC SRO A1-2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.42
APPLICABILITY:	IMPORTANCE FACTOR:		3.4
EO ☐	RO ☐	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Classroom or Simulator		
REFERENCES:	S2.OP-IO.ZZ-0010, Rev. 33 , S2.OP-SO.CH-0001 Rev 32, S2.OP-SO.SW-0005 Rev 42 S2.OP-SO.DG-0005, Rev. 6 (all rev checked 8-10-16) Salem Tech Specs		
TOOLS AND EQUIPMENT:			
VALIDATED JPM COMPLETION TIME:	30 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-10-16
Validated By:	R Boyer / P Williams SME or Instructor	Date:	9-28-16 / 10-4-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:	DATE:		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Administrative

TASK: Direct Actions for Spent Fuel Movement (Determine equipment allowed to be removed from service while SFP manipulations are in progress)

TASK NUMBER: N1120760102

INITIAL CONDITIONS: Salem Unit 2 is operating at 99% power, coasting down into a refueling outage which will start in 2 weeks.
Irradiated fuel movement is in progress in the Spent Fuel Pool IAW S2.OP-IO.ZZ-0010, Spent Fuel Pool Manipulations, and will continue without interruption for the next 8 hours, at which time it will terminate.
The unit has no active Tech Spec LCOs.
21 SW pump is C/T for emergent strainer work.
The current time is 0800.

INITIATING CUE: You have been directed to review the upcoming work for the shift and determine which activities may proceed as scheduled while continuing the fuel movement in the Spent Fuel Pool uninterrupted.

Work scheduled for today Circle whether each activity may or may not be performed

IF any of the activities cannot proceed, explain your reason(s)

0900- 22 SW pump packing replacement	MAY	MAY NOT
1000- 2A EDG tagout for maintenance	MAY	MAY NOT
1300- 21 Chiller compressor oil change	MAY	MAY NOT
1400- Delivery of New Fuel into the Fuel Handling Building for upcoming refueling outage.	MAY	MAY NOT

Successful Completion Criteria:

1. All critical steps completed
2. All sequential steps completed in order
3. All time-critical steps completed within allotted time
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Determines that 22 SW pp work, 2A EDG surveillance, and 21 Chiller work MAY be performed.
2. Determines that New Fuel delivery may NOT be performed.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Direct Actions for Spent Fuel Movement (Determine equipment allowed to be removed from service while SFP manipulations are in progress)

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide blank copies of: S2.OP-IO.ZZ-0010, Spent Fuel Pool Manipulations. S2.OP-SO.DG-0005, Preparation for Removing a Diesel Generator from Service. S2.OP-SO.SW-0005, Service Water System Operation. S2.OP-SO.CH-0001, Chilled Water System Operation. Provide copies of Tech Specs.			

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

System: ADMINISTRATIVE

Task: Direct Actions for Spent Fuel Movement (Determine equipment allowed to be removed from service while SFP manipulations are in progress)

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Notes: TSAS 3.8.1.2 applies during Modes 5 and 6 and during movement of irradiated fuel, and only requires two EDGs. Only one FHB exhaust fan is required to be backed by an operable EDG (IOP-10 P&L 3.7 and page 17) Operable SFP pumps only need one of them backed by operable EDG. (IOP-10 P&L 3.4 and page 17) TSAS 3.7.10 allows a chiller to be taken out of service if movement of irradiated fuel is already on-going, and gives 14 days to restore it. Two SW pumps are required to be operable, one of which is EDG backed. (IOP-10 page 18.) The four remaining SW pumps are powered from B and C vital and their EDGs remain operable. TSAS 3.7.4 will be entered for less than two operable SW loops, but is a 72 hour action.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

System: ADMINISTRATIVE

Task: Direct Actions for Spent Fuel Movement (Determine equipment allowed to be removed from service while SFP manipulations are in progress)

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Notes: (cont) FHB Truck Bay Door is required to be closed. (IOP-10 page 19)		
* * * *			Determines that: 22 SW pump work <u>MAY</u> be performed. 2A EDG surveillance <u>MAY</u> be performed. 21 Chiller oil replacement <u>MAY</u> be performed. New Fuel delivery may <u>NOT</u> be performed because FHB Truck Bay door is required to be closed per IOP-10, page 19. Evaluator ensure tear off sheet has candidates name and status for each of the four work activities, then terminate JPM.		

Name: Robert Boyer

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 4

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- 1 A' O 1. Task description and number, JPM description and number are identified.
- 1 B O 2. Knowledge and Abilities (K/A) references are included.
- 1 A O 3. Performance location specified. (in-plant, control room, or simulator)
- 1 A O 4. Initial setup conditions are identified.
- 1 A O 5. Initiating and terminating Cues are properly identified.
- 1 B O 6. Task standards identified and verified by SME review.
- 1 A O 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 1 A O 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. xxxxx Date 1/21/16
- 1 A O 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature]

SME/Instructor: [Signature]

SME/Instructor: _____

Date: 9/28/16

Date: 10/4/16

Date: _____

Name: _____

INITIAL CONDITIONS:

Salem Unit 2 is operating at 99% power, coasting down into a refueling outage which will start in 2 weeks.

Irradiated fuel movement is in progress in the Spent Fuel Pool IAW S2.OP-IO.ZZ-0010, Spent Fuel Pool Manipulations, and will continue without interruption for the next 8 hours, at which time it will terminate.

The unit has no active Tech Spec LCOs.

21 SW pump is C/T for emergent strainer work.

The current time is 0800.

INITIATING CUE:

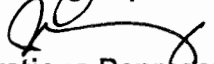
You have been directed to review the upcoming work for the shift and determine which activities may proceed as scheduled while continuing the fuel movement in the Spent Fuel Pool uninterrupted.

Work scheduled for today _____ Circle whether each activity may or may not be performed

IF any of the activities cannot proceed, explain your reason(s)

0900- 22 SW pump packing replacement	MAY	MAY NOT
1000- 2A EDG tagout for maintenance	MAY	MAY NOT
1300- 21 Chiller compressor oil change	MAY	MAY NOT
1400- Delivery of New Fuel into the Fuel Handling Building for upcoming refueling outage.	MAY	MAY NOT

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ADMINISTRATIVE		
TASK:	Review a completed surveillance to calculate Shutdown Margin		
TASK NUMBER:	1200030301		
JPM NUMBER:	15-01 NRC SRO A2		
ALTERNATE PATH:	☐	K/A NUMBER:	2.2.12
		IMPORTANCE FACTOR:	4.1
APPLICABILITY:		RO	SRO
EO ☐	RO ☐	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	SC.RE-ST.ZZ-0002, Rev. 23 Shutdown Margin Calculation S2.RE-RA.ZZ-0016, Rev. 8 Figures Salem Unit 2 Technical Specifications TSAS 3.1.3.1 Movable Control Assemblies		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	20 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	30 minutes		
Developed By:	G Gauding Instructor	Date:	8-10-16
Validated:	R Boyer / P Williams SME or Instructor	Date:	9-28-16 / 10-4-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Representative	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Administrative

TASK: Review a completed surveillance to calculate Shutdown Margin

TASK NUMBER: 1200030301

INITIAL CONDITIONS:

- Unit 2 Reactor Power is 100%.
- All Control Rod Group Demand Counters are at 227 steps.
- Current boron concentration is 300 ppm.
- Current core burnup is 10,000 EFPH.
- Control Rod 1D5 was declared INOPERABLE at **1000 today** due to not moving out during a rod exercise test. A Shutdown Margin Calculation has been performed IAW SC.RE-ST.ZZ-0002, Shutdown Margin Calculation, to satisfy the action requirement of TSAS 3.1.3.1 Action c.3.
- Control Rod 1D5 has been identified as **trippable** and is currently at 214 steps.
- The current time is 1030.

INITIATING CUE:

1. Review the attached SDM calculation procedure SC.RE-ST.ZZ-0002 for completeness and accuracy.
2. Note any discrepancies found by the review on Attachment 7.
3. Determine if SDM is SAT.

This is a time critical JPM. Time will start when procedures are given to you. Time will stop when procedures are returned to proctor.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made

Task Standard for Successful Completion:

1. Determines the incorrect value for Control Bank Worth and Shutdown Bank Worth have been used.
2. Determines SDM with correct values used is SAT.
3. Completes review within 30 minutes.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Review a completed surveillance to calculate Shutdown Margin

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
			Note to Evaluator: The time critical aspect refers to the time required when the Shutdown Margin requirement of TSAS 3.1.3.1.c must be met.		
	1	Provide the marked up copy of SC.RE-ST.ZZ-0002 and copy of S2.RE-RA.ZZ-0016, Curve Book. Time Start: _____	Ensures Sections 3 and 4 are completed.		
	2	Review Section 5 to ensure the correct Attachment was performed.	Reviews Section 5 and determines the correct Attachment (3) was performed.		
* *	3	Review the completed Shutdown Margin Calculation:	Reviews SDM and determines: 1. An incorrect value for Total Control Bank Worth (Step 4.2.1) has been used (0 EFPH value vs current 10,000 EFPH value) 2. An incorrect value for Total Shutdown Bank Worth (Step 4.2.2) has been used (0 EFPH value vs current 10,000 EFPH value)		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Review a completed surveillance to calculate Shutdown Margin

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
* * *	3 con't		3. Determines Trippable Rod Worth is incorrect based on incorrect steps 4.2.1 and 4.2.2) 4. Determines Ten Percent Rod Worth Penalty (step 4.3.2) is incorrect based on incorrect steps 4.2.1 and 4.2.2) 5. Determines SDM is incorrect based on incorrect steps 4.2.1 and 4.2.2 and 4.3.2) Correct Values are: 4.2.1 -3841 4.3.1 -5948 4.2.2 -3335 4.3.2 645 4.2.3 728 4.3.4 2742 4.2.4 0 4.3.5 -2441 4.2.5 500 4.2.6 0 4.2.7 -5948		
* *	4	Acceptance Criteria Time Stop: _____	Determines Acceptance Criteria are still met with correct values used. Turns in procedures within 30 minutes of start time. *****Total time to complete JPM MUST be 30 minutes or less.*****		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 1A 1. Task description and number, JPM description and number are identified.
- ✓ 1A 2. Knowledge and Abilities (K/A) references are included.
- ✓ 1A 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 1A 4. Initial setup conditions are identified.
- ✓ 1A 5. Initiating and terminating Cues are properly identified.
- ✓ 1A 6. Task standards identified and verified by SME review.
- ✓ 1A 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 1A 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 23, 8 Date 10/4/16
- 1A 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature]

Date: 9/28/16

SME/Instructor: [Signature]

Date: 10/4/16

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

**INITIAL
CONDITIONS:**

Name: _____

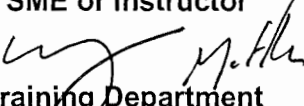
- Unit 2 Reactor Power is 100%.
- All Control Rod Group Demand Counters are at 227 steps.
- Current boron concentration is 300 ppm.
- Current core burnup is 10,000 EFPH.
- Control Rod 1D5 was declared INOPERABLE at **1000 today** due to not moving out during a rod exercise test. A Shutdown Margin Calculation has been performed IAW SC.RE-ST.ZZ-0002, Shutdown Margin Calculation, to satisfy the action requirement of TSAS 3.1.3.1 Action c.3.
- Control Rod 1D5 has been identified as **trippable** and is currently at 214 steps.
- The current time is 1030.

INITIATING CUE:

1. Review the attached SDM calculation procedure SC.RE-ST.ZZ-0002 for completeness and accuracy.
2. Note any discrepancies found by the review on Attachment 7.
3. Determine if SDM is SAT.

This is a time critical JPM. Time will start when procedures are given to you. Time will stop when procedures are returned to proctor.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ADMINISTRATIVE		
TASK:	Determine required actions upon Radioactive Effluent Monitor failure		
TASK NUMBER:	N1200010301		
JPM NUMBER:	15-01 NRC SRO Admin A3		
ALTERNATE PATH:	☐	K/A NUMBER:	2.3.11
		IMPORTANCE FACTOR:	4.3
APPLICABILITY:		RO	SRO
EO ☐	RO ☐	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	Salem ODCM, Salem Tech Specs, S1.OP-SO.WL-0001, Rev. 27, S1.OP-SO.WG-0011, Rev. 31, S1.OP-SO.CBV-0002, Rev. 21 (All checked 8-10-16)		
TOOLS AND EQUIPMENT:			
VALIDATED JPM COMPLETION TIME:	25 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-10-16
Validated By:	M Winkelspecht / P Williams SME or Instructor	Date:	9-28-16 / 10-4-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Administrative

TASK: Determine required actions upon Radioactive Effluent Monitor failure

TASK NUMBER: N1200010301

INITIAL CONDITIONS: Unit 1 is operating at 100% power.

Radiation Monitor 1R12A, Containment Noble Gas, failed its Source Check when preparing to perform a normal Containment Pressure Relief IAW S1.OP-SO.CBV-0002, Containment Pressure-Vacuum Relief System Operation.

11 CVCS Monitor Tank is ready for release.

14 Gas Decay Tank (GDT) is ready for release.

INITIATING CUE: Prior to initiating the Containment Pressure Relief, Rad Monitor 1R41D, Plant Vent Release Rate, indication fails high, and is confirmed as an instrument failure.

For the following radiological releases, determine:

- 1) Can the release be performed with the failed monitors described above, and,
- 2) If the release can be performed, what compensatory actions are required, if any.

Assume each release will be performed individually.

<u>RELEASES REQ</u>	<u>CAN BE RELEASED?</u>	<u>COMP ACTION</u>
• 14 GDT release	YES / NO	
• Containment Pressure Relief	YES / NO	
• 11 CVCS Monitor Tank release	YES / NO	

Successful Completion Criteria:

1. All critical steps completed
2. All sequential steps completed in order
3. All time-critical steps completed within allotted time
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Determines ALL the releases may be performed.
2. Determines additional compensatory measures apply for the Waste Gas Release and the Containment Pressure Relief per key, and no additional compensatory actions are required for Liquid Release.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Determine required actions upon Radioactive Effluent Monitor failure

*	STEP NO.	STEP * Denotes a Critical Step	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide blank copies of: S1.OP-SO.WG-0011 S1.OP-SO.CBV-0002 S1.OP-SO.WL-0001 Salem ODCM			
		Waste Gas Release S1.OP-SO.WG-0011	Determines ODCM 3.3.3.9 applies to Gaseous Effluent which directs actions per Table 3.3-13.		
			Determines Table 3.3-13, Instrument #1, Waste Gas Holdup System is applicable, which requires (1) 1R41A & D. Action 31 applies.		

NAME: _____
DATE: _____

Task: Determine required actions upon Radioactive Effluent Monitor failure

PSEG Restricted – Possession Requires Specific Permission from Nuclear Training
Page 4 of 8

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

System: ADMINISTRATIVE

Task: Determine required actions upon Radioactive Effluent Monitor failure

*	STEP NO.	STEP * Denotes a Critical Step	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Containment Pressure Relief S1.OP-SO.CBV-0002	Determines that Precaution and Limitation 2.3 of S1.OP-SO.CBV-0001 directs operator to notify Chemistry to comply with the contingency actions associated with the ODCM prior to performing a Containment Pressure Relief.		
			Determines Table 3.3-13, Instrument #3, applies to the Containment Pressure Relief. With both channels (1R41A & D OR 1R12A) inoperable, Action 37 applies.		

NAME: _____
DATE: _____

Task: Determine required actions upon Radioactive Effluent Monitor failure

Note to Evaluator: JPM modeled after Pilgrim Jan 2011 NRC Exam SRO Admin JPM RC.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 1 1. Task description and number, JPM description and number are identified.
- ✓ 1 2. Knowledge and Abilities (K/A) references are included.
- ✓ 1 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 1 4. Initial setup conditions are identified.
- ✓ 1 5. Initiating and terminating Cues are properly identified.
- ✓ 1 6. Task standards identified and verified by SME review.
- ✓ 1 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 1 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. _____ Date _____
All revisions checked 9/28/16
- ✓ 1 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- ✓ N/A 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- ✓ 1 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature]

Date: 9/28/16

SME/Instructor: [Signature]

Date: 10/4/16

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

Unit 1 is operating at 100% power.

Radiation Monitor 1R12A, Containment Noble Gas, failed its Source Check when preparing to perform a normal Containment Pressure Relief IAW S1.OP-SO.CBV-0002, Containment Pressure-Vacuum Relief System Operation.

11 CVCS Monitor Tank is ready for release.

14 Gas Decay Tank (GDT) is ready for release.

INITIATING CUE:

Prior to initiating the Containment Pressure Relief, Rad Monitor 1R41D, Plant Vent Release Rate, indication fails high, and is confirmed as an instrument failure.

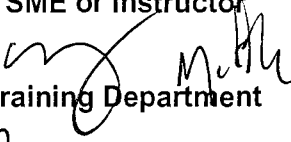
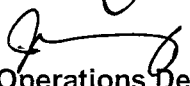
For the following radiological releases, determine:

- 1) Can the release be performed with the failed monitors described above, and,
- 2) If the release can be performed, what compensatory actions are required, if any.

Assume each release will be performed individually.

<u>RELEASES</u>	<u>CAN BE RELEASED?</u>	<u>COMP ACTION REQ</u>
• 14 GDT release	YES / NO	
• Containment Pressure Relief	YES / NO	
• 11 CVCS Monitor Tank release	YES / NO	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Emergency Plan		
TASK:	Classify an event and complete an ICMF within the regulatory committed time limit (ESG-1)		
TASK NUMBER:	1240020502		
JPM NUMBER:	15-01 NRC SRO Admin A4-1 (ESG-1)		
ALTERNATE PATH:	☐	K/A NUMBER:	2.4.41
APPLICABILITY:		IMPORTANCE FACTOR:	4.6
EO ☐	RO ☐	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulate (Simulator or Classroom)		
REFERENCES:	EP-SA-111-F2 Rev. 2 (ALERT)		
TOOLS AND EQUIPMENT:	Inform Simulator Operators – DO NOT ERASE ANY PROCEDURES UNTIL THE SRO EVALUATOR APPROVES		
VALIDATED JPM COMPLETION TIME:	12 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	15 minutes		
Developed By:	G Gauding Instructor	Date:	9-19-15
Validated By:	C Davis / P Williams SME or Instructor	Date:	9-21-16 / 10-4-16
Approved By:	 Training Department	Date:	10/14/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Emergency Plan

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-1)

TASK NUMBER: 1240020502

INITIAL CONDITIONS:

1. You have a maximum of 5 minutes to review the Emergency Operating Procedures used during this scenario to refresh your memory of all events/paths. At the end of your review you will become the Shift Manager (SM). Inform the Evaluator when you are ready to assume SM duties. You may continue to reference the procedures or to look at the control board but "the clock will be running." If there are multiple ECG calls, classify the most severe.

INITIATING CUE:

You are the Duty SM. Classify the event, complete the correct ECG Attachment and provide an ICMF to the Primary Communicator.

This is a Time Critical JPM.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Declare an **ALERT under EAL RB3.P** within 15 minutes of start time, and provide ICMF to the Primary Communicator within the next 15 minutes.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-1)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide candidate with "Tear-off sheet" and Blank Copies of EP-SA-111-F1 through F4 (UE, ALERT, SAE, GE)	Reviews initial conditions and EOP's (as necessary – 5 minute limit prior to starting)		
		*START TIME: _____ *Start time begins when candidate reports he/she is ready to assume SM duties	Cue: The regulatory commitment time clock has started.		
		Reviews ECG to classify event	Note: It is acceptable to use the laminated tables in the simulator, rather than the ECG		
		Classifies the event	Determines the classification of the event and refers to ECG Attachment 2 (Alert)		
	Att 2 A.1	CALL communicators to the Control Room	Pages communicators and initials as SM Cue: I am the Primary Communicator		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-1)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	Att 2 A.2	IF a Security Event is in progress, THEN IMPLEMENT the prompt actions of NC.EP-EP.ZZ-0102, EC Response, Attachment 10, prior to the classification.	Determines Security Event is not in progress.		
	Att 2 A.3	If time allows, DIRECT Classification Independent Verification to be performed.	Cue: No independent verification will be performed.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-1)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
* *	Att 2 A.4	After Classification Independent Verification is obtained: • DECLARE the ALERT (enter time and date on ICMF) • COMPLETE / APPROVE the ICMF	Declares an ALERT under RB3.P within 15 minutes of Start Time: _____:_____:_____ Completes and Approves the ICMF as follows: EAL#(s): RB3.P Description of Event: Potential Loss of the Reactor Coolant System Boundary Note: Description of Event is found in EAL Description Table <u>Fills out Section III:</u> Checks IS NOT for a Radiological Release in progress. <u>Fills out Section IV:</u> Retrieves wind speed (10) and direction (160) data from SPDS <u>Initials for approval to transmit</u>		
	Att 2 A.5	If time allows OBTAIN accuracy peer check of the completed ICMF	Cue: No peer check will be performed.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-1)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att 2 A.6	Continue with NOTIFICATION AND ACTIVATION as follows: • If not previously performed, ACTIVATE / DIRECT ACTIVATION of ERO Emergency Callout (EP 96-003) • DIRECT the Primary Communicator to implement ECG Attachment 6	Cue: Activation of ERO Emergency Callout is not required for this JPM. Provides ICMF to Primary Communicator within 15 minutes of event declaration _____:_____:_____		
			Terminate JPM when ICMF is given to the Evaluator.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- W 100 1. Task description and number, JPM description and number are identified.
- W 100 2. Knowledge and Abilities (K/A) references are included.
- W 100 3. Performance location specified. (in-plant, control room, or simulator)
- W 100 4. Initial setup conditions are identified.
- W 100 5. Initiating and terminating Cues are properly identified.
- W 100 6. Task standards identified and verified by SME review.
- W 100 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W 100 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 2 Date 9-21-16
- W 100 9. Pilot test the JPM:
 a. verify Cues both verbal and visual are free of conflict, and
 b. ensure performance time is accurate.
- W 100 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- W 100 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: C. D. C. P. S.Date: 9-21-16SME/Instructor: J. WilliamsDate: 10/4/16

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

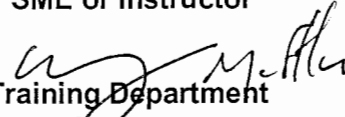
1. You have a maximum of 5 minutes to review the Emergency Operating Procedures used during this scenario to refresh your memory of all events/paths. At the end of your review you will become the Shift Manager (SM). Inform the Evaluator when you are ready to assume SM duties. You may continue to reference the procedures or to look at the control board but "the clock will be running." If there are multiple ECG calls, classify the most severe.

INITIATING CUE:

You are the Duty SM. Classify the event, complete the correct ECG Attachment and provide an ICMF to the Primary Communicator.

This is a Time Critical JPM.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Emergency Plan		
TASK:	Classify an event and complete an ICMF within the regulatory committed time limit (ESG-2)		
TASK NUMBER:	1240020502		
JPM NUMBER:	15-01 NRC SRO Admin A4-1 (ESG-2)		
ALTERNATE PATH:	☐	K/A NUMBER:	2.4.41
APPLICABILITY:	IMPORTANCE FACTOR:		4.6
EO ☐	RO ☐	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:		Simulate (Simulator or Classroom)	
REFERENCES:	EP-SA-111-F2 Rev. 2 (ALERT)		
TOOLS AND EQUIPMENT:	Inform Simulator Operators – DO NOT ERASE ANY PROCEDURES UNTIL THE SRO EVALUATOR APPROVES		
VALIDATED JPM COMPLETION TIME:	12 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	15 minutes		
Developed By:	G Gauding Instructor	Date:	9-19-15
Validated By:	C Davis / P Williams SME or Instructor	Date:	9-21-16 / 10-4-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:			DATE:

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Emergency Plan

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-2)

TASK NUMBER: 1240020502

INITIAL CONDITIONS:

1. You have a maximum of 5 minutes to review the Emergency Operating Procedures used during this scenario to refresh your memory of all events/paths. At the end of your review you will become the Shift Manager (SM). Inform the Evaluator when you are ready to assume SM duties. You may continue to reference the procedures or to look at the control board but "the clock will be running." If there are multiple ECG calls, classify the most severe.

INITIATING CUE:

You are the Duty SM. Classify the event, complete the correct ECG Attachment and provide an ICMF to the Primary Communicator.

This is a Time Critical JPM.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Declare an **ALERT under EAL RB3.L** within 15 minutes of start time, and provide ICMF to the Primary Communicator within the next 15 minutes.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-2)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide candidate with "Tear-off sheet" and Blank Copies of EP-SA-111-F1 through F4 (UE, ALERT, SAE, GE)	Reviews initial conditions and EOP's (as necessary – 5 minute limit prior to starting)		
		*START TIME: _____ *Start time begins when candidate reports he/she is ready to assume SM duties	Cue: The regulatory commitment time clock has started.		
		Reviews ECG to classify event	Note: It is acceptable to use the laminated tables in the simulator, rather than the ECG		
		Classifies the event	Determines the classification of the event and refers to ECG Attachment 2 (Alert)		
	Att 2 A.1	CALL communicators to the Control Room	Pages communicators and initials as SM Cue: I am the Primary Communicator		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-2)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	Att 2 A.2	IF a Security Event is in progress, THEN IMPLEMENT the prompt actions of NC.EP-EP.ZZ-0102, EC Response, Attachment 10, prior to the classification.	Determines Security Event is not in progress.		
	Att 2 A.3	If time allows, DIRECT Classification Independent Verification to be performed.	Cue: No independent verification will be performed.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-2)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
* *	Att 2 A.4	After Classification Independent Verification is obtained: • DECLARE the ALERT (enter time and date on ICMF) • COMPLETE / APPROVE the ICMF	Declares an ALERT under RB3.L within 15 minutes of Start Time: _____:_____:_____ Completes and Approves the ICMF as follows: EAL#(s): RB3.L Description of Event: Loss of the Reactor Coolant System Boundary Note: Description of Event is found in EAL Description Table <u>Fills out Section III:</u> Checks IS for a Radiological Release in progress. <u>Fills out Section IV:</u> Retrieves wind speed (10)and direction (160) data from SPDS <u>Initials for approval to transmit</u>		
	Att 2 A.5	If time allows OBTAIN accuracy peer check of the completed ICMF	Cue: No peer check will be performed.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Plan**

TASK: Classify an event and complete an ICMF within the regulatory committed time limit (ESG-2)

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	Att 2 A.6	Continue with NOTIFICATION AND ACTIVATION as follows: • If not previously performed, ACTIVATE / DIRECT ACTIVATION of ERO Emergency Callout (EP 96-003) • DIRECT the Primary Communicator to implement ECG Attachment 6	Cue: Activation of ERO Emergency Callout is not required for this JPM. Provides ICMF to Primary Communicator within 15 minutes of event declaration ____:____:____		
			Terminate JPM when ICMF is given to the Evaluator.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ WCP 1. Task description and number, JPM description and number are identified.
- ✓ WCP 2. Knowledge and Abilities (K/A) references are included.
- ✓ WCP 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ WCP 4. Initial setup conditions are identified.
- ✓ WCP 5. Initiating and terminating Cues are properly identified.
- ✓ WCP 6. Task standards identified and verified by SME review.
- ✓ WCP 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ WCP 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 2 Date 9.21.16
- ✓ WCP 9. Pilot test the JPM:
 a. verify Cues both verbal and visual are free of conflict, and
 b. ensure performance time is accurate.
- ✓ WCP 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- ✓ WCP 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

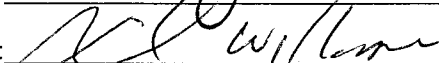
SME/Instructor:



Date:

9.21.16

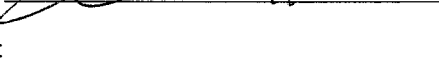
SME/Instructor:



Date:

10/7/16

SME/Instructor:



Date:

INITIAL CONDITIONS:

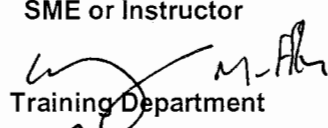
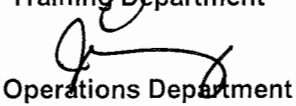
1. You have a maximum of 5 minutes to review the Emergency Operating Procedures used during this scenario to refresh your memory of all events/paths. At the end of your review you will become the Shift Manager (SM). Inform the Evaluator when you are ready to assume SM duties. You may continue to reference the procedures or to look at the control board but "the clock will be running." If there are multiple ECG calls, classify the most severe.

INITIATING CUE:

You are the Duty SM. Classify the event, complete the correct ECG Attachment and provide an ICMF to the Primary Communicator.

This is a Time Critical JPM.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Rod Control		
TASK:	Perform a Control Rod Operability Verification		
TASK NUMBER:	N0010070101		
JPM NUMBER:	15-01 NRC Sim a		
ALTERNATE PATH:	☒	K/A NUMBER:	001 A4.03
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:		Simulator	
REFERENCES:		S2.OP-ST.RCS-0001, Rev. 22 (checked 8-3-16)	
TOOLS AND EQUIPMENT:		None	
VALIDATED JPM COMPLETION TIME:		10 min	
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:		N/A	
Developed By:	G Gauding Instructor	Date:	8-3-16
Validated By:	M Lutek / W Russell SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE
JOB PERFORMANCE MEASURE
SIMULATOR SETUP INSTRUCTIONS**

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

TASK NUMBER: N0010070101

SIMULATOR IC: **IC-251** in 15-01 NRC JPMs

MALFUNCTIONS
REQUIRED: RT-1 will trip 21 SGFP
MALF: BF0105A, 21 STM GEN FEED PUMP TRIP, Final Value: 3

OVERRIDES
REQUIRED: None

SPECIAL
INSTRUCTIONS: None

Note to Evaluator: OP-AP-300-1001 required a peer check for all non-emergency manual rod motion. Do NOT correct operator if a mistake is made when performing a peer check. For all required peer checks simply state "Peer check SAT".

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

TASK NUMBER: N0010070101

INITIAL CONDITIONS:

Unit 2 is operating at steady state, 100% power. Power has been at 100% for 30 days.

INITIATING CUE:

- You are directed to perform S2.OP-ST.RCS-0001, Reactivity Control System Rod Control Assemblies.
- A Maintenance Technician is stationed at the Rod Control System Power Cabinets.
- Communications are established via plant page between the Control Room and Jeff Stevens, NEO, at the Rod Control System Power Cabinets.
- The CRS has directed that steps 5.1.2 through 5.1.9 are to be performed in order.
- The CRS directs that 15 steps of rod insertion are to be performed for each bank to ensure that each rod moves at least 10 steps.
- Initial Control Rod Positions are:

SDA		SDB		SDC		SDD	
F-04	229	G-03	229	E-03	229	C-05	229
D-10	228	C-09	227	C-11	223	E-13	223
K-12	227	J-13	227	L-13	224	N-11	224
M-06	227	N-07	229	N-05	224	L-03	227
B-04	230	J-02	223				
D-14	224	C-06	223				
P-12	229	G-13	227				
M-02	230	N-09	230				

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

Operator trips the Rx when Shutdown Bank B is inserted in the core and a SGFP trips.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up copy of S2.OP-ST.RCS-0001, Rev. 22, with Prerequisite Sections 2.1-2.8 filled out.	Reviews Prerequisites.		
		Provide copy of OP-AP-300-1001, PWR Control Rod Movement Requirements.	Reviews OP-AP-300-1001, PWR Control Rod Movement Requirements.		
			Reviews and signs Precautions and Limitations. Cue: <u>IF</u> ASKED, state that no special contingency actions have been briefed other than what is in the P&L's.		
	5.1.1	IF this surveillance is to be performed....	Determines step is N/A based on steady state 100% operation.		
	5.1.2	PERFORM Attachment 2, Rod Control System Voltage Checks as follows: ___ A. PLACE Rod Bank Selector Switch to "MANUAL". ___ B. Direct Maintenance to COMPLETE Attachment 2, Rod Control System Voltage Checks.	Performs Attachment 2 Rod Control System Voltage Checks by: A. Places Rod Bank Selector Switch to manual. B. Directs Direct Maintenance to complete Attachment 2, Rod Control System Voltage Checks. Cue: Maintenance reports Attachment 2 Rod Control System Voltage Checks are complete SAT.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	Note	— The Shutdown Rod Insertion Limit (TS 3.1.3.4) is not applicable during the performance of this test for rods actually undergoing testing. — Steps 5.1.3 through 5.1.10 may be performed in any order.	Initials both notes.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.1.3	PERFORM Shutdown Bank A testing as follows: A. <u>IF</u> the Rx is critical, <u>THEN ENSURE</u> T_{avg} is within $\pm 1^{\circ}\text{F}$ of T_{ref}. B. PLACE Bank Selector Switch in the "SBA" position. C. ENSURE GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21AC and 22AC. D. Maneuver Shutdown Bank A at least 10 steps in any one direction. E. ENSURE each rod in Shutdown Bank A indicated rod movement of at least 10 steps. F. RECORD Shutdown Bank A "Test Results" by initialing in Attachment 1. G. RESTORE Shutdown Bank A to pre-test condition.	Checks Tave/Tref recorder on 2RP4 and notes Tavg is within 1°F of Tref.. Rotates Bank Selector Switch fully counter clockwise to the SBA position. Contacts NEO and directs him to verify GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21AC and 22AC. <u>Cue:</u> GRP SELECT "C" lights are illuminated at Rod Control System Power Cabinets 21AC and 22AC. IAW OP-AA-300-1001, prior to rod movement, operator states initial Shutdown Bank A position, target control rod position, and direction of rod movement, then inserts SBA at least 10 steps to stated target position. Verifies rod movement by viewing rod position on P-250 computer. Initials SAT for Groups 1 and 2 for SBA on Att 1. Operator states initial Shutdown Bank A position, target control rod position, and direction of rod movement, then withdraws SBA to stated target ARO position. Continuous rod withdrawal is acceptable.		

OPERATOR TRAINING PROGRAM

JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Rod Control

TASK: Perform a Control Rod Operability Verification

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.1.4	PERFORM Shutdown Bank B testing as follows: A. <u>IF</u> the Rx is critical, <u>THEN ENSURE</u> T_{avg} is within $\pm 1^{\circ}\text{F}$ of T_{ref}. B. PLACE Bank Selector Switch in the "SBB" position. C. ENSURE GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21BD and 22BD. D. MANUEVER Shutdown Bank B at least 10 steps in any one direction. E. ENSURE each rod in Shutdown Bank B indicated rod movement of at least 10 steps.	Checks Tave/Tref recorder on 2RP4 and notes Tavg is within 1°F of Tref. Rotates Bank Selector Switch clockwise to the SBB position. Contacts NEO and directs him to verify GRP. SELECT "C" lights are illuminated on Rod Control System Power Cabinets 21BD and 22BD. <u>Cue:</u> GRP SELECT "C" lights are illuminated at Rod Control System Power Cabinets 21BD and 22BD. Operator states initial Shutdown Bank B position, target control rod position, and direction of rod movement, then inserts SBB at least 10 steps to stated target position. <u>SIMULATOR OPERATOR:</u> INSERT RT-1 when the operator is checking rod position after insertion. This will trip 21 SGFP. Responds to 21 SGFP trip by tripping the Rx IAW P&L 3.13. (This is Alternate Path) Movement of the Rod Bank Selector Switch to MAN or AUTO under these conditions is incorrect.		
			When the operator has either tripped the Rx or initiated rod insertion in MAN or AUTO, state JPM is complete.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

TQ-AA-106-0303
Revision 4

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 2/17 1. Task description and number, JPM description and number are identified.
- ✓ 2/17 2. Knowledge and Abilities (K/A) references are included.
- ✓ 2/17 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 2/17 4. Initial setup conditions are identified.
- ✓ 2/17 5. Initiating and terminating Cues are properly identified.
- ✓ 2/17 6. Task standards identified and verified by SME review.
- ✓ 2/17 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 2/17 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 22 Date 6/13/15
- 2/17 9. Pilot test the JPM:
 - ✓ 2/17 a. verify Cues both verbal and visual are free of conflict, and
 - 2/17 b. ensure performance time is accurate.
- 2 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: M Lutek

Date: 8/22/16

SME/Instructor: W. Russell Russell

Date: 8/22/16

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

INITIAL CONDITIONS:

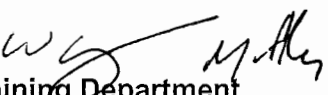
Unit 2 is operating at steady state, 100% power. Power has been at 100% for 30 days.

INITIATING CUE:

- You are directed to perform S2.OP-ST.RCS-0001, Reactivity Control System Rod Control Assemblies.
- A Maintenance Technician is stationed at the Rod Control System Power Cabinets.
- Communications are established via plant page between the Control Room and Jeff Stevens, NEO, at the Rod Control System Power Cabinets.
- The CRS has directed that steps 5.1.2 through 5.1.9 are to be performed in order.
- The CRS directs that 15 steps of rod insertion are to be performed for each bank to ensure that each rod moves at least 10 steps.
- Initial Control Rod Positions are:

SDA	SDB	SDC	SDD
F-04 229	G-03 229	E-03 229	C-05 229
D-10 228	C-09 227	C-11 223	E-13 223
K-12 227	J-13 227	L-13 224	N-11 224
M-06 227	N-07 229	N-05 224	L-03 227
B-04 230	J-02 223		
D-14 224	C-06 223		
P-12 229	G-13 227		
M-02 230	N-09 230		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

STATION:	SALEM		
SYSTEM:	Emergency Operating Procedures		
TASK:	Perform Actions for Containment Sump blockage		
TASK NUMBER:	N1150900501		
JPM NUMBER:	15-01 NRC Sim b		
ALTERNATE PATH:	☒	K/A NUMBER:	2.4.4
APPLICABILITY:		IMPORTANCE FACTOR:	4.5 4.7
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	2-EOP-LOCA-1, REV. 30, 2-EOP-LOCA-3, Rev. 30, 2-EOP-APPX-7, Rev. 0, OP-AA-101-111-1003, Rev. 6 (All rev checked 5/9/16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	5 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	5-9-16
Validated By:	D Cox / M Lutek SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/10/16
Approved By:	 Operations Department	Date:	10/10/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for Containment Sump blockage

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Simulator Operator: Insert <u>RT-1</u> to cause cavitation of both RHR pumps.		
			Determines ECCS pumps are exhibiting signs of cavitation and recognizes entry into APPX-7 is warranted from Continuous Action Summary. Evaluator: <u>IF the intention to stop the cavitating RHR pumps is voiced, THEN FOR RO APPLICANT ONLY, IMMEDIATELY CUE:</u> The CRS directs you to implement Appendix-7. (This is to prevent activation of Event Trigger when 21 RHR pump is stopped). CUE IF REQUIRED: The CRS directs you to implement Appendix-7. Note: Operator may go to EOP-LOCA-5, but first step of LOCA-5 will direct performance of APPX-7. Use above cue if required in LOCA-5.		
			Evaluator: Provide paper copy of 2-EOP-APPX-7.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for Containment Sump blockage

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	APPX-7	CAUTION - Any pump receiving suction from an RHR pump should be stopped before stopping the RHR pump. - If any Charging Pump, SI Pump, or Containment Spray Pump loses suction or shows indication of cavitation, the pump should be stopped.	Evaluator: Stopping RHR pumps here <u>WITHOUT</u> first stopping any SI or Charging pumps receiving suction from RHR pump discharge is failure criteria.		
	1	(CAS) MONITOR RHR Pump suction conditions: CHECK <u>NO</u> indications of cavitation. - Pump Amps – stable and normal for discharge pressure and flow. - Flow – stable and normal for discharge pressure. - Discharge Pressure – stable and normal for flow - Oscillations – none indicated	Determines BOTH RHR pumps exhibit signs of cavitation, as well as ALL ECCS pumps being supplied from RHR pumps.		

NAME: _____
DATE: _____

TASK: Perform Actions for Containment Sump blockage

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	1.a RNO	a. PERFORM the following: 1. ENSURE 21RH29 <u>AND</u> 22RH29 in AUTO. 2. STOP <u>ANY</u> Charging Pump or SI Pump taking suction from affected RHR Pump(s) 3. <u>IF</u> indications of cavitation continue, <u>THEN</u> CLOSE 21CS36 <u>AND</u> 22CS36, RHR TO CS VALVE. 4. <u>IF</u> indications of cavitation continue <u>AND</u> radiological conditions permit, <u>THEN</u> PERFORM the following: a. SEND an operator to release tags and restore CA to RH18(s). b. CLOSE affected RH18(s). 5. <u>IF</u> indications of cavitation continue, <u>THEN</u> REMOVE LOCKOUT(s) <u>AND</u> CLOSE affected SJ49(s), RHR TO COLD LEG. 6. <u>IF</u> indications of cavitation continue, <u>THEN</u> STOP affected RHR Pump(s).	Places 21RH29 and 22RH29 in AUTO. Stops 21 and 22 Charging pumps and 21 and 22 SI pumps. Determines pump cavitation is continuing, verifies 21CS36 and 22CS36 are shut. CUE: Radiation Protection has determined both RHR pump rooms to be unavailable for access due to high radiation levels. Determines RHR pump cavitation is continuing, and removes lockout from 21SJ49 and 22SJ49 RHR TO COLD LEG and shuts 21SJ49 and 22SJ49. Determines indications of cavitation continue, and stops 21 and 22 RHR pump. Simulator Operator: Ensure ET-1 and ET-3 are TRUE when 21 and 22 RHR pump STOP PBs are depressed. This removes the SJ44 failures.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for Containment Sump blockage

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	2	CHECK <u>ALL AVAILABLE</u> CFCUs running in Low Speed.	Determines all CFCUs are running in Low Speed.		
*	3.	SECURE Containment Spray flow path: a. STOP <u>ALL</u> Containment Spray Pumps. b. CHECK 21CS36 <u>AND</u> 22CS36 CLOSED.	Stops 21 CS pump. Determines 21CS36 and 22CS36 are shut.		
	4.	(CAS) CHECK RWST level >1.2 feet.	Determines RWST level is >1.2 feet.		
	5.	ATTEMPT to establish RHR Pump suction: a. CHECK 21SJ44 <u>AND</u> 22SJ44 OPEN. b. CHECK <u>AT LEAST ONE</u> RHR Pump running, with suction aligned to Containment Sump.	Determines 21SJ44 <u>AND</u> 22SJ44 are open. Determines NEITHER RHR Pumps is running with suction aligned to Containment Sump.		
	5.b RNO	b. PERFORM the following:			
	5.b RNO	1. REMOVE LOCKOUT(s) <u>AND</u> CLOSE affected SJ49(s) RHR DISCH TO COLD LEG.	Determines 21SJ49 and 22SJ49 previously shut.		
	5.b RNO	2. ENSURE 21RH29 <u>AND</u> 22RH29 in AUTO	Determines 21RH29 and 22RH29 are in AUTO.		
*	5.b RNO	3. START <u>ONE</u> RHR Pump.	Starts 21 OR 22 RHR pump.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for Containment Sump blockage

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.b RNO	4. IF indications of cavitation occur, THEN STOP the running RHR pump, AND START the other RHR pump.	Determines cavitation is NOT occurring.		
	5.b RNO	5. IF NO RHR Pumps can be run without cavitation, THEN STOP any running RHR Pumps AND GO TO Step 8.	Determines RHR pump can be run without cavitation.		
	5.	c) CHECK 21RH29 AND 22RH29 in AUTO. d) REMOVE LOCKOUT(s) AND CLOSE affected SJ49(s) RHR DISCH TO COLD LEG. e) CHECK ONLY ONE RHR Pump running with suction from Containment Sump.	Determines 21RH29 and 22RH29 are in AUTO. Determines BOTH SJ49's are shut. Determines only one RHR pumps is running.		
	6.	ATTEMPT to establish Charging Pump or SI Pump flow in recirculation mode: a. CHECK Charging Pump and SI Pump suction aligned to running RHR pump. b. CHECK ONLY ONE Charging Pump OR SI Pump running.	Checks Charging Pump and SI Pump suction aligned to running RHR pump by: a.1. Checks 2SJ67 and 2SJ68 SHUT a.2 Checks 22SJ45 OPEN a.3 Checks 21SJ45 OPEN a.4 Checks 21SJ113 and 22SJ113 OPEN a.5 Checks 2SJ1 and 2SJ2 OPEN a.6 Checks 2SJ30 SHUT Determine NO Charging or SI pumps are running.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for Containment Sump blockage

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	6.b RNO	START OR STOP , Charging and SI Pumps, as required, to obtain <u>ONLY ONE</u> pump running in recirculation alignment.	Starts ONLY ONE Charging or SI pump. Note: Per the NOTE at top of page 3, the intent of Step 6 is to start only a single pump (Charging or SI), and it is preferable to start a Charging pump. SAT performance of Step 6 does not require starting a Charging pump, just one of the 4 available charging or SI pumps.		
	6	c. CHECK Charging Pump or SI Pump running in recirculation alignment. d. GO TO Step 9	Determines Charging or SI pump is running in recirculation alignment.		
			Terminate JPM when operator goes to Step 9.		

JOB PERFORMANCE MEASURE

TQ-AA-106-0303

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 1. Task description and number, JPM description and number are identified.
- ✓ 2. Knowledge and Abilities (K/A) references are included.
- ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 4. Initial setup conditions are identified.
- ✓ 5. Initiating and terminating Cues are properly identified.
- ✓ 6. Task standards identified and verified by SME review.
- ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 7 Date 8.22.16
9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: D. Co. [Signature]

Date: 8.22.16

SME/Instructor: W. [Signature]

Date: 8/22/16

SME/Instructor: _____

Date: _____

JOB PERFORMANCE MEASURE

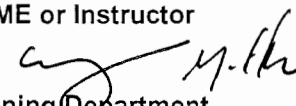
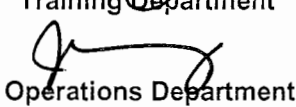
INITIAL CONDITIONS:

- A LBLOCA has occurred.
- The crew has responded by performing TRIP-1 and LOCA-1.
- The crew is performing LOCA-3 with all AC buses energized from off-site power, and is waiting at step 21 for RWST lo-lo level alarm.
- All ECCS pumps are operating except 22 CS pump which was stopped at step 8.

INITIATING CUE:

You are the Reactor Operator. Continue performing LOCA-3.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	ECCS		
TASK:	Isolate the ECCS Accumulators IAW TRIP-6		
TASK NUMBER:	1150070501		
JPM NUMBER:	15-01 NRC Sim c		
ALTERNATE PATH:	☒	K/A NUMBER:	E09 EA1.1
APPLICABILITY:	EO ☐	IMPORTANCE FACTOR:	3.5 RO 3.5 SRO
	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	2-EOP-TRIP-6 Rev. 30 (Rev checked 8-3-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	8 Minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-3-16
Validated By:	W Russell / D Cox SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: ECCS

TASK: Isolate the ECCS Accumulators IAW TRIP-6

TASK NUMBER: 1150070501

SIMULATOR SETUP:

1. Trip RCP's
2. Perform EOP's to TRIP-6, Step 12 – SI ACCUM ISOLATION
3. Override 24SJ54, Acc. Isolation Valve, close PB in the OFF position.
4. Place SPDS on Press/Temp Display
5. Put RCS Tcolds on trend on console.
6. Provide marked up copy of TRIP-6

INITIAL CONDITIONS:

A reactor trip occurred when a 500KV grid perturbation occurred, which also caused all RCPs to trip. The operating crew has progressed through the EOP's and is now in 2-EOP-TRIP-6, NATURAL CIRCULATION RAPID COOLDOWN WITH RVLIS. The RCPs will NOT be restarted.

INITIATING CUE:

Begin performing 2-EOP-TRIP-6, NATURAL CIRCULATION RAPID COOLDOWN WITH RVLIS at step 12.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Shut 21, 22, and 23 SJ54's.
2. Vent 24 Accumulator to atmospheric pressure.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Isolate the ECCS Accumulators IAW TRIP-6

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Reviews conditions and the marked up EOP-TRIP-6 Natural Circ Cooldown With RVLIS		
	12	IS RCS PRESSURE <1000 PSIG	Verifies RCS pressure <1000 psig.		
	12.1	REMOVE LOCKOUT FROM 21-24SJ54 (ACCUMULATOR OUTLET VALVES)	At 2RP4 Panel, selects VALVE OPERABLE on 21-24SJ54 ACCUMULATOR OUTLET VALVES LOCKOUT switches.		
*	12.1	CLOSE 21-24SJ54	Depresses CLOSE PB on 21-24SJ54, noting that 21, 22, and 23SJ54 begin to stroke closed.		
	12.1	ARE 21-24SJ54 CLOSED	Determines 24SJ54 is OPEN. Alternate path actions start here.		
*	12.2	VENT <u>ANY</u> AFFECTED ACCUMULATORS: • MAINTAIN RCS PRESSURE GREATER THAN ACCUMULATOR NITROGEN PRESSURE • OPEN 2NT35 (N2 HDR VALVE) • OPEN AFFECTED SJ93 (N2 SUPPLY VALVE)	Verifies RCS Pressure >24 Accumulator Pressure Opens 2NT35 (N2 HDR VALVE) Opens 24SJ93 (N2 SUPPLY VALVE) and observes 24 Accumulator pressure lowering		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: ECCS

TASK: Isolate the ECCS Accumulators IAW TRIP-6

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
* *	12.3	<u>WHEN</u> ACCUMULATOR VENTING IS <u>COMPLETE THEN</u> CLOSE: • 2NT35 • 21 THRU 24 SJ93	Cue: 24 Accumulator has been completely vented and 24 Accumulator pressure is now reading ZERO. Closes 2NT35 N2 HDR VALVE Closes 24SJ93 N2 SUPPLY VALVE Cue: JPM is complete		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ W. Russell 1. Task description and number, JPM description and number are identified.
- ✓ W. Russell 2. Knowledge and Abilities (K/A) references are included.
- ✓ W. Russell 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ W. Russell 4. Initial setup conditions are identified.
- ✓ W. Russell 5. Initiating and terminating cues are properly identified.
- ✓ W. Russell 6. Task standards identified and verified by SME review.
- W. Russell 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W. Russell 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 30 Date 8/22/16
- W. Russell 9. Pilot test the JPM:
 - a. verify cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: W. Russell

Date: 8/22/16

SME/Instructor: D. Cox [Signature]

Date: 8.22.16

SME/Instructor: _____

Date: _____

INITIAL CONDITIONS:

A reactor trip occurred when a 500KV grid perturbation occurred, which also caused all RCPs to trip. The operating crew has progressed through the EOP's and is now in 2-EOP-TRIP-6, NATURAL CIRCULATION RAPID COOLDOWN WITH RVLIS. The RCPs will NOT be restarted.

INITIATING CUE:

Begin performing 2-EOP-TRIP-6, NATURAL CIRCULATION RAPID COOLDOWN WITH RVLIS at step 12.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM 1 & 2		
SYSTEM:	Emergency Operating Procedures		
TASK:	Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)		
TASK NUMBER:	N1150290501		
JPM NUMBER:	15-01 NRC Sim d		
ALTERNATE PATH:	☒	K/A NUMBER:	EPE E05 EA1.1
APPLICABILITY:	EO ☐	IMPORTANCE FACTOR:	4.1 RO 4.0 SRO
	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	2-EOP-FRHS-1, Loss of Secondary Heat Sink, Rev. 30 31 <i>10/12/16</i>		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>6 minutes</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	8-2-16
Validated By:	S Harris / J Pierce SME or Instructor	Date:	9-28-16
Approved By:	<i>[Signature]</i> Training Department	Date:	<i>10/11/16</i>
Approved By:	<i>[Signature]</i> Operations Department	Date:	<i>10/11/16</i>
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: TCAF a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

TASK NUMBER: N1150290501

SIMULATOR SETUP: RX HEAD VENT KEYS ARE LOCATED IN SIM BOOTH

IC-254 developed by : MSL rupture (10%) downstream of MSIVs. Fail all MSIVs open. All AFW pumps failed. Fail 2PR2 shut. Performed TRIP-1 through Step 20. SG WR levels <32%.

INITIAL CONDITIONS:

- Unit 2 initiated a Rx trip from 100% power in response to a Main Steamline break at the mixing bottle.
- An automatic Safety Injection initiated.
- The Main Turbine failed to trip automatically, and was manually tripped from the control console.
- MSLI failed, and all MSIV's remain open.
- All AFW flow has been lost.
- EOP-TRIP-1 was performed and a transition to FRHS-1, Loss of Secondary Heat Sink was made at Step 20.1

INITIATING CUE:

You are the Reactor Operator. Perform FRHS-1 starting at Step 1.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Stop ALL RCPs.
2. Open 2PR1.
3. Open Rx Head Vent Valves 2RC40-2RC43.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	1	IS TOTAL AFW FLOW LESS THAN 22E04 LB/HR DUE TO OPERATOR ACTIONS	Determines it was not operator action which caused total AFW flow to be less than 22E04 lb/hr.		
	2	<u>IF AT LEAST ONE INTACT OR RUPTURED SG IS AVAILABLE, THEN DO NOT FEED A FAULTED SG</u>	Recognizes ALL SGs are faulted.		
	3	IS RCS PRESSURE GREATER THAN <u>ANY INTACT OR RUPTURED SG PRESSURE</u>	Checks RCS pressure on control console and determines it is greater than all SG pressures checked on control console.		
	3.1	ARE RCS T-HOTS GREATER THAN 350°F	Checks RCS Thot indication on control console and determines that RCS Thots are greater than 350°F.		
	4	<u>IF WR LEVELS IN AT LEAST TWO SG's ARE LESS THAN 32% (37% ADVERSE), THEN IMMEDIATELY GO TO STEP 21 TO INITIATE RCS BLEED AND FEED</u>	Determines at least 2 SG WR levels < 32% and goes to Step 21.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	21	<u>CAUTION</u> TO ESTABLISH RCS HEAT REMOVAL BY RCS BLEED AND FEED, STEPS 24 THRU 29 MUST BE PERFORMED QUICKLY AND WITHOUT INTERRUPTION	Reads Step.		
*	23	STOP <u>ALL</u> RCPS	Depresses STOP PB for 21-24 RCPs and verifies green stop light illuminates and red start light extinguishes.		
	22	INITIATE SI	Uses Safeguards key and initiates SI on at least one train of Safeguards initiation.		
	23	ARE SI VALVES IN SAFEGUARDS POSITION	Checks 2RP4 and/or console indication to determine that all valves listed in Table B are in Safeguards position. <u>Table B valves are:</u> 2SJ4 OPEN BIT INLET 2SJ5 OPEN BIT INLET 2SJ12 OPEN BIT OUTLET 2SJ13 OPEN BIT OUTLET 2CV68 CLOSED CHARGING DISCHARGE 2CV69 CLOSED CHARGING DISCHARGE (continued next page)		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			<u>Table B Valves (con't):</u> 21-24SJ54 OPEN ACCUMULATOR OUTLET 2SJ1 OPEN RWST TO CHARGING 2SJ2 OPEN RWST TO CHARGING 2CV40 CLOSED DISCHARGE STOP 2CV41 CLOSED DISCHARGE STOP		
	23.1	IS 21 OR 22 CHARGING PUMP RUNNING	Checks control console and determines 21 and 22 charging pump are running.		
		IS BIT FLOW ESTABLISHED	Checks console and determines BIT flow is established.		
	23.2	IS <u>ANY</u> SI PUMP RUNNING	Checks control console and determines 21 and 22 SI pumps are running.		
		ARE VALVES IN TABLE C OPEN FOR <u>AT LEAST</u> ONE RUNNING SI PUMP	Checks control console indication for valves listed in Table C and determines the valves are open for at least one running SI pump. <u>Table C valves are:</u> <u>21/22 SI PUMPS</u> 2SJ30 (FROM RWST) 21/22SJ33 (SI PUMP SUCTION) 2SJ135 (COLD LEG DISCHARGE) 21/22SJ134 (COLD LEG DISCHARGE)		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: **Emergency Operating Procedures**

TASK: **Respond to a Loss of Heat Sink (Initiate Bleed and Feed with SI pumps and Rx Head Vents)**

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		ARE ALL CHARGING AND SI PUMPS RUNNING AND ALIGNED TO DELIVER FLOW	Determines all charging and SI pumps are running and aligned to deliver flow.		
	24	OPEN <u>BOTH</u> PZR PORV STOP VALVES	Checks control console and determines BOTH 2PR6 and 2PR7 PORV STOP VALVES are open.		
*		OPEN <u>BOTH</u> PZR PORVS	Places 2PR1 in manual and depresses 2PR1 PZR PORV open PB and verifies green closed light extinguishes and red open light illuminates. Places 2PR2 in manual and depresses 2PR2 PZR PORV open PB and reports that 2PR2 will not open.		
	24.1	ARE <u>BOTH</u> PZR PORV STOP VALVES OPEN	Checks control console and determines BOTH 2PR6 and 2PR7 PORV STOP VALVES are open.		
		ARE <u>BOTH</u> PZR PORVS OPEN	Determines 2PR2 PZR PORV is not open. Alternate path actions start here.		
*	24.2	OPEN 2RC40 THRU 2RC43 (REACTOR HEAD VENTS)	Inserts key into each 2RC40 THRU 2RC43 (REACTOR HEAD VENTS) switch on 2RP3, turns to open, and verifies each valve opens.		
			Terminate JPM when operator has opened 2RC40 thru 2RC43 Reactor Head Vents.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- u ✓ 2 h 1. Task description and number, JPM description and number are identified.
- u ✓ 2 h 2. Knowledge and Abilities (K/A) references are included.
- u ✓ 2 h 3. Performance location specified. (in-plant, control room, or simulator)
- u ✓ 2 h 4. Initial setup conditions are identified.
- u ✓ 2 h 5. Initiating and terminating Cues are properly identified.
- u ✓ 2 h 6. Task standards identified and verified by SME review.
- u ✓ 2 h 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 2 h 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 30 Date 7/28/16
- 2 h 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: J. PIERCE JP

Date: 7/28/16

SME/Instructor: S. Harris SH

Date: 9/28/16

SME/Instructor: _____

Date: _____

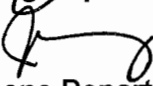
INITIAL CONDITIONS:

- Unit 2 initiated a Rx trip from 100% power in response to a Main Steamline break at the mixing bottle.
- An automatic Safety Injection initiated.
- The Main Turbine failed to trip automatically, and was manually tripped from the control console.
- MSLI failed, and all MSIV's remain open.
- All AFW flow has been lost.
- EOP-TRIP-1 was performed and a transition to FRHS-1, Loss of Secondary Heat Sink was made at Step 20.1

INITIATING CUE:

You are the Reactor Operator. Perform FRHS-1 starting at Step 1.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	Salem Generating Station		
SYSTEM:	Emergency Operating Procedures		
TASK:	TCAF Total Loss of All Service Water		
TASK NUMBER:	N1140380401		
JPM NUMBER:	15-01 NRC Sim e		
ALTERNATE PATH:	☐	K/A NUMBER:	076 A2.01
APPLICABILITY:		IMPORTANCE FACTOR:	
	EO ☐		3.5 RO
	RO ☒		3.7 SRO
	STA ☐		
	SRO ☒		
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-AB.SW-0001, Loss of SW Header Pressure, Rev. 16 S2.OP-AB.SW-0005, Loss of All Service Water Rev. 4 S2.OP-AR.ZZ-0002, Overhead Annunciator Window B Rev. 36 (All rev checked 8-3-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	5 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	3-23-16
Validated By:	W Russell / D Cox SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:	DATE:		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Abnormal Operating Procedures

TASK: TCAF a Total Loss of All Service Water

TASK: N1140380401

Simulator Setup: IC-255 100% power. **21 charging pump in service**
2 SW bay removed from service IAW S2.OP-AB.SW-003,
Service Water Bay Leak. All 4 bay pumps in service.
S2.OP-AB.SW-003 out and marked up.

MALFS: Tied to RT-1

24 SW pump trip
25 SW pump trip
26 SW pump trip

INITIAL CONDITIONS:

Unit 2 is operating at 100% power.

21 charging pump is in service.

23 charging pump is aligned for Normal operation IAW Section 4.1 of S2.OP-SO.CVC-0002,
Charging Pump Operation.

#2 SW Bay has been removed from service to isolate a leak in the bay IAW S2.OP-AB.SW-0003,
Service Water Bay Leak.

All #4 Service Water Bay pumps are in service.

INITIATING CUE:

You are the Reactor Operator. Respond to all indications and alarms.

Successful Completion Criteria

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made

Task Standard for Successful Completion:

1. Trip the Rx
2. Stop all RCPs prior to or within 3 minutes of any RCP bearing temperature reaching 175°F or any other RCP trip criteria being met in S2.OP-AB.RCP-001.
3. Place 23 Charging pump in service.
4. Reduce operating CCW pumps to one.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Abnormal Operating Procedures

TASK: TCAF a Loss of all Service Water

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
		Simulator Operator: Insert RT-1 after operator assumes watch. This trips all 4 Bay SW pumps.			
			Recognizes that all SW pumps have tripped.		
			Enters S2.OP-AB.SW-0005, Loss of All Service Water directly, or responds IAW ARP for low service water header pressures. ARP steps are after AB.SW-5 steps here in JPM. AB.SW-1 steps are after ARP steps here in JPM.		
*	AB.SW-5 3.1	TRIP the reactor, AND CONFIRM the Reactor Trip.	Trips the Rx using either trip handle, and confirms the Rx trip.		
*	AB.SW-5 3.2	STOP all RCPs.	Depresses stop PBs for all RCPs.		
		GO TO 2-EOP-TRIP-1, Reactor Trip or Safety Injection, AND CONTINUE with this procedure.	Cue: You are to continue with S2.OP-AB.SW-0005 while other operators perform TRIP-1. Evaluator: Silence OHA's as not associated with loss of SW.		
*	AB.SW-5 3.4	CLOSE the following valves to isolate letdown: • 2CV3, 45 GPM ORIFACE	Shuts the open orifice isolation valve and		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
		• 2CV4, 75 GPM ORIFICE • 2CV5, 75 GPM ORIFICE	ensures the remaining orifice isolation valves are shut.		
	AB.SW-5 3.5	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
* * *	AB.SW-5 3.6	IF a Centrifugal Charging Pump is in service <u>AND</u> the 23 Charging Pump is <u>IMMEDIATELY</u> available to be placed in service, <u>THEN</u> PERFORM the following : A. VERIFY <u>ALL</u> 21-24 RCP Seal Inlet Temperatures <225°F as indicated on 2CC1 or the Plant Computer B. ENSURE 23 Charging Pump is aligned for Normal Operation IAW S2.OP-SO.CVC-0002(Q) Charging Pump Operation. C. PLACE 23 Charging Pump in MANUAL. D. PLACE Charging Master Flow Controller in MANUAL. E. SET 23 Charging Pump Speed Demand to 10-12%. F. START 23 Charging Pump <u>AND</u> immediately INCREASE Speed Demand to approximately 20% to couple pump. G. ADJUST 23 Charging Pump Speed Demand to obtain desired flow. H. ENSURE Seal Injection Flow 6-12 gpm to each Reactor Coolant Pump not to exceed 40 gpm total Seal	Determines a centrifugal charging pump is in service and 23 charging pump is immediately available to be placed in service.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
*		Injection Flow. I. STOP 21 and 22 Centrifugal Charging Pumps. J. DISPATCH an Operator to monitor operation of 23 Charging Pump due to the potential for rising Component Cooling Water System temperature.			
	AB.SW-5 3.7	<u>IF</u> a Centrifugal Charging Pump is in service <u>AND</u> the 23 Charging Pump is <u>NOT IMMEDIATELY</u> available to be placed in service, <u>THEN STOP</u> 21 and 22 Centrifugal Charging Pumps.	Determines a Centrifugal Charging Pump is not in service		
	AB.SW-5 3.8	PLACE 2CC131, RCP THERMAL BARRIER, in MANUAL.	Places 2CC131, RCP THERMAL BARRIER, in MANUAL.		
*	AB.SW-5 3.9	PLACE Component Cooling Pumps in MANUAL <u>AND</u> REDUCE the number of operating Component Cooling Water Pumps to one.	Places all Component Cooling Pumps in MANUAL <u>AND</u> stops all but one Component Cooling Water Pumps.		
			Terminate JPM.		
	ARP OHA's B-13,14	3.1 VERIFY header low pressure using 2PA5373 on 2CC1. 3.2 <u>IF</u> low pressure conditions exists due to known system configuration, <u>THEN GO TO</u> S2.OP-SO.SW-0001(Q) Service water Pump operation, to adjust system pressure. 3.3 <u>IF</u> a valid low pressure condition exists, <u>THEN GO TO</u> S2.OP-AB.SW-0001(Q),	Determines header pressure is low. Determines loss of SW system pressure does not conditions exists due to known system configuration. Determines valid low pressure condition exists, and goes to S2.OP-AB.SW-0001, Loss of		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

*	STEP No.	STEP	STANDARD	EVAL S/U	COMMENTS
		Loss of Service Water Header Pressure.	Service Water Header Pressure.		
	AB.SW-1 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary.		
	AB.SW-1 3.2	Is there indication of a leak in a Service Water Bay per the following? • Visual indication • OHA B-29, 21-23 SW PMP SMP AREA LVL HI • OHA B-30, 24-26 SW PMP SMP AREA LVL HI	Determines there is no unisolated leak (SW Bay 2 has been isolated)		
	AB.SW-1 3.4	START Service Water pumps to maintain header pressure between 95 and 105 psig.	Determines no SW pumps are available to be started.		
	AB.SW-1 3.5	Is there a leak or valve malfunction on the turbine header as indicated by any of the following?	Determines the cause of low SW header pressure is from loss of all SW pumps.		
	AB.SW-1 3.6	Are both nuclear Service Water headers in service?	Determines neither nuclear SW header is in service.		
	AB.SW-1 3.7	<u>IF</u> the Unit is in a Service Water Header Outage...	Determines the Unit is not in a Service Water Header Outage.		
	AB.SW-1 3.8	<u>IF</u> the Unit is <u>NOT</u> in a Service Water Header Outage, <u>THEN GO TO</u> S2.OP-AB.SW-0005 (Q), Loss of All Service Water	Determines the Unit is not in a Service Water Header Outage, and goes to S2.OP-AB,SW-0005.		

TERMINATING CUE: After reduction of CCW pumps to one, terminate JPM.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

TQ-AA-106-0303
Revision 5

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- W ✓ W 1. Task description and number, JPM description and number are identified.
- W ✓ W 2. Knowledge and Abilities (K/A) references are included.
- W ✓ W 3. Performance location specified. (in-plant, control room, or simulator)
- W ✓ W 4. Initial setup conditions are identified.
- W ✓ W 5. Initiating and terminating Cues are properly identified.
- W ✓ W 6. Task standards identified and verified by SME review.
- W 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 4 Date 8/22/10
- W 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: W. Russell Russell

Date: 8/22/16

SME/Instructor: D. Cox

Date: 8-22-16

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

INITIAL CONDITIONS:

Unit 2 is operating at 100% power.

21 charging pump is in service.

23 charging pump is aligned for Normal operation IAW Section 4.1 of S2.OP-SO.CVC-0002,
Charging Pump Operation.

#2 SW Bay has been removed from service to isolate a leak in the bay IAW S2.OP-AB.SW-0003,
Service Water Bay Leak.

All #4 Service Water Bay pumps are in service.

INITIATING CUE:

You are the Reactor Operator. Respond to all indications and alarms.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Pressurizer and PRT		
TASK:	Adjust PRT level and pressure		
TASK NUMBER:	N0100110101		
JPM NUMBER:	15-01 NRC Sim f		
ALTERNATE PATH:	☐	K/A NUMBER:	007 A4.04
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☒
		RO	SRO
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-SO.PZR-0003, Rev. 15 (checked 8-5-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>8 min</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	8-5-16
Validated By:	M Lutek / W Russell SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Pressurizer and PRT

TASK: Adjust PRT level and pressure

TASK NUMBER: N0450040101

SIMULATOR SETUP: 15-01 NRC IC-256 PRT level 56.2% (or low enough for the PRT Hi-Lo Lvl alarm to be locked in) with PRT pressure 9.8 psig. Paper copies of procedure needed.

INITIAL CONDITIONS:

Unit 2 is at 100% power.

2PR1 is leaking and 2PR6 is shut with power applied. Operators have been attempting to determine the leak rate through 2PR1.

Console alarm PRT Level Hi-Lo is in alarm (lo) with PRT level at 56.2%.

INITIATING CUE:

You are directed to raise PRT level to 60% IAW S2.OP-SO.PZR-0003, Pressurizer Relief Tank Operation, Section 5.1.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Raise PRT level to > 60% while maintaining PRT pressure ≤ 10 psig.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: PZR and PRT

TASK: Adjust PRT level and pressure

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up paper copy of S2.OP-SO.PZR-0003.	Reviews Precautions and Limitations.		
	5.1.1	<u>IF</u> PRT level is low, <u>THEN</u> PERFORM the following:	Note: The step for reducing PRT pressure may be performed prior to raising level, since pressure is close to the maximum allowed. (See Step 5.1.1.D) <u>IF</u> PRT pressure reduction is performed first, <u>THEN Cue</u> if PRT pressure is lowered past 7 psig..." CRS states that the pressure reduction is sufficient to allow filling of the PRT now."		
*	5.1.1.A	<u>IF</u> required, <u>THEN</u> START 21 or 22 Primary Water Pump.	Depresses start pushbutton for 22 Primary Water Pump. (aligned for manual start) OR Places 21 Primary Water Pump in manual and depresses start pushbutton for 21 Primary Water Pump. <u>IF</u> operator does not start a Primary Water Pump and will gravity feed to makeup to the PRT, <u>THEN Cue:</u> The CRS directs you to use a Primary Water pump to makeup to the PRT.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: PZR and PRT

TASK: Adjust PRT level and pressure

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.1.1.B	OPEN 2WR80, CONT PRI WATER STOP.	Depresses open pushbutton for 2WR80 CONT PRI WATER STOP and verifies open light lit.		
*	5.1.1.C	OPEN 2WR82, PRT WATER SUPPLY.	Depresses open pushbutton for 2WR82, PRT WATER SUPPLY and verifies open light lit.		
*	5.1.1.D	<u>IF</u> PRT pressure approaches 10 psig, <u>THEN</u> OPEN 2PR15, VENT, as necessary, to maintain PRT pressure at >3 psig and ≤10 psig.	Depresses open pushbutton for 2PR15, VENT <u>prior to PRT pressure rising above 10.0 psig,</u> and depresses close pushbutton prior to PRT pressure lowering below 3.1 psig. Note: IF PRT Hi Pressure alarm is received, it prevents the 2PR15 from opening, and pressure cannot be reduced. <u>When</u> operator has demonstrated that they can raise level without exceeding 10 psig, <u>then</u> Cue: PRT level is now 60%		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: PZR and PRT

TASK: Adjust PRT level and pressure

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
* * *	5.1.1.E	When PRT level increases to between 59% and 86%: 1. CLOSE 2WR80, CONT PRI WATER STOP. 2. CLOSE 2WR82, PRT WATER SUPPLY 3. ENSURE 2PR15, VENT is closed. 4. <u>IF</u> Primary Water Pump is in service IAW Step 5.1.1.A, <u>AND</u> is <u>NOT</u> required to be in service for any other reason, <u>THEN STOP</u> running Primary Water Pump.	When PRT level rises to 60%: Closes 2WR80, CONT PRI WATER STOP. Closes 2WR82, PRT WATER SUPPLY. Ensures 2PR15, VENT is closed. Depresses stop pushbutton for running Primary Water Pump. <u>IF</u> 21 Primary Water pump was used to fill PRT, <u>THEN</u> places 21 or 22 Primary Water pump in auto.		
	5.1.1.F	DIRECT a second Operator to perform an Independent Verification of component position in Attachment 1, Section 1.0			

TERMINATING CUE: After operator reads Step 5.1.1.F, state JPM is complete.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- u ✓ 8 UM 1. Task description and number, JPM description and number are identified.
- u ✓ 8 UM 2. Knowledge and Abilities (K/A) references are included.
- u ✓ 8 UM 3. Performance location specified. (in-plant, control room, or simulator)
- b ✓ 8 UM 4. Initial setup conditions are identified.
- u ✓ 8 UM 5. Initiating and terminating Cues are properly identified.
- u ✓ 8 UM 6. Task standards identified and verified by SME review.
- u ✓ 8 UM 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- 8 UM 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 15 Date 3/22/16
- 8 UM 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: ML JTC / [Signature]

Date: 8/22/16

SME/Instructor: W. Russell Russell

Date: 8/22/16

SME/Instructor: _____

Date: _____

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

INITIAL CONDITIONS:

Unit 2 is at 100% power.

2PR1 is leaking and 2PR6 is shut with power applied. Operators have been attempting to determine the leak rate through 2PR1.

Console alarm PRT Level Hi-Lo is in alarm (lo) with PRT level at 56.2%.

INITIATING CUE:

You are directed to raise PRT level to 60% IAW S2.OP-SO.PZR-0003, Pressurizer Relief Tank Operation, Section 5.1.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	Salem Generating Station		
SYSTEM:	Emergency Operating Procedures		
TASK:	Perform Actions for a Loss of All AC Power (Energize a 4KV Vital Bus from a Station Power Transformer)		
TASK NUMBER:	1150140501		
JPM NUMBER:	15-01 NRC Sim g		
ALTERNATE PATH:	☐	K/A NUMBER:	EPE 055 EA1.07
APPLICABILITY:		IMPORTANCE FACTOR:	
			4.3 4.5
			RO SRO
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-SO.DG-0001, Rev. 39, S2.OP-SO.4KV-0001, Rev. 32		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>10 minutes</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	4-18-16
Validated By:	Hancock / Russell SME or Instructor	Date:	8-16-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Representative	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for a Loss of All AC Power (Energize a 4KV Vital Bus from a Station Power Transformer)

**TASK
NUMBER:** 1150140501

Simulator Setup: IC-256 (with 24ASD racked down and control power off)

INITIAL CONDITIONS:

1. Unit 2 is recovering from a loss of all AC Power.
2. The crew is performing step 16 of 2-EOP-LOPA-1.
3. 23 AFW pump is supplying all required AFW flow.
4. 2A EDG was started first and is providing power to 2A Vital Bus
5. 2B EDG was just started and is powering 2B 4KV vital bus.
6. 2C 4KV Vital Bus remains deenergized and unavailable.
7. 24 Station Power Transformer (SPT) has just been energized.
8. With 24 SPT now available, the CRS has directed that S2.OP-SO.DG-0001 Section 5.8 and S2.OP-SO.4KV-0001 Section 5.3.6 be utilized to swap 2A Vital Bus from 2A EDG to 24 SPT.

INITIATING CUE:

You are the Control Board Operator. Complete Section 5.8 of S2.OP-SO.DG-0001, and then utilize Section 5.3.6 of S2.OP-SO.4KV-0001 to restore power to 2A Vital Bus from 24 SPT. Prerequisites and Precautions and Limitations have been reviewed and signed.

Successful Completion Criteria

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made

Task Standard for Successful Completion:

- Energize 2A 4KV vital bus using 24 SPT

OPERATOR TRAINING PROGRAM JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Emergency Operating Procedures

TASK: Perform Actions for a Loss of All AC Power (Energize a 4KV Vital Bus from a Station Power Transformer)

	STEP No.	STEP (* denotes critical task)	STANDARD	EVAL S/U	COMMENTS
		Provide marked up copies of S2.OP-SO.DG-0001 and S2.OP-SO.4KV-0001.			
	SO.DG-1 5.8.1	IF 2A EDG is paralleled on 2A 4KV Vital Bus....	Determines 2A Diesel Generator is not paralleled on 2A 4KV Vital Bus.		
*	5.8.2	IF 2A Vital Bus is isolated and energized from 2A EDG, <u>THEN</u> : • PLACE redundant equipment in service as necessary to support deenergizing 2A Vital Bus	Places redundant equipment in service as needed: • 22 CCW pump		
*		• STOP all 2A Vital Bus loads using Attachment 1	Note: 1 SW pump with the TGA isolated is sufficient for maintaining SW header pressure Stops the following equipment (if running): • 21 CCW pump • 21 SW pump • 21 CFCU • 21 Aux Building Exhaust Fan • 21 SWGR Room Supply Fan • 21 SWGR Room Exhaust Fan • 21 BAT pump • 21 FHB Exhaust Fan • 21 AFW pump • 21 SI / 21 CS pumps		

OPERATOR TRAINING PROGRAM JOB PERFORMANCE MEASURE

	STEP No.	STEP (* denotes critical task)	STANDARD	EVAL S/U	COMMENTS
*	5.8.3	OPEN 2A DIESEL GENERATOR BREAKER by performing one of the following: • OPEN 2AD1AX6D, 2A DIESEL GENERATOR BREAKER (64' Swgr Rm), or • PRESS 2A BREAKER OPEN pushbutton (2CC3), or • PLACE 2A-DF-GCP-3, GENERATOR CIRCUIT (BCS), to TRIP (2A D/G Control Panel).	Opens 2A DIESEL GENERATOR BREAKER by performing one of the following: • Directs opening of 2AD1AX6D, 2A DIESEL GENERATOR BREAKER (64' Swgr Rm), or • Presses 2A BREAKER OPEN pushbutton (2CC3), or • Directs placing 2A-DF-GCP-3, GENERATOR CIRCUIT (BCS), to TRIP (2A D/G Control Panel).		
	5.8.4	ALLOW Diesel to run unloaded for ≥ 3 minutes prior to stopping the EDG.	Cue: 2A EDG will be shutdown by local operator.		
	5.8.5	If diesel unloading...	Determines step is N/A		
	SO.4KV-1 5.3.6	IF 2A 4KV Vital Bus is to be energized from 24 SPT, THEN:			
	5.3.6.A	Direct NEO to RACK UP 2AD1AX24ASD, 24 STATION POWER TRANSFORMER INFEED BREAKER (64' Swgr Rm).	Contacts NEO and directs them to rack up 2AD1AX24ASD, 24 STATION POWER TRANSFORMER INFEED BREAKER. Simulator Operator: MODIFY REMOTE ED21D to ON , then report 2AD1AX24ASD, 24 STATION POWER TRANSFORMER INFEED BREAKER is racked up.		
*	5.3.6.B	PRESS Mimic Bus 2A VITAL INFEED 24ASD pushbutton AND ENSURE Console Bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION illuminates.	Depresses Mimic Bus 2A VITAL INFEED 24ASD pushbutton AND determines the Console Bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION illuminates.		

OPERATOR TRAINING PROGRAM **JOB PERFORMANCE MEASURE**

*	5.3.6.C	PERFORM the following: 1. PRESS AND HOLD control console 24ASD CLOSE pushbutton. 2. RELEASE pushbutton when 24ASD indicates CLOSED. 3. ENSURE the following: a. Console bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION is extinguished. b. 2A 4KV Vital Bus voltage is 4.275 - 4.336KV (normal) OR 4.330 - 4.417KV (single source of off-site power). c. OHA J-17, 2A 4KV VTL BUS UNDRVOLT, is clear.	Presses and Holds control console 24ASD CLOSE pushbutton, then releases pushbutton when 24ASD indicates closed. Checks Console bezel 24ASD MIMIC BUS INTLK CLOSE SELECTION is extinguished. Determines 2A Vital Bus voltage is 4.275 - 4.385KV (single source of off-site power). Determines OHA J-17 2A 4KV VTL BUS UNDRVOLT is clear.		
			Terminate JPM.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ ✓ 2 1. Task description and number, JPM description and number are identified.
- ✓ ✓ 2 2. Knowledge and Abilities (K/A) references are included.
- ✓ ✓ 2 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ ✓ 2 4. Initial setup conditions are identified.
- ✓ ✓ 2 5. Initiating and terminating Cues are properly identified.
- ✓ ✓ 2 6. Task standards identified and verified by SME review.
- ✓ 2 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 2 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 39 Date 8/16/16
32
- 2 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- 2 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- 2 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: J. A. Hancock Date: 8/16/16
SME/Instructor: W. Russell Russell Date: 8/16/16
SME/Instructor: _____ Date: _____

OPERATOR TRAINING PROGRAM JOB PERFORMANCE MEASURE

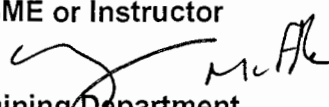
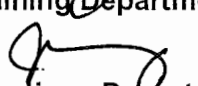
INITIAL CONDITIONS:

1. Unit 2 is recovering from a loss of all AC Power.
2. The crew is performing step 16 of 2-EOP-LOPA-1.
3. 23 AFW pump is supplying all required AFW flow.
4. 2A EDG was started first and is providing power to 2A Vital Bus
5. 2B EDG was just started and is powering 2B 4KV vital bus.
6. 2C 4KV Vital Bus remains deenergized and unavailable.
7. 24 Station Power Transformer (SPT) has just been energized.
8. With 24 SPT now available, the CRS has directed that S2.OP-SO.DG-0001 Section 5.8 and S2.OP-SO.4KV-0001 Section 5.3.6 be utilized to swap 2A Vital Bus from 2A EDG to 24 SPT.

INITIATING CUE:

You are the Control Board Operator. Complete Section 5.8 of S2.OP-SO.DG-0001, and then utilize Section 5.3.6 of S2.OP-SO.4KV-0001 to restore power to 2A Vital Bus from 24 SPT. Prerequisites and Precautions and Limitations have been reviewed and signed.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM 1 & 2		
SYSTEM:	Loss of Control Air		
TASK:	TCAF a Loss of Control Air		
TASK NUMBER:	N1140070401		
JPM NUMBER:	15-01 NRC Sim h		
ALTERNATE PATH:	☒	K/A NUMBER:	APE 065 AA2.06
APPLICABILITY:		IMPORTANCE FACTOR:	3.6 4.2
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator - Perform		
REFERENCES:	S2.OP-AB.CA-0001, Rev. 21 Loss of Control Air (both rev checked 8-5-16) S2.OP-AR.ZZ-0011, Rev. 60, pages 118-128		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	8 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-5-16
Validated By:	M Lutek / W Russell SME or Instructor	Date:	8-22-16
Approved By:	 Training Department	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Loss of Control air

TASK: TCAF Loss of Control Air

TASK NUMBER: N1140070401

SIMULATOR SETUP: Check 2WG41 is open

15-01 NRC IC-258

RT-1

MALF: CA0221 #2 STATION AIR CMPRSSR TRIP

REM : CA13D LOCKOUT #1 STATION AIR CMPRSR, Delay 1:30

I/O : AQ01 OVLO COMP 1 TROUBLE ALARM, Fin Val: ON, Delay 1:30

Other setup:

REM: CA15D LOCKOUT #3 STATION AIR CMPRSR

CA11D #1 EMERGENCY AIR CMPRSR IN AUTO Fin Val: MANUAL

CA12M #1 EMRGENCY AIR CMPRSR SWITCH Fin Val: Stop

I/O : AQ01 OVLO COMP 3 TROUBLE ALARM, Fin Val: ON

INITIAL CONDITIONS:

- Unit 2 is operating at 100% power.
- A WG release is in progress from 21 GDT.
- #3 SAC is C/T for scheduled maintenance.
- Unit 1 is operating at 100% power.

INITIATING CUE:

You are the Unit 2 Reactor Operator. Respond to all indications and alarms.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Isolate Letdown
2. Terminate Gaseous Release
3. Trip the Rx prior to auto trip on lo lo SG NR level.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note: The #1 and #3 Compressor Trouble alarms on 2CC1 do NOT have audible or flash capabilities (S2.OP-AR.ZZ-0011 page 120).		
		Simulator Operator: Insert <u>RT-1</u> after operator assumes watch.			
			Determines #2 Station Air Compressor (SAC) has tripped, and the Unit 2 Emergency Control Air Compressor (ECAC) has automatically started. Note: All 3 SAC supply breakers open is an auto start signal for both ECAC's. The Unit 2 ECAC will auto start, (Unit 1 will not). By design, the ECAC will not load until its control air header pressure lowers to 85 psig. 2A control air header is supplied from #2 ECAC, 2B control air header is supplied from #1 ECAC.		
			Note: The next-to-load SAC (follow) will not immediately start. It requires Station Air header pressure to drop to 5 psig below the follow setpoint (105 psig) for ~ 5 seconds.		
			Refers to the S2.OP-AR.ZZ-0011, Control Console 2CC1, ARP for COMPR 2 TROUBLE.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	ARP 3.1	<u>IF</u> SAC 2 trips, <u>THEN</u> : A. COORDINATE (as necessary) with Unit 1 B. ENSURE operating <u>OR</u> START at least one SAC IAW SC.OP- SO.SA-0001(Z), Station Air System Operation.	Coordinates with Unit 1 to start at least one SAC IAW SC.OP-SO.SA-0001(Z), Station Air System Operation. Cue when contacted as Unit 1: Unit 1 will ensure #1 SAC is placed in service.		
	ARP 3.2	<u>IF AT ANY TIME</u> Station Air pressure cannot be maintained ≥ 100 psig, <u>THEN</u> GO TO S2.OP-AB.CA-0001(Q), Loss of Control Air.	GOES TO S2.OP-AB.CA-0001(Q), Loss of Control Air, when Station Air pressure lowers to 100 psig if AB not entered previously.		
			Simulator Operator: Ensure #1 SAC locks out 1:30 after insertion of RT-1 , then announce twice on plant page: “#1 Station Air Compressor Trip.”		
			Recognizes that no SAC's are running, and enters S2.OP-AB.CA-0001 if not entered previously.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
			Note: Approximate event times are listed below <u>from insertion of RT-1</u>. 1 min – 1 SAC auto starts. 1 min 30 sec – 1 SAC trips. 3 min 50 sec – 2 ECAC begins loading. 4 min 10 sec - 2B Control Air header pressure reaches 80 PSIG, and SA low pressure alarm. 4 min 25 sec – CA low pressure alarm. 6.5 min – 21-24BF19 Feed Reg Valves begin closing. 7 min 10 sec – G-15 OHA ADFCS TRBL.		
	AB.CA 3.1	INITIATE Attachment 1, Continuous Action Summary.	Initiates Attachment 1, Continuous Action Summary. Cue IF Required: If CAS Step 8.0 action is being taken to refer to Attachment 2 for valve fail positions and for further action to be taken, then: Cue: The CRS will refer to Attachment 2		
	3.2	EITHER of the following conditions exist? • ALL Station Air Compressors stopped • EITHER Station Air Header <100 psig	Determines all SAC's are stopped or determines Station Air Header pressure is <100 psig.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.3	IF a loss of 1F Group Bus has occurred, AND 1H Group Bus is available, THEN SEND an Operator to restore power to #1 SAC Auxiliary Oil pump by transferring 11 Turbine West 230V Control Center IAW Attachment 14, Section 1.0.	Determines a loss of 1F Group Bus has not occurred.		
	3.4	IF a loss of 2F Group Bus has occurred, AND 2H Group Bus is available, THEN SEND an Operator to restore power to #2 SAC Auxiliary Oil pump by transferring 21 Turbine West 230V Control Center IAW Attachment 14, Section 2.0.	Determines a loss of 2F Group Bus has not occurred.		
	3.5	IF a loss of 1H Group Bus has occurred, AND 1F Group Bus is available, THEN SEND an Operator to restore power to #3 SAC Auxiliary Oil pump by transferring 11 Turbine East 230V Control Center IAW Attachment 14, Section 3.0.	Determines a loss of 1H Group Bus has not occurred.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.6	START or have Unit One Control Room START <u>Next to Load</u> Station Air Compressor.	Determines no SAC's are available. Cue if required: If Unit 1 is contacted, report that #3 Station Air Compressor is tagged out and #1 Station Air Compressor has tripped, and that an operator has been dispatched to the Station Air Compressors.		
	3.7	Either Station Air Header indicating less than 100 psig?	Determines both Station Air Headers are indicating less than 100 psig.		
	3.8	Attempt to START , or have Unit One Control Room START , remaining Station Air Compressor.	Determines no SAC's can be started.		
	3.9	NOTIFY the CRS/SM to evaluate use of temporary air compressors.	Cue: CRS will evaluate use of temporary air compressors.		
	3.10/3.11	2A Control Air Header less than or equal to 88 psig?	Determines 2A control air header pressure is: less than or equal to 88 psig and determines #2 ECAC is in service OR >88 psig.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.12/3.13	2B Control Air Header less than or equal to 88 psig?	Determines 2B control air header pressure is: less than or equal to 88 psig and contacts Unit 1 to place #1 ECAC in service OR >88 psig. Cue: When Unit 1 is contacted to start #1 ECAC, state: "# 1 ECAC has tripped."		
	3.14	SEND operators to: • LOCATE AND ISOLATE ANY air system leaks. • INVESTIGATE cause of ANY air compressor trip/trouble.	Dispatches operators to: • LOCATE AND ISOLATE any air system leaks. • INVESTIGATE cause of air compressor trip/trouble.		
	3.15	<u>IF</u> Station Air is lost, <u>AND</u> EITHER of the following are in service: • Spent Fuel Pool Transfer Pool Weir Gate Seals • SG Nozzle Dams <u>THEN:</u> A. EVACUATE personnel from areas subject to leakage/flooding. B. INITIATE monitoring of Fuel Pool or Reactor Vessel levels for the need to makeup due to leakage.	Determines Station Air is lost, and <u>neither</u> of the following are in service: • Spent Fuel Pool Transfer Pool Weir Gate Seals • SG Nozzle Dams		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	3.16	IF cooling water to the Station or Emergency Air Compressors is lost, <u>THEN INITIATE</u> the applicable portion of Attachment 6.	Determines cooling water to the Station or Emergency Air Compressors is not lost.		
	3.17	BOTH 2A and 2B Control Air Headers less than 80 psig?	Determines both 2A and 2B Control Air Headers are not <80 psig.		
	3.18	ANY Excess Flow Check Valve closed as indicated by at least one of the following? (Attachment 5, Excess Flow Check Valve Table) [INPO IER L2 15-16] - Groups of valves repositioning with normal air pressure indicated in Control Room - Local observation of low pressure downstream of an Excess Flow Check Valve	Determines NO Excess Flow Check Valve closed.		
	3.19	2B Control Air Header less than 80 psig?	Determines 2B Control Air Header is <80 psig. Alternate Path actions start here.		
	3.57	CONTINUE			
*	3.58	CLOSE the following valves: A. 2CV3, LTDWN ORIFICE ISOL VALVE B. 2CV4, LTDWN ORIFICE ISOL VALVE C. 2CV5, LTDWN ORIFICE ISOL VALVE D. 2CV2 LETDOWN LINE ISOL VALVE E. 2CV277 LTDWN LINE ISOL VALVE F. 2CV7 LTDWN HX INLET VALVE	Closes the following valves: A. 2CV3, LTDWN ORIFICE ISOL VALVE B. 2CV4, LTDWN ORIFICE ISOL VALVE C. 2CV5, LTDWN ORIFICE ISOL VALVE D. 2CV2 LETDOWN LINE ISOL VALVE E. 2CV277 LTDWN LINE ISOL VALVE F. 2CV7 LTDWN HX INLET VALVE		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	3.57	VERIFY ANY Liquid or Gaseous release is stopped by ensuring the following valves CLOSED: • 2WL51, TO CIRC WTR DISCHARGE • 2WG41, GAS DECAY TK TO PLANT VENT	Determines 2WG41, GAS DECAY TK TO PLANT VENT is open, and shuts 2WG41, GAS DECAY TK TO PLANT VENT. Verifies 2WL51 is shut.		
*	CAS 6.0	<u>IF AT ANY TIME</u> Station Air is lost, <u>THEN:</u> • INITIATE monitoring of 21-24BF19 SG FW CONT VALVE, operation. • <u>IF</u> ANY BF19 SG FW CONT VALVE closes <u>AND</u> Applicable SG water level CANNOT be maintained, <u>THEN:</u> TRIP the Reactor CONTINUE with this procedure GO TO 2-EOP-TRIP-1, Reactor Trip Or Safety Injection.	Identifies all BF19 valves closing. Trips the Reactor.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Loss of Control Air

TASK: TCAF a Loss of Control Air

*	STEP NO.	STEP (* Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	TRIP-1		Performs Immediate Actions of EOP-TRIP-1: Confirms the Rx trip Trips the Main turbine Checks at least 1 4KV vital bus energized Determines SI is not initiated or required. Cue after Immediate Actions have been performed: The CRS and another operator will continue performing the EOPs. You are to continue in current procedure.		
			Terminate JPM once Rx has been tripped, letdown isolated, and the WG release terminated.		

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ JUN 1. Task description and number, JPM description and number are identified.
- ✓ JUN 2. Knowledge and Abilities (K/A) references are included.
- ✓ JUN 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ JUN 4. Initial setup conditions are identified.
- ✓ JUN 5. Initiating and terminating Cues are properly identified.
- ✓ JUN 6. Task standards identified and verified by SME review.
- JUN 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- JUN 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 21 Date 8/22/16
- JUN 9. Pilot test the JPM:
 - a. verify Cues both verbal and visual are free of conflict, and
 - b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: MLUTEV JUN

Date: 8/22/16

SME/Instructor: W. Russell

Date: 8/22/16

SME/Instructor: _____

Date: _____

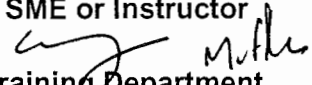
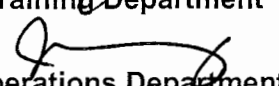
INITIAL CONDITIONS:

- Unit 2 is operating at 100% power.
- A WG release is in progress from 21 GDT.
- #3 SAC is C/T for scheduled maintenance.
- Unit 1 is operating at 100% power.

INITIATING CUE:

You are the Unit 2 Reactor Operator. Respond to all indications and alarms.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Pressurizer/Pressure Relief		
TASK:	Transfer Pressurizer 22 Backup Heaters to Emergency Power Supply		
TASK NUMBER:	115 043 05 01		
JPM NUMBER:	15-01 NRC Annual IP-i		
ALTERNATE PATH:	☐	K/A NUMBER:	010 A4.02
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:		In Plant, Simulate	
REFERENCES:		S2.OP-SO.PZR-0010 Rev. 10 (Rev. checked 8-5-16)	
TOOLS AND EQUIPMENT:		None	
VALIDATED JPM COMPLETION TIME:		<u>20 minutes</u>	
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:		<u>N/A</u>	
Developed By:	G Gauding Instructor	Date:	8-5-16
Validated By:	S Bickhart / J Hancock SME or Instructor	Date:	9-30-16
Approved By:	 Training Department	Date:	10/10/16
Approved By:	 Operations Department	Date:	10/10/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

SYSTEM: Pressurizer/Pressure Relief

TASK: Transfer Pressurizer 22 Backup Heaters to Emergency Power Supply

TASK 115 043 05 01

NUMBER:

INITIAL CONDITIONS:

Due to a loss of power to the group buses it is necessary to energize the PZR Backup Heaters from their emergency power supply.

INITIATING CUE:

The CRS directs you to perform S2.OP-SO.PZR-0010(Q), Section 5.3, Transferring Pzr Htr 22 BU Group To The Emergency Power Supply.

All prerequisites and Precautions and Limitations have been reviewed and completed.

2A EDG is carrying 2A Vital Bus and is loaded to 2400 KW.

You have a JAM key.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Simulate opening all but three PZR heater disconnects.
2. Simulate aligning emergency power to PZR heater bus from 2A vital bus.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: Pressurizer/Pressure Relief

TASK: Transfer 22 Backup Heaters to Emergency Power Supply

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Operator obtains procedure S2.OP-SO.PZR-0010, Pressurizer Backup Heaters Power Supply Transfer.	Provide marked up copy of S2.OP-SO.PZR-0010, Pressurizer Backup Heaters Power Supply Transfer		
	5.3.1	ENSURE 2AX1AX14X, 2EP PRESSURIZER HEATER BUS FEED (EMERGENCY) is OPEN (84' Swgr Rm).	Locates breaker 2AX1AX14X 2EP PRESSURIZER HEATER BUS FEED (EMERGENCY) on 2A 460V Vital Bus, Elev. 84' Swgr Rm and determines current breaker position. Cue: 2AX1AX14X Breaker indicates OPEN.		
	5.3.2	Request NCO to PERFORM the following: A. PLACE 22 Backup Group in MANUAL. B. PRESS the 22 BACKUP OFF pushbutton	Locates nearest means of communications and requests NCO to place 22 B/U Htrs in MANUAL and press the 22 BACKUP OFF pushbutton. Cue: NCO reports 22 B/U Htrs are in MANUAL and 22 BACKUP OFF pushbutton has been depressed.		
*	5.3.3	ENSURE 2EX1EP2EPX, 2EP 480V PRESSURIZER HEATER BUS MAIN BREAKER is OPEN (78' Electrical Penetration).	Locates 2EX1EP2EPX, 2EP 480V PRESSURIZER HEATER BUS MAIN BREAKER on El. 78 Electrical Pen. Cue: 2EX1EP2EPX, 2EP 480V PRESSURIZER HEATER BUS MAIN BREAKER is OPEN.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

SYSTEM: Pressurizer/Pressure Relief

TASK: Transfer 22 Backup Heaters to Emergency Power Supply

NAME: _____

DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.3.4	REMOVE interlock key from breaker 2EX1EP2EPX.	Describes turning key and removing interlock key from breaker 2EX1EP2EPX. Note: Tab must be pushed in to rotate key to allow its removal.		
*	5.3.5	PLACE any eleven of the following disconnects in OFF (only three disconnects are to remain ON) <u>AND</u> MARK the remaining disconnects as N/A:	Operator locates individual heater disconnects, places all but three disconnects in OFF and notes disconnects left ON as N/A in procedure.		
*	5.3.6	PLACE , 2AX1AX14X-1, 2EP PZR HTR BUS EMERGENCY FEED DISCONNECT SWITCH, in the ON position (Elev 78' Electrical Penetration).	Operator locates 2AX1AX14X-1, 2EP PZR HTR BUS EMERGENCY FEED DISCONNECT SWITCH, and places it in the ON position (Elev 78' Electrical Penetration).		
*	5.3.7	INSERT interlock key, <u>AND</u> UNLOCK breaker 2AX1AX14X.	Locates Corry Key Interlock on 2AX1AX14X breaker, inserts and turns key.		
	5.3.8	NOTIFY NCO that PZR Htr 22 B/U Group has been transferred to the emergency power supply (2A 460V Vital Bus).	Candidate makes report to the Control Room. Repeat back notification and state: JPM is complete.		

**OPERATIONS TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- h 2 5 / 1. Task description and number, JPM description and number are identified.
- h 2 5 / 2. Knowledge and Abilities (K/A) references are included.
- h 2 5 / 3. Performance location specified. (in-plant, control room, or simulator)
- u 2 5 / 4. Initial setup conditions are identified.
- h 2 5 / 5. Initiating and terminating Cues are properly identified.
- h 2 5 / 6. Task standards identified and verified by SME review.
- h 2 5 / 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- h 2 5 / 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 4 Date 11/4/15
- h 2 5 / 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature] Hanner

Date: 9/30/16

SME/Instructor: [Signature] BK

Date: 9/30/16

SME/Instructor: _____

Date: _____

**OPERATIONS TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

**INITIAL
CONDITIONS:**

Due to a loss of power to the group buses it is necessary to energize the PZR Backup Heaters from their emergency power supply.

INITIATING CUE:

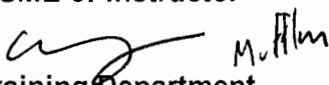
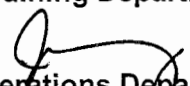
The CRS directs you to perform S2.OP-SO.PZR-0010(Q), Section 5.3, Transferring Pzr Htr 22 BU Group To The Emergency Power Supply.

All prerequisites and Precautions and Limitations have been reviewed and completed.

2A EDG is carrying 2A Vital Bus and is loaded to 2400 KW.

You have a JAM key.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

STATION:	SALEM		
SYSTEM:	Auxiliary Feedwater		
TASK:	Perform Auxiliary Feedwater Back Leakage Test		
TASK NUMBER:	N0610160204		
JPM NUMBER:	15-01 NRC IP-j		
ALTERNATE PATH:	☐	K/A NUMBER:	061 K1.02
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
		RO	3.4
		SRO	3.7
EVALUATION SETTING/METHOD:	Inplant- Simulate		
REFERENCES:	S1.OP-PT.AF-0002, Rev. 7, Auxiliary Feedwater Backleakage (checked 8-5-16)		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	5 minutes		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-5-16
Validated By:	S Bickhart / J Hancock SME or Instructor	Date:	9-30-16
Approved By:	 Training Department	Date:	10/10/16
Approved By:	 Operations Department	Date:	10/10/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE:

NAME: _____

DATE: _____

SYSTEM: Auxiliary Feedwater System

TASK: Perform Auxiliary Feedwater Back Leakage Test

TASK NUMBER: N0610160204

INITIAL CONDITIONS:

Unit 1 is operating at 100% power. 11 AFW pump was secured 30 minutes ago after its normally scheduled quarterly surveillance run.

INITIATING CUE:

You are directed to perform S1.OP-PT.AF-0002, Auxiliary Feedwater Backleakage.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made.

Task Standard for Successful Completion:

1. Determine 11 AFW pump and piping requires venting.
2. Vents 11 AFW pump and piping.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: AFW

TASK: Perform Auxiliary Feedwater Back Leakage Test

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
		Provide operator with blank copy of S1.OP-PT.AF-0002, Rev. 7, and a calculator.	Reviews and initial all Prerequisites and Precautions and Limitations.		
* * *	5.1.1	RECORD following on Attachment 1, Section 1.0: • AFW Line Temperatures (1TA16555I) • Highest line temperature • Average line temperature • Temperature difference	After locating 1TA165551, provide the following AFW line temperatures: 11- 104 12- 106 13- 161 14- 108 On Attachment 1, records AFW temperatures, calculates total of 479 , divides that by 4 and determines highest line temp is 161 and average line temp is 119.75 . Determines Temp Diff 161- 119.75 = 41.25 .		
	5.1.2	IF temperature in each line is <120°F, AND temperature difference between highest and average line temp is ≤10°F, THEN RECORD Test Results in Attachment 1, Section 5.0.	Determines highest temp is >120°F		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: AFW

TASK: Perform Auxiliary Feedwater Back Leakage Test

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
	5.1.3	IF temperature in any line is $\geq 120^{\circ}\text{F}$ AND $< 160^{\circ}\text{F}$, THEN: A. RECORD Auxiliary Feedwater System piping temperatures every hour in Control Room Narrative Log. B. RECORD Test Results in Attachment 1, Section 5.0.	Determines highest temp is $> 160^{\circ}\text{F}$		
*	5.1.4	IF temperature difference between highest and average line temp is $> 10^{\circ}\text{F}$, THEN RECORD Test Results in Attachment 1, Section 5.0.	Determines temperature difference between highest and average line temp is $> 10^{\circ}\text{F}$, and records Test Results in Attachment 1, Section 5.0., as UNSAT for all three categories.		
	5.1.6	IF temperature in any line is $\geq 160^{\circ}\text{F}$ THEN: A. RECORD findings in Control Room Narrative Log. B. VENT Auxiliary Feedwater pumps/piping IAW Section 5.2.			

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____
DATE: _____

SYSTEM: AFW

TASK: Perform Auxiliary Feedwater Back Leakage Test

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT evaluation)
*	5.2.1	IF 11 Auxiliary Feedwater Pump and piping is affected, THEN: A. SLOWLY THROTTLE OPEN 11AF83, AF PUMP VENT, until a solid stream of water is observed. B. CLOSE 11AF83. C. SLOWLY THROTTLE OPEN 11AF114, AF PUMP DISCH VENT, until a solid stream of water is observed. D. CLOSE 11AF114.	Locates and opens 11AF83, AF PUMP VENT, until a solid stream of water is observed. Cue: A solid stream of water is observed. Closes 11AF83 Locates and opens 11AF114, AF PUMP DISCH VENT, until a solid stream of water is observed. Cue: A solid stream of water is observed. Closes 11AF114.		
			Terminate JPM		

This page intentionally blank.

JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 1. Task description and number, JPM description and number are identified.
- ✓ 2. Knowledge and Abilities (K/A) references are included.
- ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 4. Initial setup conditions are identified.
- ✓ 5. Initiating and terminating Cues are properly identified.
- ✓ 6. Task standards identified and verified by SME review.
- ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 7 Date 1/6/10
- 2 ✓ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature] HANCOCK

Date: 9/30/16

SME/Instructor: [Signature] BICKHART

Date: 9/30/16

SME/Instructor: _____

Date: _____

JOB PERFORMANCE MEASURE

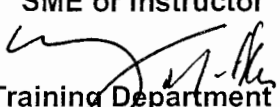
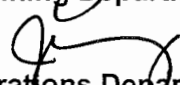
INITIAL CONDITIONS:

Unit 1 is operating at 100% power. 11 AFW pump was secured 30 minutes ago after its normally scheduled quarterly surveillance run.

INITIATING CUE:

You are directed to perform S1.OP-PT.AF-0002, Auxiliary Feedwater Backleakage.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM		
SYSTEM:	Emergency Diesel Generator		
TASK:	Start and synchronize an Emergency Diesel Generator		
TASK NUMBER:	113 001 05 01		
JPM NUMBER:	15-01 NRC IP-k		
ALTERNATE PATH:	☐	K/A NUMBER:	064 A2.09
APPLICABILITY:	IMPORTANCE FACTOR:		
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	In Plant / Simulate		
REFERENCES:	2A Diesel Generator Operation, S2.OP-SO.DG-0001(Q)Rev. 39 (checked 8-5-16)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	20 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-5-16
Validated By:	S Bickhart / J Hancock SME or Instructor	Date:	9-30-16
Approved By:	 Training Department	Date:	10/10/16
Approved By:	 Operations Department	Date:	10/10/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: EDG

TASK: Start and synchronize an Emergency Diesel Generator

**TASK
NUMBER:** 113 001 05 01

**INITIAL
CONDITIONS:** Unit 2 is operating at 100% power, with no equipment OOS. Engineering has requested a loaded run of 2A EDG.

INITIATING CUE:

The Unit 2 CRS has directed you to locally start and parallel load 2A D/G IAW S2.OP-SO.DG-0001(Q), Section 5.2. & 5.4

Prerequisites, Precautions and Limitations, and Section 5.1, Diesel Generator Startup Checks have been performed.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

Perform S2.OP-SO.DG-0001 Sections 5.2 and 5.4 in correct order and to completion.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide marked up copy of S2.OP-SO.DG-0001 2A Diesel Generator Operation			
	5.2.1	NOTIFY NCO that 2A Diesel Generator is to be locally started.	Locates nearest page. Cue: NCO acknowledges.		
	5.2.2	CHECK voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel is OFF	Points out voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel. Cue: Light is off		
	5.2.3	CHECK speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel is OFF.	Points out speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel. Cue: Light is off		
*	5.2.4	<u>IF</u> 2A 4KV Vital Bus is currently energized, (Diesel Generator is to be parallel loaded), <u>THEN:</u> A. PLACE 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW in MANUAL (DROOP). B. ENSURE B-9, GENERATOR LOADING IN DROOP MODE, is in alarm.	Points out 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW., and simulates rotating to DROOP position. Cue: Annunciator B-9, Generator loading in Droop Mode annunciates.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.5	PLACE 2A-DF-SS, 2A DG STOP/START SWITCH in START.	Points out 2A-DF-SS, 2A DG STOP/START SWITCH, and simulates placing in START. Cue: 2A Diesel is accelerating.		
*	5.2.6	<u>IF</u> Diesel Generator Speed is <u>NOT</u> 900 rpm, <u>THEN SET</u> speed to 900 rpm using the SPEED CONTROL SWITCH (GS).	Cue when Speed indicator is checked: Speed is 880 rpm. Points out SPEED CONTROL SWITCH (GS) and simulates turning in RAISE direction. Cue: 2A Diesel Speed is 900 rpm.		
	5.2.7	<u>IF</u> 2AD1AX6D*, 2A Diesel Generator 125VDC breaker is closed, <u>THEN</u> : CHECK voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE, on Generator Control Panel is ON. CHECK speed permissive indicator light 2DAE4-LT3, EDG SPEED, on Generator Control Panel is ON.	Points out voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE. Cue: Voltage permissive indicator light 2DAE4-LT2, EDG VOLTAGE Light is on Points out speed permissive indicator light 2DAE4-LT3, EDG SPEED Cue: Speed permissive indicator light 2DAE4-LT3, EDG SPEED light is on		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.2.8	IF Field Ground Relay 64/G white indicating light is OFF, <u>AND</u> C-6, GENERATOR FIELD GROUND, is clear, <u>THEN</u> : A. RESET 64/G relay B. ENSURE 64/G white indicating light is illuminated.	Points out Field Ground Relay 64/G white indicating light. Cue: Field Ground Relay 64/G white indicating light is illuminated.		
	5.2.9	ENSURE 2A Diesel Generator K1C Field Flashing Relay Supervisory Light is OFF.	Points out 2A Diesel Generator K1C Field Flashing Relay Supervisory Light Cue: 2A Diesel Generator K1C Field Flashing Relay Supervisory Light is off.		
	5.2.10	RECORD the following Diesel Generator Start Readings:	Points out gages and switches to determine readings. Cue when proper gauge is pointed out : <u>2VM189</u> Gen. Volts - 4160 Volts on all 3 phases <u>2FM186</u> Gen. Frequency - 60 Hz <u>2PL6429</u> LO Hdr. Press - 80 psig <u>2PL6449</u> JW Hdr. Press - 45 psig <u>2PL7209</u> Air Manifold Press - 0 psig <u>2TA16524</u> Gen. Stator Temp - 187°		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.2.11	IF 2A Diesel Generator is to be operated unloaded for an extended period of time(>30 minutes), THEN INITIATE Section 5.7, Diesel Generator Running Checks.	Determines from initiating cue that 2A EDG is to be parallel loaded.		
	5.2.12	IF Diesel Generator is to be parallel loaded, THEN INITIATE Section 5.4, Diesel Generator Parallel Loading.	Initiates Section 5.4, Diesel Generator Parallel Loading.		
	5.4.1	IF Section 5.1 Diesel Generator Startup Checks, was NOT performed, THEN at 2A DG 4KV cabinet cubicle breaker 2AD1AX6D, PLACE selector switch 2A DG SYNC ENABLE in ENABLE.	Verifies Section 5.1 is complete.		
	5.4.2	NOTIFY NCO that 2A EDG is to be synchronized and loaded locally.	Cue: NCO acknowledges 2A DG is to be synchronized and loaded locally.		
	5.4.3	ENSURE 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW, in MANUAL (DROOP)	Verifies 2A-DF-GCP-1, 2A DIESEL GEN LOADING SW, is in MANUAL (DROOP)		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.4.4	ADJUST 2A EDG output voltage, as indicated on VOLTMETER-GEN (2VM189) to 50 - 100 Volts higher than that of 2A 4KV Vital Bus voltage, as indicated by VOLTMETER-BUS (2VM190), using the VOLTAGE CONTROL SWITCH (VCS).	When operator points out 2VM189 and 2VM190, then Cue: 2VM189 reads 4150, and 2VM190 reads 4150. Points out VOLTAGE CONTROL SWITCH (VCS) and simulates going to RAISE. Cue: 2VM189 reads 4210, and 2VM190 reads 4150.		
	5.4.5	ENSURE generator terminal voltage is present on all 3 phases by rotating VOLTMETER SWITCH - GEN (VS-G) through each position <u>AND</u> OBSERVING voltmeter 2VM189.	Points out VOLTMETER SWITCH - GEN (VS-G), simulates rotating through each position, while observing 2VM189 Cue: Voltage is present on all three phases.		
	5.4.6	ENSURE 2A 4KV Vital Bus voltage is present on all 3 phases by rotating VOLTMETER SWITCH - BUS (VS-B) through each position <u>AND</u> OBSERVING voltmeter 2VM190.	Points out VOLTMETER SWITCH - BUS (VS-B), simulates rotating through each position, while observing 2VM190 Cue: Voltage is present on all three phases.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.4.7	PLACE the following switches in "1-2": - VOLTMETER SWITCH - GEN (VS-G) - VOLTMETER SWITCH - BUS (VS-B)	Points out VOLTMETER SWITCH - GEN (VS-G) and VOLTMETER SWITCH - BUS (VS-B) Cue: Switches are in 1-2.		
	5.4.8	ENSURE 2DAE4-LT2, EDG Voltage indication light is ON.	Points out 2DAE4-LT2, EDG Voltage indication light. Cue: 2DAE4-LT2, EDG Voltage indication light is on.		
	5.4.9	ENSURE 2DAE4-LT3, EDG Speed indication light is ON.	Points out 2DAE4-LT3, EDG Speed indication light. Cue: 2DAE4-LT2, EDG Voltage indication light is on.		
	5.4.10	SYNCHRONIZE 2A Diesel Generator to 2A 4KV Vital Bus as follows:			
*	5.4.10 A.	PLACE 2A-DF-SYNCH, 2A DG SYNC SWITCH (SS) to ON.	Points out 2A-DF-SYNCH, 2A DG SYNC SWITCH (SS), and simulates placing in ON position.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.4.10 B.	ADJUST Diesel speed using the SPEED CONTROL SWITCH (GS) such that Synchroscope pointer rotates slowly in the FAST (clockwise) direction.	When operator points out synchroscope, then Cue: Scope is going in the slow direction. Points out SPEED CONTROL SWITCH (GS) and simulates going to RAISE. Cue Synchroscope pointer rotating slowly in the FAST direction.		
	5.4.10 C.	ENSURE the following four SYNC CHECK RELAY 25 UPPER AND LOWER VOLTAGE LIMIT LEDS are ON. "Upper Voltage Limit" "L OK" "B OK" "Lower Voltage Limit" "L OK" "B OK"	Points out four SYNC CHECK RELAY 25 UPPER AND LOWER VOLTAGE LIMIT LEDS. Cue as each LED is located: Light is on.		
	5.4.10 D.	<u>IF</u> the SYNC CHK RELAY 25 ΔF OK LED is OFF, <u>THEN</u> ADJUST 2A Diesel Generator speed using the SPEED CONTROL SWITCH (GS) until the ΔF OK LED is ON.	Points out SYNC CHK RELAY 25 ΔF OK LED. Cue Light is on.		
	5.4.10 E.	ENSURE DG SYNC PERMISSIVE green indicating light is ON each time the synchroscope indicator is near 12 o'clock position (+/- approximately 3 minutes), <u>AND</u> is OFF in any other position of the synchroscope.	Cue DG SYNC PERMISSIVE green indicating light is ON each time the synchroscope indicator is near 12 o'clock position (+/- approximately 3 minutes), <u>AND</u> is OFF in any other position of the synchroscope.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

SYSTEM: Emergency Diesel Generator
TASK: Locally start and load an Emergency Diesel Generator

NAME: _____
DATE: _____

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.4.10 F	COORDINATE the following when the synchroscope is at "12 o'clock" (+zero, -2 minutes): CLOSE 2A-DF-GCP-3 GENERATOR CIRCUIT BREAKER SWITCH (BCS) RAISE 2A Diesel Generator load to $\geq 500\text{KW}$ using SPEED CONTROL SWITCH (GS) to prevent tripping the Diesel Generator Breaker on reverse power.	Points out 2A-DF-GCP-3 GENERATOR CIRCUIT BREAKER SWITCH (BCS) and simulates taking to the CLOSE position when the synch scope is at 12 o'clock. Cue 2A Diesel Generator Breaker has closed. Points out SPEED CONTROL SWITCH (GS) simulates going to RAISE while observing KW meter. Cue Load is 500 KW.		
	5.4.10 G	PLACE 2A-DF-SYNCH, 2A DG SYNCH SWITCH (SS) to OFF.	Points out 2A-DF-SYNCH, 2A DG SYNCH SWITCH (SS), and simulates placing it in OFF position. Cue: JPM is complete.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- ✓ 25 ✓ 1. Task description and number, JPM description and number are identified.
- ✓ 25 ✓ 2. Knowledge and Abilities (K/A) references are included.
- ✓ 25 ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- ✓ 25 ✓ 4. Initial setup conditions are identified.
- ✓ 25 ✓ 5. Initiating and terminating Cues are properly identified.
- ✓ 25 ✓ 6. Task standards identified and verified by SME review.
- ✓ 25 ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- ✓ 25 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 37 Date 9/30/16
- 25 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature] Hancock

Date: 9/30/16

SME/Instructor: [Signature] Bickart

Date: 9/30/16

SME/Instructor: _____

Date: _____

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

Unit 2 is operating at 100% power, with no equipment OOS. Engineering has requested a loaded run of 2A EDG.

INITIATING CUE:

The Unit 2 CRS has directed you to locally start and parallel load 2A D/G IAW S2.OP-SO.DG-0001(Q), Section 5.2. & 5.4

Prerequisites, Precautions and Limitations, and Section 5.1, Diesel Generator Startup Checks have been performed.

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 15-01 NRC ESG-1

SCENARIO NUMBER: 15-01 NRC ESG-1

EFFECTIVE DATE: See Approval Dates

EXPECTED DURATION: 60 minutes

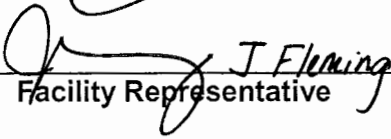
REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

Revision Summary
New issue for 15-01 ILOT NRC exam

PREPARED BY: G Gauding 7-18-16
Lead Regulatory Exam Author Date

APPROVED BY:  10/11/16
Operations Training Manager Date

APPROVED BY:  10/11/16
Facility Representative Date

I. OBJECTIVES

- A. Given the order or indications of a charging system malfunction, perform actions as the nuclear control operator to RESPOND to the loss or malfunction of the charging system in accordance with S2.OP-AB.CVC-0001.
- B. Given indication of a loss or malfunction of the Charging system DIRECT corrective action for a Charging System malfunction in accordance with S2.OP-AB.CVC-0001.
- C. Given the indications of a reactor coolant system (RCS) malfunction or leak, perform actions as the nuclear control operator to RESPOND to the malfunctioning accordance with S2.OP-AB.RC-0001.
- D. Given the indications of a reactor coolant system (RCS) malfunction or leak, DIRECT the response to the malfunction in accordance with S2.OP-AB.RC-0001.
- E. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with 2-EOP-TRIP-1.
- F. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with 2-EOP-TRIP-1.
- G. Given the order or indications of a safety injection, perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- H. Given indication of a safety injection, DIRECT the response to the safety injection in accordance with the approved station procedures.
- I. Given a safety injection has occurred and inadequate ECCS flow exists, ALIGN equipment to obtain required ECCS flow in accordance with station procedures.
- J. Given the order or indications of a loss of coolant accident (LOCA), complete actions as the nuclear control operator to PERFORM the immediate response to the LOCA in accordance with the approved station procedures.
- K. Given indication of a loss of coolant accident (LOCA), DIRECT the immediate response to the LOCA in accordance with the approved station procedures.
- L. Given conditions that allow SI termination, perform actions as the nuclear control operator to TERMINATE safety injection in accordance with the approved station procedures.
- M. Given conditions that allow SI termination, DIRECT the response to the safety injection termination in accordance with the approved station procedures.

II. MAJOR EVENTS

- A. Controlling PZR Level Channel fails low
- B. 20 gpm RCS leak
- C. 150 gpm SBLOCA
- D. SBLOCA escalation during SI termination

III. SCENARIO SUMMARY

- A. The crew will take the watch with the unit at 100% power, MOL. PZR Pressure Channel III is selected for Control.
- B. After the crew assumes the watch, the controlling PZR level channel will fail low. CVCS Letdown will isolate, and the Master Flow Controller will raise charging flow. The crew will enter S2.OP-AB.CVC-0001, Loss of Charging. The crew will take manual control of charging flow and return PZR level to program. The crew will remove the failed channel from service. The crew will restore normal letdown. The CRS will identify Tech Specs.
- C. After the PZR level channel failure has been addressed, a 20 gpm RCS leak will occur. The crew will diagnose the leak. The crew will swap to a centrifugal charging pump and a 45 gpm orifice, and estimate the leak rate. The CRS will identify Tech Specs.
- D. The CRS will determine a unit shutdown is required based on the RCS leak. The crew will initiate a controlled shutdown.
- E. After the downpower has commenced, the RCS leak rate will rise to 150 gpm. The crew will diagnose deteriorating primary conditions. When it is determined that PZR level will be unable to be maintained >11%, the crew will take the CAS action to trip the reactor and initiate Safety Injection.
- F. The crew will perform plant stabilization and diagnostics in EOP-TRIP-1, Reactor Trip or Safety Injection. The crew will identify no AFW flow. 2A SEC will not reset, and will require deenergization to start 21 AFW pump. The crew will establish at least 22E4 lbm/hr AFW flow. (CT#1)
- G. The crew will determine there is no High Head ECCS flow, and open the BIT Isolation Valves to establish High Head ECCS Injection (CT#2).
- H. The crew will transition to EOP-LOCA-1, Loss of Reactor Coolant when directed in TRIP-1.
- I. The crew will reset Safeguards in LOCA-1, and determine SI termination criteria are met to transition to EOP-TRIP-3, SI Termination.
- J. The crew will stop all but one Charging pump and align CVCS for normal charging. When evaluating the leak, the SBLOCA will get significantly larger. The crew will diagnose degrading conditions and take the CAS action to start ECCS pumps as necessary, and go to EOP-LOCA-2.
- K. The scenario will terminate when the transition to LOCA-2 has been made.

IV. INITIAL CONDITIONS

___ IC-231

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
___ 1	VC1and VC4 C/T
___ 2	RCPs (SELF CHECK)
___ 3	RTBs (SELF CHECK)
___ 4	MS167s (SELF CHECK)
___ 5	500 KV SWYD (SELF CHECK)
___ 6	SGFP Trip (SELF CHECK)
___ 7	23 CV PP (SELF CHECK)
___ 8	PZR Pressure Channel III selected for Control.
___ 9	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
		EVENT ACTION:
		COMMAND:
		PURPOSE:

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
___ 01	PR0017A PZR LEVEL CH I (LT459) FAILS H/L	N/A	N/A	N/A	RT-1	0
___ 02	RC0002 RCS LEAK INTO CONTAINMENT (equiv to 0-4 inches)	N/A	N/A	N/A	RT-3	20
___ 03	RP0342 2SJ12 FAILS TO AUTO OPEN ON SEC	N/A	N/A	N/A	N/A	
___ 04	RP0343 2SJ13 FAILS TO AUTO OPEN ON SEC	N/A	N/A	N/A	N/A	
___ 05	RP318E1 Aux Feed Pump 21 Fails to Start on SEC	N/A	N/A	N/A	N/A	
___ 06	RP318E2 Aux Feed Pump 22 Fails to Start on SEC	N/A	N/A	N/A	N/A	
___ 07	AF0353A 21 AFP FAILURE TO AUTO START ON ANY (ALL) SIGNALS	N/A	N/A	N/A	N/A	
___ 08	AF0353B 22 AFP FAILURE TO AUTO START ON ANY (ALL) SIGNALS	N/A	N/A	N/A	N/A	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
___ 01	DG01D DEENERGIZE "A" SEC CABINET	N/A	N/A	N/A	RT-9	YES
___ 02	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10
___ 03	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160
___ 04	MS05A 21MS45 21 STM GEN STM SUP-23 AFP	N/A	N/A	N/A	N/A	0
___ 05	MS06A 23MS45 23 STM GEN STM SUP-23 AFP	N/A	N/A	N/A	N/A	0

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
___ 01	CB09 A DI KCB09ASE 2A SFGD EMER OPR-EMERG LOADING RESET	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:

Description

1.

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Controlling PZR Level Channel Fails LOW Simulator Operator: Insert RT-1 on direction from Lead Evaluator. MALF: PR0017A PZR LVL CH I (LT459) FAILS H/L Severity: 0			
	RO reports unexpected OHA E-36, PZR HTR OFF LVL LO.		
	RO diagnoses controlling PZR level channel failure.		
	RO may request permission to place the Master Flow Controller in manual prior to direction in AB.CVC to do so.		
	CRS enters S2.OP-AB.CVC-0001, Loss of Charging.		
	CRS directs initiation of Attachment 1 CAS.		
	RO reports charging pumps are running, and no indication of cavitation.		
	RO reports PZR level channel I has failed low.		
	If not previously performed, RO takes manual control of charging and maintains PZR level within band assigned by CRS.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Proceed to next event after Tech Spec call at direction from Lead evaluator.	RO selects operable channel for Control, Alarm, and Recorders.		
	RO restores PZR heaters to pre-event condition.		
	RO announces OHA E-20 PZR HTR ON LVL HI as expected when it annunciates due to no letdown with charging flow established.		
	PO restores letdown by: - Ensuring open 2CV7 - Placing 2CV18 in manual and closing, then opening until close PB extinguishes. - Opening 2CV2 and 2CV277, then placing both in auto. - Ensuring charging flow is ~85-90 gpm. - Opening a 75 gpm orifice isolation valve while adjusting 2CV18 to control letdown pressure ~300 psig. - Placing 2CV18 in auto.		
	RO returns Master Flow Controller to auto when PZR level returns to program.		
	CRS contacts IC for assistance in removing channel from service.		
	CRS enter LCO 3.3.1.1 action 6.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
3. 20 gpm RCS leak			
Simulator Operator: Insert RT-3 on direction from Lead Evaluator			
MALF: RC0002 – RCS Leak into Containment. Final Value: 20 gpm			
	RO reports that charging flow is rising (if MFC has been returned to auto) and PZR level is lowering slowly.		
	Crew reports reading on 2R11A containment radiation monitor is rising.		
	RO reports unexpected OHA C-2 CNTMT SUMP PMP START when it occurs.		
	CRS enters S2.OP-AB.RC-0001, Reactor Coolant System Leak.		
	CRS directs initiation of Attachment 1 CAS.		
	RO reports RCS temperature > 350°F in Mode 1.		
	RO reports PZR level is lowering.		
Note: PZR level is above program, and charging flow will be lower than normal to reduce PZR level. If charging flow is raised here to stabilize level, then step to place a centrifugal charging pump in			

Role Play: If contacted state Rad Pro recommends placing 2 CFCU's in Low Speed and 2 CFCU's in High Speed.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Role Play: After RCS leakage Tech Spec has been identified, if CRS does not initiate a power reduction to bring the unit offline, call Control Room as Operations Director and instruct the CRS to initiate a 30% per hour load reduction to 20% in preparation for taking Unit 2 offline to investigate and repair the RCS leak. CRS may enter S2.OP-AB.LOAD-0001 Rapid Load Reduction to perform the power reduction. Proceed to next event on direction from Lead Evaluator. 3. SBLOCA	CRS enters TSAS 3.4.7.2.b action b for RCS leakage, and 3.6.1.4 for containment pressure if it has reached 0.3 psig.		
	CRS briefs power reduction.		
	RO initiates boration.		
	PO initiates turbine load reduction at 30% per hour.		
Simulator Operator: MODIFY MALF RC0002 from 20 to 150 on direction from Lead Evaluator.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
4. AFW fails to initiate	RO reports PZR level trending down and adjusts charging flow to maintain PZR level.		
	RO reports PZR level will not be able to be maintained >17%.		
	CRS directs Rx trip and Safety Injection		
	RO trips the reactor and confirms the reactor trip.		
	RO initiates Safety Injection.		
	RO continues Immediate Actions of TRIP-1 - Trips the Main Turbine - Reports at least one 4KV vital bus energized. - Reports SI initiated.		
	CRS and RO verify performance of immediate actions.		
	PO reports SEC loading is not complete for 2A and 2B vital buses, 21 and 22 AFW pumps have failed to start. Reports 23 AFW pump started but is not providing any AFW flow.		
	RO blocks 2A and 2B SECs.		
	PO resets 2A and 2B SECs.		
	PO reports 2A SEC will not reset.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	CRS dispatches operator to deenergize 2A SEC.		
Simulator Operator: 2 minutes after being dispatched, insert RT-9 to deenergize 2A SEC.			
	RO starts 22 AFW pump.		
CT#1 (CT-5) Establish at least 22E4 lbm/hr AFW flow to the SG's prior to a transition to FRHS-1 being required. SAT _____ UNSAT _____			
	PO reports AFW pump status, and starts 21 AFW pump when 2A SEC is deenergized.		
	PO reports not all valve groups in Table B are in safeguards position.		
	PO reports 2SJ12 and 2SJ13 BIT isolation valves have not opened.		
	RO opens 2SJ12 and 2SJ13 and reports BIT flow is established.		
CT#2 (CT-6) Establish flow from at least one charging pump prior to transition out of TRIP-1. SAT _____ UNSAT _____			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports 21 and 22CA330s are shut.		
	RO reports containment pressure has remained <15 psig.		
	PO reports no high steam flow condition.		
	CRS directs implementation of the ECG.		
	PO reports all 4KV vital buses energized.		
	RO reports CAV in AP mode.		
	RO reports 2 switchgear supply and one exhaust fan are running.		
	RO reports 2 or more CCW pumps running.		
	RO reports RCS not aligned for Cold Leg Recirc.		
	RO reports charging flow is at least 100 gpm on SI systems charging flow meter.		
	RO reports RCS pressure is not <1540 psig.		
	PO reports total AFW flow is 22E4 lbm/hr or 9% level in at least one SG NR level, then maintains 19-33% level.		
	RO reports all RCPs running, and RCS temperature is stable at or trending to 547°F.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: CFST monitoring will commence after the transition out of TRIP-1. STA will report to the control room 10 minutes after being paged.	RO reports both RTBs are open.		
	RO reports both PZR PORVs are shut with their Block Valves open.		
	RO reports all RCPs in service and PZR spray operating as expected for current plant conditions.		
	RO reports RCS pressure >1350 psig and maintains seal injection flow to all RCPs.		
	PO reports no indication of faulted SGs.		
	PO/RO report no indications of a steam generator tube rupture.		
	RO reports radiation monitor channel status in Table F.		
	PO reports 2 or more channels rising or in warning or alarm.		
	CRS transitions to EOP-LOCA-1, Loss of Reactor Coolant.		
	PO reports no faulted SGs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports total AFW flow is 22E4 lbm/hr or 9% level in at least one SG NR level, then maintains 19-33% level.		
	PO reports no ruptured SGs.		
	RO resets SI, Phase A, and Phase B isolations.		
	RO opens 21 and 22CA330s.		
	PO resets 2C SEC, and reports 230V control centers reset.		
	RO resets SGBD Sample Isolation Bypass and opens 21-24SS94s.		
	CRS directs chemistry to sample SGs for activity and boron.		
	RO reports PORVs are shut with Block valves open.		
	RO reports subcooling > 0°F.		
	PO reports AFW flow >22E4 lbm/hr or SG NR level >9%.		
	RO reports RCS pressure is stable or rising.		
	RO reports PZR level >11%.		
	CRS transitions to EOP-TRIP-3, SI Termination.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO resets SI, Phase A, and Phase B isolations if not previously reset.		
	RO opens 21 and 22CA330s if not previously performed.		
	RO reports spray valves are not failed open.		
	PO resets 2C SEC and reports 230V control centers reset if not previously performed.		
	RO stops all but one charging pump.		
	RO reports RCS pressure is stable or rising.		
	RO reports charging pump suction is aligned to the RWST.		
	RO opens 2CV139 and 2CV140 if shut.		
	RO shuts BIT isolation valves 2SJ4, 2SJ5, 2SJ12, 2SJ13.		
Simulator Operator: MODIFY MALF RC0002 from 150 to 2000 when 2CV68 and 2CV69 are open.			
	RO shuts 2CV55 Charging Flow Control valve.		
	RO opens 2CV68 and 2CV69 Charging Discharge valves.		
	RO adjusts 2CV55 to maintain PZR level		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Terminate scenario on direction from Lead Evaluator after the transition to LOCA-2 is made.	>25%.		
	RO reports PZR level is not stable or rising.		
	PO reports no faulted SGs.		
	RO opens BIT isolation valves 2SJ4,5,12,13, and shuts normal charging line isolation valves 2CV68 and 69.		
	CRS goes to EOP-LOCA-2, Post LOCA Cooldown and Depressurization.		

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-IO.ZZ-0004, Power Operation
- F. S2.OP-AB.CVC-0001, Loss of Charging
- G. S2.OP-AB.RC-0001, Reactor Coolant System Leak
- H. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- I. 2-EOP-TRIP-3, Safety Injection Termination
- J. 2-EOP-LOCA-1, Loss of Reactor Coolant
- K. 2-EOP-LOCA-2, Post LOCA Cooldown and Depressurization

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 100% RCS BORON: 857 MWe 1220

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

ABNORMAL PLANT CONFIGURATIONS:

PZR Pressure Channel III is selected for Control

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.
No penalty minutes in the last 24 hrs.

PRIMARY:

SECONDARY:

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
with Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 15-01 ILOT NRC ESG-1 **REVIEWER:** C Recchione

Initials Qualitative Attributes

- | | | |
|----|-----|---|
| CR | 1. | The scenario has clearly stated objectives in the scenario. |
| CR | 2. | The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events. |
| CR | 3. | The scenario consists mostly of related events. |
| CR | 4. | Each event description consists of: <ul style="list-style-type: none"> • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point |
| CR | 5. | No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event. |
| CR | 6. | The events are valid with regard to physics and thermodynamics. |
| CR | 7. | Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives. |
| CR | 8. | The simulator modeling is not altered. |
| CR | 9. | All crew competencies can be evaluated. |
| CR | 10. | The scenario has been validated. |
| CR | 11. | If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario. |
| CR | 12. | ESG-PSA Evaluation Form is completed for the scenario at the applicable facility. |

ATTACHMENT 4
SIMULATOR SCENARIO REVIEW CHECKLIST

Initial Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)

GG	3	Malfunctions after EOP entry: 1-2
GG	2	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	EOP based Critical tasks: 2-3

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

15-01 NRC ESG-1

CT#1 (CT-4) Establish the minimum required feed flow prior to a transition to FRHS-1 being required.

Basis: Failure to establish the required minimum feed flow rate under the postulated plant conditions, results in “adverse consequence(s) or significant degradation in the mitigative capability of the plant.”

Failure to perform the critical task represents a “demonstrated inability of the crew to:

- Take an action or combination of actions that would prevent a challenge to plant safety
- Recognize a failure / incorrect auto actuation of an ESF system or component.”

CT#2 (CT-6) Establish flow from at least one high-head ECCS pump before transition out of TRIP-1.

Basis: Failure to manually start at least one Charging pump under the postulated conditions constitutes misoperation or incorrect crew performance in which the crew does not prevent “degraded emergency core cooling system (ECCS) ... capacity.” In this case, at least one Charging/SI pump (high-head SI pump for LP plants) can be manually started from the control room. Therefore, failure to manually start a Charging/SI pump (high-head SI pump for LP plants) also represents a failure by the crew to “demonstrate the following abilities:

- Effectively direct or manipulate engineered safety feature (ESF) controls that would prevent a significant reduction of safety margin beyond that irreparably introduced by the scenario
- Recognize a failure or an incorrect automatic actuation of an ESF system or component”

Additionally, under the postulated plant conditions, failure to manually start a Charging/SI pump (high-head SI pump for LP plants) (when it is possible to do so) is a “violation of the facility license condition.”

Note to Evaluators: CT numbers in parentheses are the corresponding Westinghouse ERG Rev. 2- based Critical Task procedure WCAP-17711-NP

ATTACHMENT 6**ESG-PRA RELATIONSHIP EVALUATION****EVENTS LEADING TO CORE DAMAGE**

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
N	TRANSIENTS with PCS Unavailable	N	Loss of Service Water
N	Steam Generator Tube Rupture	N	Loss of CCW
N	Loss of Offsite Power	N	Loss of Control Air
N	Loss of Switchgear and Pen Area Ventilation	N	Station Black Out
Y	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
N	Containment Sump Strainers	N	Gas Turbine
N	SSWS Valves to Turbine Generator Area	N	Any Diesel Generator
N	RHR Suction Line valves from Hot Leg	Y	Auxiliary Feed Pump
N	CVCS Letdown line Control and Isolation Valves	N	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
N	Restore AC power during SBO
N	Connect to gas turbine
N	Trip Reactor and RCPs after loss of component cooling system
N	Re-align RHR system for re-circulation
N	Un-isolate the available CCW Heat Exchanger
N	Isolate the CVCS letdown path and transfer charging suction to RWST
N	Cooldown the RCS and depressurize the system
N	Isolate the affected Steam Generator that has the tube rupture(s)
N	Early depressurize the RCS
N	Initiate feed and bleed

Complete this evaluation form for each ESG.

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 15-01 NRC ESG-2
SCENARIO NUMBER: 15-01 NRC ESG-2
EFFECTIVE DATE: See Approval Dates
EXPECTED DURATION: 60 minutes
REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

Revision Summary
New issue for 15-01 ILOT NRC exam

PREPARED BY:	<u>G Gauding</u> Lead Regulatory Exam Author	<u>8-3-16</u> Date
APPROVED BY:	<u></u> Operations Training Manager	<u>10/12/16</u> Date
APPROVED BY:	<u></u> Facility Representative	<u>10/11/16</u> Date

SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

- A. Given a loss of a heater drain pump or condensate pump, take corrective action IAW S2.OP-AB.CN-0001.
- B. Given the order or indications of a feedwater or condensate system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S2.OP-AB.CN-0001.
- C. Given indication of a feedwater or condensate system malfunction, DIRECT the response to the malfunction in accordance with S2.OP-AB.CN-0001.
- D. Given a steam generator tube leak, take corrective action, IAW S2.OP-AB.SG-0001.
- E. Given the order or indications of a steam generator tube leak (SGTL), perform actions as the nuclear control operator to RESPOND to the tube leak in accordance with S1/S2.OP-AB.SG-0001.
- F. Given the order or indications of a steam generator tube leak (SGTL), DIRECT the response to the tube leak, in accordance with S2.OP-AB.SG-0001.
- G. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with 2-EOP-TRIP-1.
- H. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with 2-EOP-TRIP-1.
- I. Given the order or indications of a safety injection, perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- J. Given indication of a safety injection, DIRECT the response to the safety injection in accordance with the approved station procedures.
- K. Given the order or indications of a steam generator tube rupture (SGTR), perform actions as the nuclear control operator to RESPOND to the tube rupture in accordance with the approved station procedures.
- L. Given indication of a steam generator tube rupture (SGTR), DIRECT the response to the SGTR in accordance with the approved station procedures.

II. MAJOR EVENTS

- A. 2C EDG Pre-lube pump failure (TS)
- B. 23 Condensate pump trip /Power Reduction
- C. 21 SGTL / 21 SGTR
- D. Loss of Spray during RCS depressurization
- E. Stuck open PZR PORV during RCS depressurization

III. SCENARIO SUMMARY

- A. The crew will take the watch with the unit at 89.4% power, BOL. Power was reduced last shift in preparation for performing Main Turbine Valve testing later this shift. 2C EDG Jacket Water heater failed last shift. Condensate Polisher is in service.
- B. After the crew assumes the watch, the 2C EDG Pre-lube pump will fail. The CRS will identify 2C EDG is now inoperable, and enter the appropriate LCO. The CRS will initiate actions to ensure S2.OP-ST.500-0001, Electrical Power Systems AC Sources Alignment is performed within one hour.
- C. After the EDG failure has been addressed, 23 Condensate pump will trip. The crew will enter S2.OP-AB.CN-0001, Main Feedwater / Condensate System Abnormality, and perform actions to bypass the polisher and feedwater heaters strings as required to maintain SGFP suction pressure.
- D. The crew will perform a power reduction to $\leq 85\%$ power and ensure SGFP suction pressure remains above 320 psig.
- E. During the power reduction, a SGTL will occur on 21 SG. The crew will enter S2.OP-AB.SG-0001, Steam Generator Tube Leak. The crew will perform actions to minimize the spread of secondary contamination and minimize the potential for steam release from the affected SG. The crew will isolate the steam supply to 23 AFW pump from 21 SG, and the CRS will identify 23 AFW is inoperable and enter the appropriate LCO.
- F. After the AFW pump LCO is identified, the SG tube will rupture. The crew will identify the SGTR, and initiate a Rx trip and Safety Injection based on the uncontrolled rise in 21 SG NR or WR level.
- G. The crew will perform plant stabilization and diagnostics in EOP-TRIP-1, Reactor Trip or Safety Injection. 21 AFW pump will trip shortly after starting. The crew will isolate AFW flow to and steam flow from 21 SG. (CT-1).
- H. The crew will transition to EOP-SGTR-1, Steam Generator Tube Rupture to address the tube rupture. The crew will perform a cooldown to target temperature and demonstrate the ability to maintain RCS temperature < Target temperature (CT-2). The crew will initiate a RCS depressurization. During the depressurization the RCP providing driving head for Spray flow will trip. The crew will open a PZR PORV to continue the depressurization. If 2PR1 is the selected PORV, it will not open, and 2PR2 will be used.
- I. The PZR PORV opened to depressurize the RCS will fail to shut when demanded after the depressurization is complete. The crew will attempt to terminate the depressurization by closing the PORV Block valve, which will not shut.
- J. The scenario will be terminated after the CRS transitions to EOP-SGTR-3, SGTR with LOCA – Subcooled Recovery, to address the loss of RCS pressure with the stuck open PORV.

IV. INITIAL CONDITIONS

____ IC-232

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
____ 1	VC1and VC4 C/T
____ 2	RCPs (SELF CHECK)
____ 3	RTBs (SELF CHECK)
____ 4	MS167s (SELF CHECK)
____ 5	500 KV SWYD (SELF CHECK)
____ 6	SGFP Trip (SELF CHECK)
____ 7	23 CV PP (SELF CHECK)
____ 8	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
	1	EVENT ACTION: MONP254 <10. //CONTROL BANK C GROUP POSITION COMMAND: PURPOSE: <update as needed>
	3	EVENT ACTION: KB202LCI //2PR7 RELIEF VALVE-22 STOP VALVE COMMAND: PURPOSE: <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	AN0266 SER 266 FAILS - :J20 2C DIESEL GENERATOR URGENT TROUBLE	N/A	N/A	N/A	RT-1	SER POINT FAILS/OVRD TO ON
02	CN0117C 23 CONDENSATE PUMP TRIP	N/A	N/A	N/A	RT-3	
03	SG0078A 21 STEAM GENERATOR TUBE RUPTURE	N/A	N/A	N/A	RT-5	5
04	AN3735 AAS 735 FAILS - :21 TGA SUMP LEVEL HIGH	00:03:00	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
05	AN3736 AAS 736 FAILS - :22 TGA SUMP LEVEL HIGH	00:03:10	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
06	AN3737 AAS 737 FAILS - :23 TGA SUMP LEVEL HIGH	00:03:20	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
07	AN3738 AAS 738 FAILS - :24 TGA SUMP LEVEL HIGH	00:03:30	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
08	AN3739 AAS 739 FAILS - :25 TGA SUMP LEVEL HIGH	00:03:50	N/A	N/A	RT-11	AAS POINT FAILS/OVRD TO ON
09	AF0181A 21 AUX FEEDWATER PUMP TRIP	00:04:00	N/A	N/A	ET-1	
10	VL0297 2PR1 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	0
11	VL0298 2PR2 Fails to Position (0-100%)	N/A	N/A	N/A	RT-13	100
12	RC0003C 23 RC PUMP ELECTRICAL TRIP	N/A	N/A	N/A	RT-7	
13	RC0003A 21 RC PUMP ELECTRICAL TRIP	N/A	N/A	N/A	RT-9	
14	VL0123 2PR7 Fails to Position (0-100%)	N/A	N/A	N/A	ET-3	90
15	CV0035 CHRG MASTER FLO CNTRLR FAILS H/L	N/A	N/A	N/A	RT-15	26

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
01	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10

02	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160
----	----------------------------------	-----	-----	-----	-----	-----

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
01	C201 A LO QC201AY3 2C DIESEL GEN-TROUBLE ALARM	N/A	N/A	N/A	N/A	ON

OTHER CONDITIONS:

Description

1. None

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. 2C EDG Pre-Lube pump failure Simulator Operator: Insert RT-1 on direction from Lead Evaluator. MALF: AN0266 SER 266 Fails – J20 2C DIESEL GENERATOR URGENT TROUBLE Role play: 2 minutes after being dispatched call control room and report 2C EDG Alarm Window A-7 PRE-LUBE PUMP FAILURE is in alarm. You have checked the Pre-lube pump and it is NOT running. The Pre-lube pump breaker remains shut.			
	PO announces unexpected OHA J-20, 2C DG URGENT TRBL.		
	The crew refers to the ARP and dispatches an operator to investigate.		
	CRS refers to S2.OP-SO.DG-0003, 2C Diesel Generator Operation and recognizes 2C EDG is now inoperable with the Jacket Water Heater and Pre-lube pump inoperable, and enters LCO 3.8.1.1.b, action b.		
	CRS determines that S2.OP-ST.500-0001, Electrical Power Systems AC Sources Alignment must be performed within one hour		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Proceed to next event on direction from Lead Evaluator. 2. 23 Condensate pump trip / downpower Simulator Operator: Insert RT-3 on direction from Lead Evaluator. MALF:CN0117C 23 Condensate pump trip	to demonstrate the operability of the independent AC Sources (action b.1)		
Role Play: If dispatched, after 3 minutes report as NEO that you can't see anything wrong with 21 condensate pump. If dispatched to 4KV breaker, report after 2 minutes that there is an overcurrent relay flag dropped.	PO announces 23 Condensate pump trip.		
	CRS enters S2.OP-AB.CN-0001, Main Feedwater / Condensate System Abnormality.		
	PO reports no SGFP tripped.		
	PO reports 23 Condensate pump tripped.		
	PO bypasses condensate polisher by opening 21 thru 23 CN108 if SGFP suction pressure is <320 psig.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Crew should discuss power increase expected when the CN47 is opened. (Note in AB.CN-1).	PO reports SGFP suction pressure.		
	CRS directs 2CN47 opened if suction pressure remains < 320 psig.		
	PO isolates letdown by closing 21-24BG4 and 21-24GB185 if 2CN47 is open.		
	CRS directs power reduction to 85% at ≤5%/min with 2 Condensate pumps and 3 Heater Drain pumps in service IAW Attachment 4.		
	CRS enters S2.OP-AB.LOAD-0001, Rapid Load Reduction, to perform the load reduction.		
	CRS directs initiation of AB.LOAD CAS.		
	RO calculates boron addition and rate for power reduction.		
	RO initiates boration.		
	PO initiates turbine load reduction to 85% at rate directed by CRS.		
3. Master Flow Controller failure			
Simulator Operator: Once the			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
power reduction is underway, insert RT-15 to fail the Master Flow Controller. MALF: CV0035 CHRG Master Flow Controller fails H/L Severity: 26			
	RO reports unexpected low seal injection flow alarms on all RCP's.		
	RO reports Master Flow Controller is failed based on lower than expected Charging flow (75 gpm vs 87 gpm)		
	CRS enters S2.OP-AB.CVC-0001, Loss of Charging.		
	RO may use Alarm Response Procedure initially in response to Low Seal Injection Flow to adjust 2CV71 to restore seal injection flow to above the alarm setpoint, but this will not restore normal charging flow.		
	RO reports: - A charging pump is running with no indication of cavitation - RO reports neither a PZR level nor a VCT level channel has failed. - No indication of charging system leak		
	RO reports the Charging Master Flow Controller has failed.		
	RO takes manual control of 23 charging		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Proceed to next event on direction from Lead Evaluator. 4. 21 SG Tube Leak	pump, and adjusts speed to maintain PZR level on program IAW Att. 2.		
	RO reports PZR level can be maintained stable or rising.		
	RO announces auto rod motion when it occurs.		
Simulator Operator: Insert RT-5 on direction from Lead evaluator. MALF: SG00078A 21 STEAM GENERATOR TUBE RUPTURE Severity: 5			
	RO reports unexpected OHA A-6 and identifies 2R53A 21 Main Steamline N2 monitor from CRT indication in alarm.		
	RO reports 2R53A level >1000 gpd.		
	CRS enters S2.OP-AB.SG-0001, Steam Generator Tube Leak, and S2.OP-AB.RAD-0001, Abnormal Radiation.		
	RO reports OHA A-6 reflash is 2R15 Condenser Air Ejector monitor in alarm.		
	CRS directs initiation of AB.SG CAS.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports when 2R19A is in warning and alarm.		
	PO reports when SGBD isolation occurs on high radiation.		
	CRS dispatches operator to isolate TGA and Condensate Polisher area sump pumps.		
Simulator Operator: When dispatched to isolate TGA and polisher building sumps, insert RT-11 . This deenergizes 21-25 TGA sump pumps after a 3 minute time delay.			
	RO reports PZR level being maintained stable by rising charging flow.		
Note: Initial charging flow with stable PZR level is ~89 gpm. 23 charging pump will deliver ~99 gpm at full flow and seal injection of ~26 gpm. IF RO reports PZR level is lowering (due to lag in Master Flow control response,) then RO will swap to a centrifugal charging pump at Step 3.6.			
	RO reports PZR level can be maintained stable or rising.		
Note: SGBD will isolate ~2.5 minutes after 2R53D alarms. This			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
will cause RCS temp and PZR level to rise. Note: Do NOT report the 21MS45 is shut until AFTER 23 AFW pump has been tripped at step 4.7 of SGTR-1. Proceed to next event on direction from Lead Evaluator after Tech Specs have been addressed.			
	PO reports when load reduction is complete.		
	RO reports unit 2 is in MODE 1.		
	RO reports 21 SG is affected SG.		
	PO sets 21MS10 to 1045 psig, and ensures 21GB4, 21MS18, and 21MS7 are shut.		
	CRS dispatches an operator to shut 21MS45 steam supply to 23 AFW pump.		
	CRS enters LCO 3.7.1.2 for less than 3 operable AFW pumps, and LCO 3.4.7.2.c for >150 gpd primary to secondary leakage though 21 SG.		
	CRS determines (Action Level 3) the unit must be less than 50% in less than one hour, the unit placed in Hot Standby within 6 hours.		
	CRS dispatches an operator to realign SGBD and Sampling.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
5. SGTR			
Simulator Operator: MODIFY MALF SG00078A from 5 to 650 on direction from Lead Evaluator.			
	RO reports indications of worsening tube leak/rupture, with lowering PZR level and pressure.		
	CRS directs RO to trip the Rx, confirm the Rx trip, and initiate a Safety Injection, then enter TRIP-1.		
	RO trips the Rx, confirms the Rx trip, and initiates Safety Injection.		
Simulator Operator: Ensure <u>ET-1</u> is true when the rx is tripped. This trips 21 AFW pump after a 4 minute delay.			
	RO backs up the Main Turbine trip, reports 4KV vital buses energized, and reports SI manually initiated.		
	CRS and RO verify TRIP-1 immediate actions.		
	PO receives CRS concurrence and isolates AFW to 21 SG by closing 21AF11 and 21AF21.		
CT#1 (CT-18) Isolate feed flow into (part 1) and steam flow from			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
(part 2) 21 SG prior to a transition to SGTR-3 being required. (steam flow isolation will be performed on page 19) Part 1 SAT _____ UNSAT _____			
6. 21 AFW pump trip Note: 21 AFW pp supplies 23/24 SGs 22 AFW pp supplies 21/22 SGs 23 AFW pump supplies all SGs	PO receives CRS concurrence and reduces total AFW flow to no less than 22E04 lbm/hr until at least one SG NR level is > 9%, then maintains NR levels 19-33%.		
	PO reports when 21 AFW pump trips, and adjusts flow from 22 and 23 AFW pumps as required based on current SG NR levels.		
Simulator Operator: Use REMOTE AF20 to OFF 2 minutes after being dispatched to remove control power from 21 AFW pump breaker.			
	PO reports all available equipment started on SEC loading.		
	PO reports which AFW pumps are running.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports correct safeguards valve alignment.		
	RO reports 21 and 22CA330s are shut.		
	RO reports containment pressure remains less than 15 psig.		
	PO reports no indication of hi steam flow.		
	PO reports all 4KV vital buses energized.		
	RO reports ventilation systems lineups correct for current plant conditions.		
	RO reports at least 2 CCW pumps are running.		
	RO reports proper ECCS flow indicated for current plant conditions.		
	PO reports AFW status, and maintains at least 22E4 lbm/hr flow until at least one SG NR level is >9%, then maintains SG NR levels 19-33%.		
	RO reports RCP status, and RCS temperature stable at or trending to 547°.		
	RO reports both RTBs are open.		
	RO reports PZR PORVs are shut with Block valves open.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: STA will report to Control Room 10 minutes after being paged to report, and initiate monitoring of CFSTs.	RO reports PZR spray status for current plant conditions.		
	RO reports RCS pressure is >1350 psig.		
	RO maintains seal injection flow to RCPs.		
	PO reports no SGs are faulted.		
	PO reports NR or WR level is rising in 21 SG.		
	CRS transitions to EOP-SGTR-1, Steam Generator Tube Rupture.		
	RO maintains seal injection flow to all RCPs.		
Note: These steps complete CT#1	PO reports 21 SG is ruptured, and ensures 21MS10 is set at 1045 psig.		
	PO reports 21MS10 operating as expected for current pressure.		
	PO shuts 21MS167, and verifies 21MS7, 21MS18, and 21GB4 are shut.		
CT#1 (CT-18) Isolate feed flow into and steam flow from 21 SG prior to a transition to SGTR-3			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
being required. SAT _____ UNSAT _____			
	PO reports 21 SG is ruptured.		
	PO reports 23 AFW pump is not the only source of AFW.		
	PO ensures 23 AFW pump speed at minimum and trips 23 AFW pump.		
	CRS dispatches operator to shut 21MS45 if not previously performed.		
	CRS dispatches operator to shut 2SS321 sample valve.		
	PO reports 21 SG is isolated from all intact SGs.		
	PO controls AFW flow if required to maintain 21 SG NR level >9%.		
	PO reports when 21MS10 opens for pressure control @ 1045 psig.		
	PO reports 21 SG pressure is >375 psig.		
	CRS determines target temperature from Table D. (503°F for ruptured SG pressure ≥1000 psig.)		
	PO reports intact SGs are available for cooldown.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: When hottest CET is <503°F, crew will stop cooldown with steps on next page.	PO reports steam dumps are available, and places in MS Pressure Control-Manual at 25% to perform cooldown.		
	PO bypasses Tavg when Tavg Low-Low is reached.		
	Crew monitors hottest CET temperature while continuing in procedure.		
	RO reports power is available to both PZR PORV Stop valves, both PZR PORVs are shut, and both PZR PORV Stop valves are open.		
	RO resets SI and Phase A isolation, and reports Phase B isolation reset.		
	RO opens 21 and 22CA330.		
	PO resets each SEC and reports 230V control centers are reset.		
	RO reports RHR suction is aligned to the RWST and no RHR pump discharge flow.		
	RO stops both RHR pumps.		
	PO reports 21 SG is ruptured and 21MS167 is shut.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO shuts charging pump mini flows when RCS pressure is < 1500 psig.		
	RO reports when hottest CET is less than RCS cooldown target temp.		
	PO stops the cooldown and dumps steam to maintain RCS temp < target temp.		
CT#2 (CT-19) Establish and <u>maintain</u> an RCS temperature so that transition from SGTR-1 does not occur because temperature is too high to maintain minimum subcooling, or so low it causes transition to FRTS or FRSM. SAT _____ UNSAT _____			
7. 23 RCP trip			
	PO reports 21 SG pressure is stable or rising.		
	RO reports RCS subcooling >20 deg.		
	RO reports normal PZR spray is available.		
	RO fully opens both PZR Spray Valves.		
	RO reports PZR Spray is reducing RCS pressure.		
	Crew continues depressurization.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Simulator Operator: Insert RT-7 on direction from Lead Evaluator once RO reports spray effectiveness. MALF: 23 RCP trip Note: The crew is in a "do loop" when reducing RCS pressure using spray, and will return to Step 19.2			
	RO reports 23 RCP trip.		
	RO reports with loss of 23 RCP, spray is no longer effectively reducing RCS pressure.		
Simulator Operator: IF crew continues to use spray flow to lower RCS pressure, THEN insert RT-9 on direction from Lead Evaluator to trip 21 RCP.			
	RO shuts both PZR Spray valves.		
	RO reports both PZR PORVs are available.		
	Crew reviews depressurization criteria.		
	CRS directs RO to open only one PORV.		
	IF 2PR1 is selected as the PORV to depressurize, RO reports 2PR1 will not open.		
	RO opens 2PR2.		
Simulator Operator: Insert RT-13			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
as soon as 2PR2 is open. MALF: 2PR2 fails to 100%.			
8. PZR PORV fails to shut	RO reports when condition in Table E is met.		
	RO reports 2PR2 will not shut.		
	CRS directs closure of 2PR7 PZR PORV Block Valve.		
Simulator Operator: Ensure ET-3 is TRUE when 2PR7 shut PB is depressed. This fails 2PR7 90% open.			
Terminate the scenario when the transition to SGTR-3 is announced.	RO reports that after initiating closed on 2PR7, the open limit extinguished but the closed limit has not illuminated.		
	RO reports RCS pressure is NOT rising.		
	CRS transitions to EOP-SGTR-3, SGTR with LOCA – Subcooled Recovery.		

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-IO.ZZ-0004, Power Operation
- F. S2.OP-SO.DG-0003, 2C Diesel Generator Operation
- G. S2.OP-ST.500-0001, Electrical Power Systems AC Sources Alignment
- H. S2.OP-AB.SG-0001, Steam Generator Tube Leak
- I. S2.OP-AB.LOAD-0001, Rapid Load Reduction
- J. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- K. 2-EOP-SGTR-1, Steam Generator Tube Rupture

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 89.4% RCS BORON: 1114 MWe 1080

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Control Bank D is at 193 steps.

Xenon is building in at 25 pcm / hr following downpower last shift

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

S2.OP-PT.TRB-0003, Main Turbine Valve testing is being briefed by a dedicated test team to be performed in one hour.

ABNORMAL PLANT CONFIGURATIONS:

2C EDG Jacket Water heater is failed and C/T.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

SECONDARY:

Condensate Polisher is in service

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
with Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation.

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 15-01 ILOT NRC ESG-2 **REVIEWER:** C Recchione

Initials	Qualitative Attributes
CR	1. The scenario has clearly stated objectives in the scenario.
CR	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
CR	3. The scenario consists mostly of related events.
CR	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
CR	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
CR	6. The events are valid with regard to physics and thermodynamics.
CR	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
CR	8. The simulator modeling is not altered.
CR	9. All crew competencies can be evaluated.
CR	10. The scenario has been validated.
CR	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
CR	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4
SIMULATOR SCENARIO REVIEW CHECKLIST

Initial Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)

GG	2	Malfunctions after EOP entry: 1-2
GG	3	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	1	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	EOP based Critical tasks: 2-3

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

15-01 ILOT NRC ESG-2

CT#1 (CT-18) Isolate feed flow into and steam flow from 21 SG prior to a transition to SGTR-3 being required.

Basis: Failure to isolate the ruptured SG causes a loss of differential pressure between the ruptured SG and the intact SGs. The fact that the crew allows the differential pressure to dissipate and, as a result, are then forced to transition to a contingency ERG constitutes an incorrect performance that "necessitates the crew taking compensating action that would complicate the event mitigation strategy...."

CT #2: (CT-19) Establish and maintain an RCS temperature so that transition from SGTR does not occur because temperature is too high to maintain minimum subcooling, or so low it causes transition to FRTS or FRSM.

Basis: Failure to establish and maintain the correct RCS temperature during a SGTR leads to a transition from E-3 to a contingency ERG. This failure constitutes an incorrect performance that "necessitates the crew taking compensating action that would complicate the event mitigation strategy...."

Note to Evaluator: CT numbers in parentheses are the corresponding Westinghouse ERG Rev. 2- based Critical Tasks procedure WCAP-17711-NP

ATTACHMENT 6**ESG-PRA RELATIONSHIP EVALUATION****EVENTS LEADING TO CORE DAMAGE**

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
<u>N</u>	TRANSIENTS with PCS Unavailable	<u>N</u>	Loss of Service Water
<u>Y</u>	Steam Generator Tube Rupture	<u>N</u>	Loss of CCW
<u>N</u>	Loss of Offsite Power	<u>N</u>	Loss of Control Air
<u>N</u>	Loss of Switchgear and Pen Area Ventilation	<u>N</u>	Station Black Out
<u>N</u>	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
<u>N</u>	Containment Sump Strainers	<u>N</u>	Gas Turbine
<u>N</u>	SSWS Valves to Turbine Generator Area	<u>Y</u>	Any Diesel Generator
<u>N</u>	RHR Suction Line valves from Hot Leg	<u>Y</u>	Auxiliary Feed Pump
<u>N</u>	CVCS Letdown line Control and Isolation Valves	<u>N</u>	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
<u>N</u>	Restore AC power during SBO
<u>N</u>	Connect to gas turbine
<u>N</u>	Trip Reactor and RCPs after loss of component cooling system
<u>N</u>	Re-align RHR system for re-circulation
<u>N</u>	Un-isolate the available CCW Heat Exchanger
<u>N</u>	Isolate the CVCS letdown path and transfer charging suction to RWST
<u>Y</u>	Cooldown the RCS and depressurize the system
<u>Y</u>	Isolate the affected Steam Generator that has the tube rupture(s)
<u>N</u>	Early depressurize the RCS
<u>N</u>	Initiate feed and bleed

Complete this evaluation form for each ESG.

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 15-01 NRC ESG-3

SCENARIO NUMBER: 15-01 NRC ESG-3

EFFECTIVE DATE: See Approval Dates

EXPECTED DURATION: 60 minutes

REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

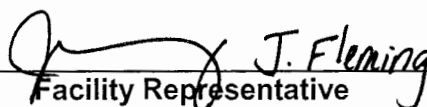
Revision Summary
New issue for 15-01 ILOT NRC exam

PREPARED BY: G Gauding
Lead Regulatory Exam Author

8-11-16
Date

APPROVED BY: 
Operations Training Manager

10/12/16
Date

APPROVED BY:  J. Fleming
Facility Representative

10/12/16
Date

SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

- A. Given the order or indications of a pressurizer control system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S1/S2.OP-S2.OP-AB.PZR-0001.
- B. Given indication of a pressurizer control system malfunction, DIRECT the response to the malfunction in accordance with S2.OP-AB.PZR-0001.
- C. Given indication of a dropped control rod perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S2.OP-AB.ROD-0002.
- D. Given indication of a dropped control rod DIRECT the response to the malfunction in accordance with S2.OP-AB.ROD-0002.
- E. Given the order or indications of a loss of secondary coolant, perform actions as the nuclear control operator to RESPOND to the coolant loss in accordance with EOP-LOSC-1.
- F. Given indication of a loss of secondary coolant, DIRECT the response to the loss of secondary coolant, in accordance with EOP-LOSC-1.
- G. Given the order or indications of a reactor trip, perform actions as the nuclear control operator to RESPOND to the reactor trip in accordance with 2-EOP-TRIP-1.
- H. Given indication of a reactor trip, DIRECT the response to the reactor trip in accordance with 2-EOP-TRIP-1.
- I. Given the order or indications of a safety injection, perform actions as the nuclear control operator to RESPOND to the safety injection in accordance with the approved station procedures.
- J. Given indication of a safety injection, DIRECT the response to the safety injection in accordance with the approved station procedures.
- K. Given the order or indications of a multiple steam generator depressurization, perform actions as the nuclear control operator to RESPOND to the generator depressurization in accordance with EOP-LOSC-2.
- L. Given indication of a multiple steam generator depressurization, DIRECT the response to the generator depressurization in accordance with EOP-LOSC-2.

II. MAJOR EVENTS

- A. Power Ascension
- B. Controlling PZR Pressure Instrument fails high
- C. Dropped control rod
- D. Steam Leak/Rupture in containment

III. SCENARIO SUMMARY

- A. The crew will take the watch with the unit stable at 1×10^{-8} Amps during a plant startup, BOL. 21 SGFP is in service and 22 SGFP is O/S. Steam dumps are in MS Pressure Control – Auto @ 1,000 psig. 25 CFCU is C/T for maintenance. 23 charging pump is C/T for emergent maintenance. The crew will be instructed to raise power to 2%.
- B. The crew will initiate a power ascension to 2% using Main Steam Dumps and control rods.
- C. After the power ascension is started the controlling PZR Pressure instrument will fail high, deenergizing PZR heaters and causing both PZR Spray valves to open fully. The crew will respond IAW S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction to establish manual control of PZR pressure control and restore pressure to normal, swap to an operable channel, and remove the failed channel from service. The CRS will identify applicable LCO's.
- D. After the Pressurizer Pressure control system failure has been addressed, a control rod will drop fully into the core. The crew will respond IAW S2.OP-AB.ROD-0002, Dropped Rod. The crew will identify the Rx is subcritical, and initiate control rod insertion to fully insert all Control and Shutdown rods. The CRS will identify applicable LCO's.
- E. During the rod insertion, a steam leak in containment will occur on 21 SG. The crew will diagnose changing containment conditions, and determine a Rx trip is required and MSLI will be required in an attempt to isolate the steam leak.
- F. The crew will trip the Rx. The steamline will rupture. MSLI (auto and manual) will fail to shut any MSIV. The crew will initiate Safety Injection when it is identified that the source of the steam leak is not isolated.
- G. The crew will perform plant stabilization actions in EOP-TRIP-1, Reactor Trip or Safety Injection. With all SGs faulted, the crew will maintain AFW flow to each SG. 21 AFW pump will trip during TRIP-1 performance. Containment Spray pumps will not automatically start when demanded on hi-hi containment pressure, and the crew will manually start at least one CS pump (CT-1). The crew will transition to EOP-LOSC-1, Loss of Secondary Coolant.
- H. The crew will transition to EOP-LOSC-2, Multiple Steam Generator Depressurization with all SG's faulted.
- I. The crew will identify the RCS cooldown rate of $>100^{\circ}\text{F} / \text{hr}$, and control AFW flow to minimize RCS cooldown (CT-2). The crew will reset Safeguards. When evaluating SI Flow Reduction Criteria, local operators will successfully shut a MSIV. The crew will recognize rising pressure in that SG, and transition to LOSC-1.
- J. The scenario will be terminated upon transition to LOSC-1.

IV. INITIAL CONDITIONS

___ IC-233

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers ,tagged equipment)

Initial	Description
___ 1	VC1and VC4 C/T
___ 2	RCPs (SELF CHECK)
___ 3	RTBs (SELF CHECK)
___ 4	MS167s (SELF CHECK)
___ 5	500 KV SWYD (SELF CHECK)
___ 6	SGFP Trip (SELF CHECK)
___ 7	23 CV PP (SELF CHECK)
___ 8	21 Charging pump in service
___ 9	25 CFCU C/T
___ 10	23 Charging pump C/T
___ 11	IOP-3 open and complete up to Section 4.3, Power Operation.
___ 12	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
	1	EVENT ACTION: MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) COMMAND: PURPOSE: <update as needed>

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
___ 01	PR0016A PZR PRESS CH I (PT455) FAILS H/L	N/A	N/A	N/A	RT-1	2500
___ 02	RD0267 ANY ROD(S) INADVERTENTLY DROPS	N/A	N/A	N/A	RT-3	3
___ 03	MS0090Ar 21 Main Steam Line Leak Inside Cont AFTER orifice	N/A	N/A	N/A	RT-5	2
___ 04	RP0279A AUTO MSLIS FAILS TO ACT, TRN A	N/A	N/A	N/A	N/A	
___ 05	RP0279B AUTO MSLIS FAILS TO ACT, TRN B	N/A	N/A	N/A	N/A	
___ 06	MS0092E 21MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
___ 07	MS0092F 22MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
___ 08	MS0092G 23MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
___ 09	MS0092H 24MS167 FAILS OPEN	N/A	N/A	N/A	N/A	
___ 10	RP0274A AUTO SI FAILS TO ACT, TRN A	N/A	N/A	N/A	N/A	
___ 11	RP0274B AUTO SI FAILS TO ACT, TRN B	N/A	N/A	N/A	N/A	
___ 12	AF0181A 21 AUX FEEDWATER PUMP TRIP	00:05:00	N/A	N/A	ET-1	
___ 13	RP318L1 21 CS Pump Fails to Start on SEC	N/A	N/A	N/A	N/A	
___ 14	RP318L2 22 CS Pump Fails to Start on SEC	N/A	N/A	N/A	N/A	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
___ 01	CV52D 23 CHG PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF
___ 02	PR34D PORV STOP VALVE 2PR6 TAGGED	N/A	N/A	N/A	RT-7	TAGGED
___ 03	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10
___ 04	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160
___ 05	CT195-1D 25 CFCU BKR #1 High Speed 125VDC	N/A	N/A	N/A	N/A	OFF
___ 06	CT195-2D 25 CFCU BKR #2 High Speed 125VDC	N/A	N/A	N/A	N/A	OFF

07	CT195-3D 25 CFCU BKR #3 Low Speed 125VDC	N/A	N/A	N/A	N/A	OFF
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OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
01	CI11 C LO QCI11HR2 TURNING GEAR-ENGAGED	N/A	N/A	N/A	N/A	ON
02	CI11 E LO QCI11PR2 TURNING GEAR-STARTED	N/A	N/A	N/A	N/A	ON
03	CI11 F LO QCI11TG2 TURNING GEAR-STOPPED	N/A	N/A	N/A	N/A	OFF
04	CI11 D LO QCI11LG2 TURNING GEAR-DISENGAGED	N/A	N/A	N/A	N/A	OFF

OTHER CONDITIONS:

Description

1. None

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. Power ascension Continue to next event on direction from Lead evaluator.			
	CRS directs power ascension using Main Steam Dumps and control rods.		
	PO raises steam dump demand.		
	RO adjusts control rods to maintain Tave on program.		
	RO announces when POAH is observed.		
2. Controlling PZR pressure instrument fails HIGH			
Simulator Operator: Insert RT-1 on direction from Lead Evaluator. MALF: MALF: PR0016A PZR Press CH I (PT-455) fails H/L Final Value: 2500			
	RO announces unexpected OHA alarms D-8 RC Press HI, and E-42 2PR1 ½ Trip.		
	RO determines actual pressure is not high and reports spray valves open, and recommends placing Master Pressure Controller in manual.		
	CRS directs RO to place Master Pressure Controller in manual.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: The requirement to shut 2PR6 and deenergize it is within one hour of the PZR Pressure Channel I instrument failure IAW Tech Specs.	RO takes manual control of Master Pressure controller, and lowers demand (increase pressure) to close sprays.		
	CRS enters S2.OP-AB.PZR-001, Pressurizer Pressure Malfunction.		
	CRS directs initiation of AB.PZR CAS.		
	CRS gives band for control of PZR pressure.		
	RO identifies PZR Press Channel I failed high.		
	RO selects Channel III for control.		
	RO matches master pressure controller demand to current pressure and returns Master Pressure Controller to AUTO.		
	RO shuts 2PR6.		
	CRS directs WCC to remove power from 2PR6.		
Simulator Operator: Insert RT-7 2 minutes after being directed to remove power from 2PR6. REMOTE: PR34D PORV Stop			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Valve 2PR6 tagged			
Note: Crew may wait for I&C response before performing any actions in S2.OP-SO.RPS-0003.	CRS directs PO to initiate removing failed channel from service and contact I&C for support IAW S2.OP-SO.RPS-003.		
	PO checks that tripping associated bistables will not result in an ESF or RPS actuation.		
	PO verifies that Channel III is selected for Master Pressure Control.		
	PO selects PZR Pressure recorder to channel other than I.		
	CRS enters TSAS 3.3.1.1 Action 6, 3.3.2.1.b Action 19*, 3.4.5.b, and 3.2.5.		
Proceed to next event on direction from Lead Evaluator.			
3. Dropped control rod			
Simulator Operator: Insert RT-3 on direction from Lead Evaluator. MALF: RD0064 ANY ROD INADVERTENTLY DROPS Severity: 3			
	RO reports unexpected OHA E-48 ROD		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	BOTTOM, and reports Shutdown Bank rod 1SA3 rod bottom light is illuminated.		
	CRS enters S2.OP-AB.ROD-0002, Dropped Rod.		
	RO reports control rods are in manual.		
	PO reports Main Turbine is S/D.		
	RO reports the reactor is subcritical as a result of the dropped rod.		
	CRS directs RO to insert all control and shutdown bank rods.		
	RO initiates insertion of control rods in manual.		
Simulator Operator: Once rod insertion has commenced, insert RT-5 on direction from Lead Evaluator MALF: MS0090Ar Severity: 2			
4. 21 SG steam leak / rupture in containment			
	PO announces OHA C-38, CFCU LK DET HI as unexpected, and shortly after, OHA C-30, CFCU LK DET HI-HI as well.		
	PO refers to ARP for OHA C-38 and C-30.		
	PO reports indications of a steam leak inside		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: A MSLI will be required to attempt to isolate the steam leak in containment. The CRS can use the CAS of AB.STM for direction, or will use Step 11 in TRIP-1 when containment pressure exceeds 15 psig. Auto and Manual MSLI are	containment including: - Containment pressure rising - Tavg slowly lowering - Rising steam flows on all loops - Steam dump demand lowering		
	CRS enters S2.OP-AB.STM-0001, Excessive Steam Flow.		
	CRS directs monitoring of AB.STM CAS.		
	CRS notifies Emergency Services of the steam leak.		
	PO reports that the Main Turbine is not latched.		
	PO reports no indication of MS10 or steam dump valves malfunction.		
	RO reports no indication of a SG Safety Valve partially open or leaking.		
	RO reports Rx power and RCS temperature are NOT stable.		
	CRS directs the RO to trip the Rx.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
failed and will not actuate.			
Simulator Operator: After the Rx is tripped, then MODIFY MALF MS0090Ar to 100 with a 5 minute ramp.			
	RO trips the Rx and confirms the Rx trip.		
Simulator Operator: Ensure ET-1 is true upon the Rx trip. This trips 21 AFW pump after a 5 minute delay.			
5. All MSIVs fail to shut			
	IF using AB.STM CAS, then RO will attempt a MSLI, and reports it did not actuate.		
	IF using AB.STM CAS, RO reports the steam leak is NOT isolated.		
	IF using AB.STM CAS, RO initiates SI and begins performing TRIP-1 immediate actions.		
	RO continues Immediate Actions of TRIP-1: - Reports the Main Turbine is tripped. - Reports all 4KV vital buses are energized. - Reports a demand for Safety Injection is present and Safety Injection has NOT actuated (or reports Safety injection was manually initiated as described in		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: With <u>all</u> SGs faulted, AFW flow should not be isolated to <u>any</u> SG.	previous step).		
	RO initiates Safety Injection (if not previously performed.)		
	PO reports SEC loading is not complete.		
	PO reports all available equipment started.		
	PO reports 21 and 22 AFW pumps are running, and if not previously performed, receives permission to throttle total AFW flow to no less than 22E4 lbm/hr.		
	PO reports valve groups in Table B are in Safeguards position.		
6. 21 AFW pump trip	RO reports 21 and 22CA330 are shut.		
	PO reports when 21 AFW pump trips, and adjusts AFW flow if required.		
7. Containment Spray pumps fail to start			
	RO reports containment pressure has not remained less than 15 psig.		
	RO initiates Phase B and Spray actuation.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports neither Containment Spray pump has started.		
	RO blocks 2A and 2C SECs.		
	PO resets 2A and 2C SEC's.		
	RO starts 21 and 22 CS pumps.		
CT#1 (CT-3) Manually start at least one Containment Spray pump before a red path challenge develops on the containment CSF. SAT UNSAT			
	RO reports MSLI has failed to close any MSIV.		
	RO stops all RCPs.		
	RO reports MSLI has failed to close any MSIV.		
	CRS dispatches operators to place valves from Table D (locally close MSIVs) in safeguards position if not previously performed.		
	PO reports all 4KV vital buses are energized.		
	RO reports control room ventilation is in Accident Pressurized mode.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports 2 switchgear supply and 1 exhaust fan are running.		
	RO reports 2 CCW pumps running.		
	RO reports ECCS flow as expected for current RCS pressure.		
	PO maintains total AFW flow greater than 22E4 lbm/hr until at least one SG NR level is >15% (adverse), then maintains SG NR level 19-33%.		
	RO reports no RCPs are running.		
	RO reports RCS Tcolds are dropping.		
	PO reports no dumping steam.		
	RO reports MSLI failed to isolate any MSIV.		
	RO reports both RTBs are open.		
	RO reports both PZR PORVs are closed.		
	RO reports 2PR6 is shut, and 2PR7 is open.		
	CRS dispatches operator to restore power to 2PR6 if deenergized.		
	RO reports no RCPs are running.		
	RO maintains seal injection flow to all RCPs.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: CFSTs are in effect when transition out of TRIP-1 occurs. STA will report to control room 10 minutes after being summoned via page to monitor CFSTs.	PO reports all SG pressures are dropping in an uncontrolled manner or completely depressurized with MSLI failure.		
	CRS transitions to EOP-LOSC-1, Loss of Secondary Coolant.		
	RO reports MSLI failed to shut any MSIV, all MS7 and MS18 valves are shut, and action has been directed to locally shut MSIVs.		
	PO reports all SG pressures are dropping in an uncontrolled manner.		
	CRS transitions to EOP-LOSC-2, Multiple Steam Generator Depressurization.		
	PO reports all valves in table A are shut except for all MSIVs.		
	CRS dispatches operators, if not previously performed, to locally shut valves in Table A one loop at a time.		
	CRS determines 23 AFW pump in service, and either stops 23 AFW pump or leaves it in service.		

Note: 23 AFW pump may be left in

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
service if it is determined it is necessary to maintain minimum AFW flow to keep SG tubes wet.			
	IF required, PO lowers 23 AFW pump speed to minimum, trips then stops 23 AFW pump.		
	IF required, CRS dispatches operator to shut 21 and 23MS45 steam supplies to 23 AFW pump.		
	RO reports RCS cooldown rate is $>100^{\circ}\text{F} / \text{hr}$.		
	PO reduces AFW flow to each SG to no less than $1.0\text{E}4 \text{ lbm} / \text{hr}$.		
CT#2 (CT-33) Control the AFW rate to not less than $1\text{E}4 \text{ lbm/hr}$ to minimize the cooldown rate.			
SAT <u> </u> UNSAT <u> </u>			
	When STA reports RED path on Heat Sink, CRS transitions to FRHS-1, verifies operator action was cause of RED path, and returns to procedure in effect.		
	RO reports RCS Thots are dropping.		
	RO reports RCS pressure, and reports a cooldown is in progress or RCPs are secured.		
	RO reports both PZR PORVs are closed.		
	RO reports 2PR6 is shut, and 2PR7 is open.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO resets SI, Phase A, and Phase B.		
	RO opens 21 and 22CA330.		
	PO reports all SECs are reset.		
	PO reports all 230V Control Centers reset.		
	RO resets SGBD Sample Isolation Bypass, and opens 21-24SS94s.		
	CRS directs Chemistry to sample SGs for activity and boron.		
	PO reports no indication of any SGTR.		
	RO reports RHR pumps running with suction from RWST, and no discharge flow.		
	RO reports RCS pressure is stable or rising, and stops both RHR pumps..		
	RO reports both CS pumps are running.		
	RO reports containment pressure.		
	If/when containment pressure is <13 psig, then RO: • Resets Spray actuation • Stops both CS pumps • Shuts 21 and 22CS2 CS pump discharge valves		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	RO reports RWST lo level alarm has not actuated.		
	RO reports RCS Thots status. IF <375, PO removes lockouts for 21-24SJ54's, and RO shuts 21-24SJ54's.		
	RO reports subcooling is >0°F.		
	RO reports RCS pressure and trend.		
	RO reports RCS pressure is stable or rising and: • Realigns charging pump through normal charging line and adjusts 2CV55 to maintain PZR level >33%. • Reports PZR level trend.		
Simulator Operator: When Step 20 is complete then shut 23MS167 by deleting MALF MS0092G and ensuring 23MS167 shuts.			
	PO reports indication of 23MS167 being shut.		
	CRS determines pressure rise in 23 SG requires transition to LOSC-1.		
	CRS transitions to LOSC-1.		
Terminate the scenario when the transition to LOSC-1 is announced.			

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load
- F. S2.OP-AB.PZR-0001, Pressurizer Pressure Malfunction
- G. S2.OP-AB.ROD-0002, Dropped Rod
- H. S2.OP-AB.STM-0001, Excessive Steam Flow
- I. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- J. 2-EOP-LOSC-1 Loss of Secondary Coolant
- K. 2-EOP-LOSC-2 Multiple Steam Generator Depressurization

**ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY**

MODE: 1 POWER: 1×10^{-8} A RCS BORON: 1711 MWe 0

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Control Bank D at 130 steps withdrawn.

Core burnup 500 EFPH.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

3.6.2.3 for 25 CFCU action a expires in 165 hours.

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

S2.OP-IO.ZZ-0003, Hot Standby to Minimum Load complete up to Section 4.3, Power Operation.

ABNORMAL PLANT CONFIGURATIONS:

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

PRIMARY:

21 Charging pump in service

23 Charging pump C/T for emergent repair, expected return in 2 hours

25 CFCU C/T for breaker replacement

SECONDARY:

21 SGFP in service

22 SGFP O/S with steam supply valves shut

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
with Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 15-01 NRC ESG-3 **REVIEWER:** C Recchione

Initials	Qualitative Attributes
CR	1. The scenario has clearly stated objectives in the scenario.
CR	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
CR	3. The scenario consists mostly of related events.
CR	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
CR	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
CR	6. The events are valid with regard to physics and thermodynamics.
CR	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
CR	8. The simulator modeling is not altered.
CR	9. All crew competencies can be evaluated.
CR	10. The scenario has been validated.
CR	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
CR	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4
SIMULATOR SCENARIO REVIEW CHECKLIST

Initial Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)

Initial	Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)	
GG	2	Malfunctions after EOP entry: 1-2
GG	3	Abnormal Events: 2-4
GG	1	Major Transients: 1-2
GG	2	EOPs entered/requiring substantive actions: 1-2
GG	0	EOP contingencies requiring substantive actions: 0-2
GG	2	EOP based Critical tasks: 2-3

COMMENTS:

ATTACHMENT 5
ESG CRITICAL TASKS

15-01 NRC ESG-3

CT#1 (CT-3) Manually actuate at least one Containment Spray pump before an extreme challenge develops to the containment CSF.

Basis: Failure to manually actuate the minimum required complement of containment cooling equipment under the postulated conditions demonstrates the inability of the crew to "recognize a failure or an incorrect automatic actuation of an ESF system or component."

CT#2 (CT-33) Control the AFW rate to not less than 1E4 lbm/hr to minimize the cooldown rate.

Basis: Failure to control the AFW flow rate to the SGs leads to an unnecessary and avoidable severe challenge to the integrity CSF. Also, failure to perform the critical task increases the challenges to the subcriticality and the containment CSFs beyond those irreparably introduced by the postulated plant conditions. Thus, failure to perform the critical task constitutes a demonstrated inability by the crew to "take one or more actions that would prevent a challenge to plant safety." It also fails to prevent "a significant reduction of safety margin beyond that irreparably introduced by the scenario."

Note to Evaluator: CT numbers in parentheses are the corresponding Westinghouse ERG Rev. 2- based Critical Tasks procedure WCAP-17711-NP

ATTACHMENT 6**ESG-PRA RELATIONSHIP EVALUATION****EVENTS LEADING TO CORE DAMAGE**

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
<u>N</u>	TRANSIENTS with PCS Unavailable	<u>N</u>	Loss of Service Water
<u>N</u>	Steam Generator Tube Rupture	<u>N</u>	Loss of CCW
<u>N</u>	Loss of Offsite Power	<u>N</u>	Loss of Control Air
<u>N</u>	Loss of Switchgear and Pen Area Ventilation	<u>N</u>	Station Black Out
<u>N</u>	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
<u>N</u>	Containment Sump Strainers	<u>N</u>	Gas Turbine
<u>N</u>	SSWS Valves to Turbine Generator Area	<u>N</u>	Any Diesel Generator
<u>N</u>	RHR Suction Line valves from Hot Leg	<u>Y</u>	Auxiliary Feed Pump
<u>N</u>	CVCS Letdown line Control and Isolation Valves	<u>N</u>	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
<u>N</u>	Restore AC power during SBO
<u>N</u>	Connect to gas turbine
<u>N</u>	Trip Reactor and RCPs after loss of component cooling system
<u>N</u>	Re-align RHR system for re-circulation
<u>N</u>	Un-isolate the available CCW Heat Exchanger
<u>N</u>	Isolate the CVCS letdown path and transfer charging suction to RWST
<u>N</u>	Cooldown the RCS and depressurize the system
<u>N</u>	Isolate the affected Steam Generator that has the tube rupture(s)
<u>N</u>	Early depressurize the RCS
<u>N</u>	Initiate feed and bleed

Complete this evaluation form for each ESG.